Aging and Structural Change: Cost Disease in the Care Sector

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Abstract

For most of the service sector, market productivity growth has pulled production out of the home. We show that formal nursing care is an exception because its productivity growth has been negative. A two-sector general equilibrium overlapping generations model with informal care attributes the rise in formal care's relative price to this decline, not demography. Care/non-care complementarity dominates formal/informal complementarity. As formal prices rise, elders substitute non-care consumption for formal care, shifting care onto informal caregivers. Population aging need not slow growth if non-care productivity rises or entering cohorts grow and its effects are offset by declining care needs.

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1 Introduction

The United States spends more than 1% of total outlay on formal nursing and residential care, about $201 billion in 2024 (in 2017 dollars), while the relative price of that care has more than tripled since the 1950s. Informal caregivers add unpaid time worth an estimated 1.4% to 4.0% of GDP.[1] Population aging raises the need for both kinds of care, but it also changes the input and output prices that determine what each costs, so how aging affects elder care, and how elder care affects output, are general equilibrium questions whose answers require a general equilibrium framework.

Trends in the care sector, however, run against the usual structural change patterns. In most services productivity growth in the market has pulled production out of the home: cooking, cleaning and child care have all moved into the market (Ngai and Pissarides 2008; Bridgman, Duernecker, and Herrendorf 2018). For elder nursing care, though, we estimate that negative productivity growth in the formal sector has driven up relative prices thus supporting substitution from formal to informal care. The cost disease channel a la Baumol (1967) thus pushes care toward the home, because of the structure of demand: we estimate that the care is a much stronger complement to other, non-care consumption than formal care is to informal care within the care composite. While demographic effects due population aging have shifted demand for elder care, we find that technology, not demography, drives the rise in the relative price of formal care and so is the primary exogenous force driving the sector's structural transformation.

It is well established that aging moves input prices, namely wages and rates of return (Krueger and Ludwig 2007; Backus, Cooley, and Henriksen 2014; Cooley and Henriksen 2018), though for goods and services which feature strongly age-dependent demand, output prices may also be sensitive to demographic changes (Pretnar and Feldman 2026). We bring such a general-equilibrium (GE) approach to elder care to better understand why the relative price of formal nursing care has increased, while both carefully separating the roles of technological change, market power, and demographic change and accounting for the relative substitutability of formal and informal care as well as care and non-care consumption. We seek to ascertain to what extent technological change, pricing power, and demographic effects have each contributed to structural change in the care sector, both formally and informally. We also seek to use our framework to understand how the economy might evolve in the face of population aging when there is differential technology growth between the care sector and the economy at large and aging itself impacts demand. Along the way we will take a holistic approach to studying the care sector, quantifying the resources elder care absorbs once accounting for informal care.[2]

Our endeavor uses a workhorse two-sector (non-care and nursing care) overlapping-generations

[1] Chari et al. (2015) value the informal elder care reported in the 2011–2012 American Time Use Survey at $221 billion a year at the cost of unskilled paid care, $522 billion at caregivers' forgone wages and $642 billion at the cost of skilled care. Broader estimates that include care for adults of all ages are of similar size (Arno, Levine, and Memmott 1999).

[2] Our valuation of informal elder care complements the household-production accounts (Bridgman et al. 2012; Bridgman 2016a; Bridgman, Craig, and Kanal 2022; Bridgman 2023; Bridgman 2025), which value housework, cooking, shopping and child care and have only recently separated care for adults from other activities (Bridgman 2026).

(OLG) model which is to our knowledge the first to place formal care, informal care, dependency risk, and population aging together in general equilibrium. Following the setup in Pretnar and Feldman (2026), a symmetric, monopolistically competitive nursing sector a la Dixit and Stiglitz (1977) produces formal care from capital and labor alongside a non-nursing sector producing consumption and investment goods. Retirees face idiosyncratic survival and dependency risk against for which insurance operates only by personal saving, government transfers, and their children's care; thus, markets for dependency risk are incomplete in the sense that relying on children's care is imperfect insurance since the parents cannot directly choose how much time their children will help them. Working-age households care about the welfare of elderly dependents and supply unpaid care that those dependents combine with the care they themselves buy. Dependents combine purchased and informal care in a constant elasticity of substitution (CES) care composite to generate total care demanded. The informal care each working-age cohort supplies goes to the dependents of the associated elder generation, and nursing labor is paid a fixed fraction of the economy-wide wage. This care aggregate then enters the dependent elder's utility function in a second CES aggregate with ordinary consumption. The difference between the two composite's elasticities of substitution determines how a change in the price of formal care moves care between the market and the home. This turns out to be novel, as we are the first to microfound the tri-partite tradeoff between formal care, informal care, and non-care consumption.

We calibrate the model to the 1990–2020 transition path of national-accounts prices, expenditure shares and sectoral input shares, and to the life-cycle profile of off-market time spent caring for elders from the American Time Use Survey (BLS 2026a). Four results stand out. First, negative productivity growth in the nursing sector accounts for almost all of the rise in the relative price of formal care, falling nursing markups slightly restrain it, and demographic change alone barely moves it. Second, falling age-specific dependency nearly offset population aging over 1990–2020 and raised the number of working-age adults per dependent, which, together with the rising price of formal care, raised informal care relative to formal care at every age. Third, informal care made up between 25% and 50% of the total cost of elder care over our sample, depending on how an hour is priced, adding potentially 0.5% to 1.8% to elder care's share of GDP. Fourth, in future projections only faster nursing productivity growth halts the rise in the relative price of formal care, while nursing wages catching up to economy-wide wages raise prices further, driving care substitution toward the informal sector. With dependency incidence held fixed, elder care's claim on output roughly triples by 2100, and the informal share of elder care falls; if incidence keeps declining, the informal share rises instead. Long-run GDP growth depends mainly on non-nursing productivity and the size of entering cohorts.

We contribute to four literatures. The first is structural change. Sectors with slow productivity growth suffer the cost disease of Baumol (1967) and Baumol, Blackman, and Wolff (1985) and absorb a growing share of spending and inputs, including time (Ngai and Pissarides 2007; Acemoglu and Guerrieri 2008; Herrendorf, Rogerson, and Valentinyi 2013; Herrendorf, Rogerson, and Valentinyi 2014; Herrendorf, Herrington, and Valentinyi 2015; Horenstein and Santos 2019;

Boppart and Ngai 2021). We apply the productivity-and-markup decomposition that Pretnar and Feldman (2026) use for health care to nursing care, whose market power has heretofore mostly only been studied at the facility level (Nyman 1994; Ching, Hayashi, and Wang 2015; Hackmann 2019). Market productivity growth has moved household services into the market (Ngai and Pissarides 2008; Ngai and Petrongolo 2017; Moro, Moslehi, and Tanaka 2017; Bridgman, Duernecker, and Herrendorf 2018; Bridgman 2016b; Bridgman and Valentinyi 2026), while we find that cost disease actually pushes elder care from the market toward the home.

The second literature involves the aggregate implications of aging, particularly how the size and age structure of the population move wages, returns, output and public finances (Storesletten, Telmer, and Yaron 1999; Huggett and Ventura 1999; Krueger and Ludwig 2007; Backus, Cooley, and Henriksen 2014; Kitao 2014; Acemoglu and Restrepo 2017; Cooley and Henriksen 2018).[3] We add the need for care that aging induces, which households can get from the market or off-market. Third, we add to the literature on the economics of informal care. Caregiving takes time from market work (Becker 1965; Muurinen 1986; Leigh 2010; Van Houtven, Coe, and Skira 2013), micro evidence finds formal and informal care to be substitutes (Orsini 2010; Mommaerts 2018; Mommaerts 2025), and care risk shapes late-life saving and insurance (Kopecky and Koreshkova 2014; Braun, Kopecky, and Koreshkova 2019). We add to Koreshkova and Lee (2024), who close the nursing-home market in equilibrium with local market power, by letting the interest rate, nursing homes' costs and the population, which they hold fixed, move in general equilibrium, and we show why micro data find substitution: formal and informal care are weak complements, but care and other consumption are much stronger ones. The fourth literature to which we add is the medical and public-health literature that measures care costs (Hickenbottom et al. 2002; Hayman et al. 2001; Langa et al. 2002; Plassman et al. 2007; Langa et al. 2001; Hurd et al. 2013; Kelley et al. 2015; Jutkowitz et al. 2017; Dunbar et al. 2018) and projects it forward at fixed age-specific rates (Brookmeyer, Gray, and Kawas 1998; Alzheimer's Association 2026; Zissimopoulos, Crimmins, and St. Clair 2014; Nandi et al. 2024). Incidence rates at each age has been falling (Satizabal et al. 2016; Langa et al. 2017; Wolters, Chibnik, Waziry, et al. 2020; Hudomiet, Hurd, and Rohwedder 2022), which lowers projected costs (Hurd, Martorell, and Langa 2015; Brück et al. 2023). Our additions are to ensure the price of formal care and the mix of care respond to aging in equilibrium and to answer how the choice between formal and informal care responds to prices in the aggregate, a question Chandra, Coile, and Mommaerts (2023) leave open.

The paper proceeds as follows. Section 2 decomposes the relative price of formal care, Section 3 presents the household model, Section 4 the calibration, and Section 5 the quantitative results. Section 6 concludes.

[3] These build on Fougère and Mérette (1999) and Ríos-Rull (2001). Further, Miller and Bairoliya (2023) stress how unequally health and longevity are distributed among the elderly.

2 The Formal Nursing-care Sector

To our knowledge there exist no proper investigations as to the underlying causes of nursing sector structural transformation in the U.S., despite the sector having grown to represent over 1% of total U.S. outlay, while seeing relative prices rise by a factor > 3 since 1950. It is natural to ask to what degree is the nursing-home sector plagued by cost disease a la Baumol (1967)?[4] To answer this question we must have an idea of how the sector's TFP has evolved relative to the rest of the economy. Further, estimates of TFP leveraged strictly from prices may be contaminated by changing sectoral markups, so we must introduce a framework that can control for this. While the focus of the structural change literature is typically on how very broadly defined sectors, like services and manufacturing,[5] have evolved over time, the principles of analysis can be applied to any sectoral partition.[6] Whether cost disease has driven up relative prices or other factors, like increasing markups or changes to the composition of demand, matters for policymakers concerned with the sector's structural evolution in the face of population aging.

In this section we specify a structural change model of sectoral production that can be used to cursorily measure how relative sectoral variation in markups and TFP's has affected prices, while also providing preliminary insights as to the magnitude of demand-driven GE effects. There exists some micro evidence that local market concentration within the nursing-care sector leads to higher prices (Nyman 1994) and possibly lower care quality (Hackmann 2019), suggesting a role for markups in driving up prices, absent cost disease. Some policymakers in recent years have expressed concerns that the nature of sectoral concentration (i.e., private equity firms acquiring nursing homes) has driven up nursing-sector prices (Braun et al. 2021; The White House 2022; Gupta et al. 2024), though there lacks evidence that such phenomena are adversely affecting aggregate outcomes. Indeed, firms may have relatively high pricing power in the nursing sector because demand for nursing-home services, like other acute health-care services, is rather inelastic (Ching, Hayashi, and Wang 2015), suggesting a role for studying both *why* and *how* factors that impact demand may also affect inferences regarding concentration via GE channels. Thus, to understand the role of pricing power versus productivity versus demand effects in driving changes to the aggregate share of nursing consumption expenditure, a more sophisticated macroeconomic framework is needed: a flexible modification of the two-sector growth accounting decomposition of Pretnar and Feldman (2026) provides just such a framework. We thus proceed by first showing how nursing-sector aggregates have evolved before engaging in a value-added (VA) growth accounting exercise designed to identify the roles of markups, TFP's, and GE effects in driving structural change. We summarize the VA growth accounting exercise here in the main text, leav-

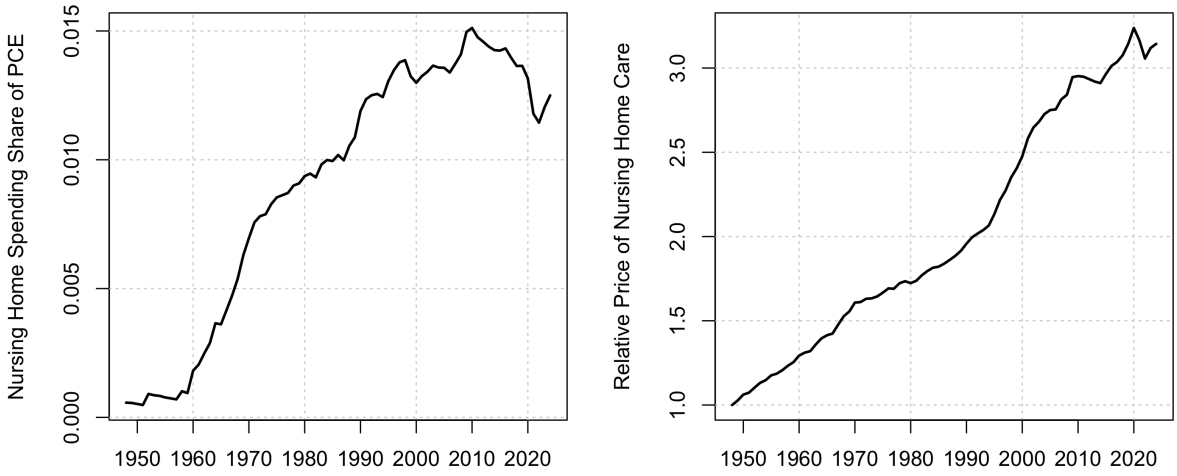
[4] See also Baumol, Blackman, and Wolff (1985), Ngai and Pissarides (2007), Herrendorf, Rogerson, and Valentinyi (2014), Herrendorf, Herrington, and Valentinyi (2015), and Pretnar and Feldman (2026).

[5] See Herrendorf, Rogerson, and Valentinyi (2014) for a high-level treatment as to the factors that lead to an economy's structural transformation.

[6] For example, in Pretnar and Feldman (2026) the authors look at what forces have contributed to the simultaneous rise in both the U.S. health-services share of consumption and the relative price of U.S. health services to all other consumption.

ing the details for Technical Appendix B.1.

2.1 Aggregate Nursing Sector Prices and Expenditure



(a) Expenditure Share of PCE (b) Relative Price, \$1948 = 1

Figure 1: Panel (a) shows the nursing-home expenditure share of all PCE, while panel (b) presents the relative price of nursing-home care to all other consumption.

Figure 1a presents the share of personal consumption expenditure (PCE) devoted to nursing-home services from National Income and Product Accounts (NIPA) Table 2.5.5., while Figure 1b shows the relative price of nursing-home services to all other consumption.[7] Note that the numbers presented in Figure 1 are inclusive of governmental expenditure from Medicaid and Medicare that cover nursing-home services outlay, as well as expenditure made by other non-governmental non-profit organizations on behalf of consumers using nursing-home services.[8] In the U.S. outlay on nursing-home services increased from 0.05% of total personal consumption expenditure (PCE) in 1950 to 1.25% in 2024 (BEA 2026e). Even within the health sector itself, nursing-home (and related) services expenditure increased from 1.2% of all health spending in 1950 to 6% in 2024 (BEA 2026e).[9] From 1948 to 2024 spending on nursing-home services grew 22 times faster than all other spending in the U.S. economy. Meanwhile, the relative price of nursing-home services to all other

[7] To compute the relative price of nursing-home to non-nursing-home services consumption, we follow the procedures outlined in Whelan (2002) and Whelan (2003) for unwinding and then reconstructing chain-weighted quantity and price indices for newly-defined sectoral aggregates. Herrendorf, Rogerson, and Valentinyi (2013), Pretnar and Feldman (2026), and Bednar and Pretnar (2026) also use the procedure to construct time series of relative prices for their various structural change analyses. The data required to compute new relative price indices are the quantity indices from NIPA Table 2.5.3, price indices from Table 2.5.4, and expenditure from Table 2.5.5 (BEA 2026e).

[8] See *NIPA Handbook* (2022).

[9] In Figure A.1b of Technical Appendix A.2, we also show that within the broader health sector, the nursing-home share of total outlay rose from $< 2\%$ in 1950 to over 8% by 1980, before settling to 6% today.

consumption has grown by 214% since the mid-twentieth century (BEA 2026e).[10]

2.2 Two-sector Production Model

We now present a sectoral production model with sector-specific monopolistic competition. Following a CES generalization of the two-sector Cobb-Douglas decomposition procedure in Pretnar and Feldman (2026), we decompose variation in the relative price of nursing-home services into three drivers: 1) relative TFP; 2) relative sectoral markups; 3) other GE effects. The GE effects on prices can result from both changes to the composition of demand due to population aging and the reallocation of sectoral inputs in response to exogenous variation in relative TFP and relative markups. We thus caution that this exercise should be seen as a first pass: it fails to control for how exogenous TFP, markup variation, and population will impact the allocation of capital and labor inputs, and thus prices, between sectors once accounting for changing household behavior over time. Later, in Section 3.3 we will close the model with a household sector in order to fully simulate general equilibrium outcomes.

The economy is characterized by two sectors — one which produces non-nursing-home consumption plus investment and another producing nursing-home services. Each sector is characterized by a mass of firms, indexed on a unit-continuum, $i \in [0, 1]$. Each firm in a sector produces imperfectly substitutable varieties, y_{it}. Let y_t be a sector-specific output index a la Dixit and Stiglitz (1977): $y_t = \left(\int_0^1 y_{it}^{1/\mu_{yt}} \mathrm{d}i \right)^{\mu_{yt}}$. In this setup μ_{yt} is the sectoral average markup, so that variations in sector-specific markups over time entirely result from variations in the relative substitutability of within-sector varieties, where the elasticity of substitution is $\mu_{yt}/(\mu_{yt} - 1)$. When varieties are relatively more substitutable, markups within a sector are relatively lower and vice-versa.

Let the population level be given by M_t (i.e., the mass of consumers). Aggregate output for each sector is denoted with capital letters such that $Y_t = M_t y_t$, where y_t is the output index. The non-nursing-home sector retains a Cobb-Douglas production technology:

$$C_t + I_t = Z_{ct} K_{ct}^{\alpha_c} L_{ct}^{1-\alpha_c}. \tag{1}$$

The nursing-home sector, however, is generalized to a constant elasticity of substitution (CES) production function:

$$N_t = Z_{nt} \left[\alpha_n K_{nt}^{\frac{\epsilon-1}{\epsilon}} + (1 - \alpha_n) L_{nt}^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \tag{2}$$

where ϵ is the elasticity of substitution between capital and labor in the nursing sector. When $\epsilon = 1$, the CES aggregator in (2) collapses to the Cobb-Douglas form $N_t = Z_{nt} K_{nt}^{\alpha_n} L_{nt}^{1-\alpha_n}$, recovering a version of the specification in Pretnar and Feldman (2026).[11] When $\epsilon < 1$, capital and labor are

[10] Figure A.1a of Technical Appendix A.2 shows that the run up in nursing sector expenditure shares is not entirely a result of price increases: in 2024 the relative real $1948 quantity of nursing-home consumption was greater than seven times that in 1948.

[11] In our full GE calibration we found through benchmarking that a Cobb-Douglas nursing-home sector specification could not match variation in the nursing-sector factor-input shares over time — hence our choice to generalize here.

gross complements in nursing production; when $\epsilon > 1$, they are gross substitutes.

Subscript c denotes the non-nursing-home consumption and investment sector, while subscript n denotes the nursing-home services sector, outlay from which is summarized by N_t. Note that the non-nursing-home sector is assumed to produce all physical capital. TFP's are Z_{yt}, where $y \in \{c, n\}$. K_{yt} are the sector-specific stocks of productive capital. L_{yt} are sector-specific labor allocations. The parameter α_c is the capital intensity in the consumption sector. In the nursing sector, α_n governs the relative weight of capital versus labor in the CES aggregator, coinciding with the sector-specific capital share only when $\epsilon = 1$.

Assume that the average capital rental rate, r_t, is constant across sectors. Let w_t be the wage paid by the non-nursing sector and let $\omega_t = w_{nt}/w_t$ be the ratio of the nursing-sector wage to the non-nursing-sector wage, so that nursing firms pay $\omega_t w_t$ per unit of labor. We take the investment good as the numeraire, normalizing $p_{It} = 1$. We also assume that investment is sold at marginal cost. The non-nursing sector's marginal cost, MC_{ct}, is therefore tied to the numeraire,

$$MC_{ct} \equiv \frac{1}{Z_{ct}} \left(\frac{r_t}{\alpha_c} \right)^{\alpha_c} \left(\frac{w_t}{1 - \alpha_c} \right)^{1-\alpha_c} = p_{It} = 1, \tag{3}$$

so that the non-nursing sector's consumer price is the markup itself, $p_{ct} = \mu_{ct} MC_{ct} = \mu_{ct}$. Equation (3) is not a normalization beyond the choice of numeraire. It is the non-nursing sector's factor price frontier, and it removes one degree of freedom from factor prices, so that given Z_{ct} and r_t, its wage is

$$w_t = (1 - \alpha_c) \, Z_{ct}^{\frac{1}{1-\alpha_c}} \left(\frac{\alpha_c}{r_t} \right)^{\frac{\alpha_c}{1-\alpha_c}}. \tag{4}$$

The nursing sector's price under CES production is

$$p_{nt} = \frac{\mu_{nt}}{Z_{nt}} \left[\alpha_n^{\epsilon} r_t^{1-\epsilon} + (1 - \alpha_n)^{\epsilon} (\omega_t w_t)^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}, \tag{5}$$

where the marginal cost is $MC_{nt} = (1/Z_{nt}) \left[\alpha_n^{\epsilon} r_t^{1-\epsilon} + (1 - \alpha_n)^{\epsilon} (\omega_t w_t)^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}$ and the sector-specific markup is $\mu_{nt} \geq 1$. When $\epsilon = 1$, the CES cost function in (5) reduces to $p_{nt} = \frac{\mu_{nt}}{Z_{nt}} \left(\frac{r_t}{\alpha_n} \right)^{\alpha_n} \left(\frac{\omega_t w_t}{1-\alpha_n} \right)^{1-\alpha_n}$.

Let $p_t = p_{nt}/p_{ct}$ be the relative price of nursing-home consumption, $\mu_t = \mu_{nt}/\mu_{ct}$ be the relative markup, and $Z_t = Z_{nt}/Z_{ct}$ be the relative productivity.[12] Dividing (5) by $p_{ct} = \mu_{ct} MC_{ct}$, with MC_{ct} written out as in (3), and re-arranging gives an expression for the relative price of nursing-home services:

$$p_t = \frac{\mu_t}{Z_t} \left(\frac{\alpha_c}{r_t} \right)^{\alpha_c} \left(\frac{1 - \alpha_c}{w_t} \right)^{1-\alpha_c} \left[\alpha_n^{\epsilon} r_t^{1-\epsilon} + (1 - \alpha_n)^{\epsilon} (\omega_t w_t)^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}}. \tag{6}$$

[12] Relative variables are always placed in units of sector c.

When $\epsilon \to 1$, (6) simplifies to the Cobb-Douglas relative price expression:

$$p_t = \frac{\mu_t}{Z_t}\omega_t^{1-\alpha_n}\left(\frac{r_t}{w_t}\right)^{\alpha_n-\alpha_c}\left(\frac{\alpha_c^{\alpha_c}(1-\alpha_c)^{1-\alpha_c}}{\alpha_n^{\alpha_n}(1-\alpha_n)^{1-\alpha_n}}\right), \tag{7}$$

recovering equation (4) from Pretnar and Feldman (2026).

Finally, note that firms make positive economic profits and distribute these as dividends. Sectoral profits are $\Pi_{nt} = (\mu_{nt} - 1)MC_{nt}N_t$ and $\Pi_{ct} = (\mu_{ct} - 1)MC_{ct}C_t$, the latter applying to consumption sales only, since investment is sold at cost. Total profits are $\Pi_t = \Pi_{ct} + \Pi_{nt}$.

The CES generalization on the nursing sector introduces flexibility while sacrificing a bit of tractability relative to a case where both sectors are Cobb-Douglas. When $\epsilon \neq 1$, the nursing-sector cost function is no longer log-linear in factor prices, so a growth-accounting decomposition of (6) along the lines of Pretnar and Feldman (2026) no longer yields a clean additive structure. In the nursing sector CES production implies that the capital-labor ratio responds to factor prices according to

$$\frac{K_{nt}}{L_{nt}} = \left(\frac{\alpha_n}{1-\alpha_n}\right)^{\epsilon}\left(\frac{\omega_t w_t}{r_t}\right)^{\epsilon}. \tag{8}$$

Time-varying GE effects are embedded in the ratio of input prices, which are fully captured by changes in the allocation of inputs in either sector *only* in the dual Cobb-Douglas case. When $\epsilon \neq 1$, the nursing sector's capital-labor ratio responds more ($\epsilon > 1$) or less ($\epsilon < 1$) elastically to factor price changes than does the consumption sector's, breaking the one-to-one correspondence.

The same cost structure implies within-sector factor cost shares for the nursing sector,

$$\theta_{K_n,t} = \frac{\alpha_n^{\epsilon} r_t^{1-\epsilon}}{\alpha_n^{\epsilon} r_t^{1-\epsilon} + (1-\alpha_n)^{\epsilon}(\omega_t w_t)^{1-\epsilon}} \qquad \text{and} \qquad \theta_{L_n,t} = 1 - \theta_{K_n,t}, \tag{9}$$

which are the fractions of nursing-sector income accruing to capital and to labor. Unlike the Cobb-Douglas case, in which they are the constants $\theta_{K_n,t} = \alpha_n$ and $\theta_{L_n,t} = 1 - \alpha_n$, in the general CES case these objects vary with factor prices.[13]

Technical Appendix B.1 carries out the naive growth accounting decomposition, which operates on a log-linearized and first-differenced version of (6) using the factor-share representations in (9) and an inversion of the non-nursing-sector FOC in (4) to extract r_t. The decomposition assumes an average capital share within the nursing sector, $\overline{\theta}_{K_n}$, and is conducted with data for real wages by sector, sectoral prices, sectoral expenditure shares, sectoral labor inputs, sectoral capital inputs, and non-nursing-sector TFP to back out estimates for $\{\widetilde{g}_{\mu,t}\}_t$ (logged growth rate of rela-

[13] A note on notation, since two different kinds of share appear in what follows. A *subscripted* factor index denotes a within-sector share, so that $\theta_{K_n,t}$ is the fraction of sector-n income accruing to capital, and $\theta_{K_n,t} + \theta_{L_n,t} = 1$ by construction. When presented unindexed (e.g., θ_{Lt}) this is an economy-wide total income share. A *superscripted* factor index denotes the split of an economy-wide factor across sectors, so that $\theta_{nt}^K = K_{nt}/K_t$ is the fraction of the aggregate capital stock employed in nursing, and $\theta_{nt}^K + \theta_{ct}^K = 1$. The superscript thus names the factor being divided between sectors, while the subscript names the sector receiving the share.

tive markups) and $\{\widetilde{g}_{Z_n,t}\}_t$ (logged growth rate of nursing-sector TFP), from which we can then glean how structural transformation in the form of a rising nursing-sector expenditure share has been driven (in the data) by various allocative forces. Such a decomposition is inherently "naive" because in decomposing how allocative changes in the data affect inferences regarding the degree to which markups and productivities have driven structural change, we must take factor and output prices as given, even though such prices will embed GE effects resulting from changes to the age-composition of households, for example.[14]

Nonetheless, we find two results attain. Differential sectoral productivity growth is the dominant force, so that Z_{ct} is growing faster than Z_{nt}, but so is μ_{ct} growing faster than μ_{nt}. The growth accounting suggests that relative markups, $\mu_t = \mu_{nt}/\mu_{ct}$ fell by between 9.4% and 10.5% from 1991 to 2022, so that concentration in the nursing sector is *not* driving relative price growth. Over the same period, non-nursing sector TFP grew by a whopping 66.1% to 68.5% *more* than nursing-sector TFP, suggesting cost disease is almost entirely the driver of rising $p_t = p_{nt}/p_{ct}$.

The growth accounting findings also suggest that GE effects, operating through factor prices, are weak, and the sectoral components largely offset one another. The identification is not sharp in one respect that we will show later matters: the decomposition uses a single average nursing capital cost share rather than its time series, and through (9) a single share restricts the nursing capital weight α_n and the elasticity ϵ only to a locus of pairs, so the decomposition must assume a value for that average share to proceed.[15]

Yet, we should expect that as the population ages aggregate demand for nursing-sector services grows, but nothing in the decomposition reported in Technical Appendix B.1 accounts for how changing household preferences on aggregate affects GE outcomes. Output and factor prices summarize the GE effects in the data, but they are endogenous and dependent on variation in household demand. This decomposition fails to control for such dependencies, and as Pretnar and Feldman (2026) show, appropriately modeling the household side of the economy matters for estimating the underlying contribution of relative markup and relative TFP variation to structural change.

Also, heretofore, all data and associated decompositions are that of the *formal* (i.e., market, measured) nursing-care sector — paid long-term care services recorded in the national accounts. But, some old-age nursing care is not necessarily provisioned by market forces, which means it is thus not directly measured in things like aggregate expenditure shares or aggregate output. By turning to time-use data and the literature, we will now document the role of informal caregiving in the market for elder care. We then introduce a model of household behavior with features uniquely relevant to the market for old-age nursing-care services and which accounts *both* for resources allocated to nursing care that are not directly measured by market outcomes and more

[14] It is for this reason that we reserve the data decomposition exercise for an appendix, using it for comparison purposes only after we engage in a full decomposition of a fully calibrated GE model with households later in Section 5.2.

[15] In Technical Appendix B.1 we assume that this cost share is equivalent to that for the broader health-care sector, which Donahoe (2000) report to be 0.26. We relax this assumption in our full GE calibration in Section 4 and find that the capital cost share within the nursing sector is considerably lower.

formal, marketized nursing-care services.

3 Household Demand for Formal and Informal Nursing-care Services

Section 2 documents that the formal nursing-care sector — i.e., paid long-term care services produced in the market and measured in the national accounts — is plagued by relatively slow TFP growth such that, despite increases to the sector's capital input, firms within the sector are actually becoming *less* efficient at production over time. This phenomenon causes relative prices to grow at an accelerated rate, independent of any reallocative effects. However, formal outlay to market-provided care services is not the only kind of outlay for elder care: younger, working-age adults also spend time caring informally (off-market) for their elders. The degree to which formal- and informal-care are substitutable (or even complementary) may matter for determining measured GE outcomes and the relative rise of the formal nursing-care sector.

In this section we show how informal care is provided. We then discuss patterns of substitution between formal and informal care discussed in the literature. After documenting some data facts, we then close the full model economy with a household sector where the children of elderly retirees are motivated to care for their elders.

3.1 Informal Outlay for Elder Care

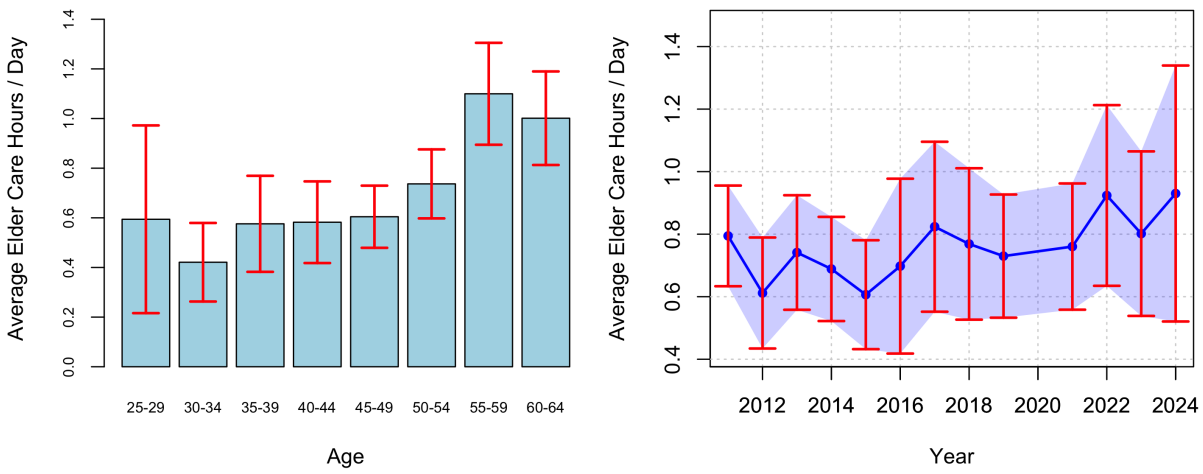
In 2011 the American Time Use Survey (ATUS) began asking respondents whether or not they provided care/help for elders any time over the previous three to four months (BLS 2026a).[16] For individuals who responded in the affirmative, ATUS records the time the individual respondent spent on the day of record engaging in care/help for elders. Figure 2a presents weighted averages of off-market elder care hours supplied per day conditional upon the care provider's age. Figure 2b presents the weighted average over care providers of all ages from 2011 to 2024.[17] In both plots of Figure 2, the weighted averages are conditional upon the respondent participating in the informal market for elder care.[18] In the ATUS the number of respondents who selected into the elder care supplement was around 18% in 2011, falling to 16.6% by 2019 then 15.6% in 2023, before ticking up to 16.0% in 2024, suggesting that the extensive margin of informal care provision is itself flat (i.e., adults are *not* increasingly more likely to be care providers as the population ages).[19] Over 37% of caregivers report providing care to either a mother or father. When parents-in-law are included, this number rises to 48%, and when grandparents, aunts, and uncles are included it rises to 65%. Thus, a substantial majority of care providers are providing cross-generational care where the provider is approximately one generation younger than the receiver. The remainder are

[16] The time range depends on when, during the month (i.e., beginning or end), an individual filled out the ATUS diary survey.

[17] Note that data for 2020 are excluded since the ATUS does not feature sampling weights for that year.

[18] The extensive margin here is observed when the respondent answers in the affirmative that they provided care/help to at least one elder over the previous three to four months.

[19] There were 15,697 elder-care providers between the age of 25 and 64 in the ATUS over the entire period, 2011-2024, not including 2020, a year for which ATUS provides no sampling weights.



(a) Care Time Provided by Age

(b) Care Time Provided over Time

Figure 2: This figure shows how the intensive margin of informal elder care supply varies across age groups and over time for those in the ATUS who affirmed that they engaged in some elder care in the previous 3-4 months (i.e., conditional upon the extensive margin holding). Panel (a) shows the weighted average elder care hours per day provided by each age group, while panel (b) shows the overall average supply for ATUS respondents age 25-64 in each year of the sample. 90% confidence intervals are included. 2020 is excluded from the sample because of issues with sampling weights in that year (BLS 2021). For unconditional plots computed over the entire sample (i.e., the fractions of elder care time provided where the denominator includes those who *do not* provide elder care), see Technical Appendix A.4.

spread out amongst neighbors, friends, other relatives and non-relatives, siblings, and partners both married and unmarried. The average age difference between care provider and care receiver is 27.6 years (i.e., roughly one generation gap). When parents or parents-in-law are involved, the average age difference is between 25 and 28 years. The average age of care providers conditional upon the age of the care receiver is 54.0 years for care receivers over 80, 47.7 years for care receivers between 70 and 80, and 44.1 years for care receivers between 65 and 70. In the face of population aging, as the ratio of young to old declines, one question is how much the demand for elder care (particularly through the time use channel) will impact other economic aggregates (e.g., the intensive margin of labor, other home production activities, etc.).[20]

Although the ATUS intensive-margin estimates in Figure 2 imply roughly 0.6–1.0 hours of elder care time per day among those who report providing any care (i.e., on the order of 4–7 hours per week if care were provided daily), retrospective caregiver surveys typically report even larger "usual" weekly hours among caregivers. The *Caregiving in the U.S.* reports, for example, generally place average unpaid caregiving intensity on the order of tens of hours per week among active caregivers (National Alliance for Caregiving and AARP 2004; National Alliance for Caregiving and AARP 2009). Similarly, profiles of chronically-disabled elders and their informal helpers as

[20]See Technical Appendix A.3 and particularly Figure A.4 which shows how the ratio of working-age adults to those over age 65 has declined over the last 70 years.

recorded in the National Long-Term Care Survey tend to imply high-intensity care arrangements (again, often on the order of tens of hours per week) for the subset of elders receiving substantial assistance (Manton 2010; Center on an Aging Society, Georgetown University 2005).[21]

3.2 Formal and Informal Care Substitutability

In July of 2022 there were over 1.2 million U.S. residents of over 15,000 commercial nursing home facilities (Office of Inspector General, HHS 2024). There is evidence that coresidence (ostensibly a proxy for off-market care time) and nursing-home utilization are substitutes (Mommaerts 2018), but exactly measuring the elasticity of substitution between off-market care *time* and nursing-home residence is difficult, especially at the micro level. Mommaerts (2025) does provide evidence, though, that the availability of informal caregivers affects elders' decisions to buy long-term care insurance. Further, the availability of informal care seems to also suppress aggregate Medicaid spending on formal care (Mommaerts 2025). Micro evidence from Van Houtven, Coe, and Skira (2013) and Mommaerts (2018) that formal and informal care are substitutes is also further supported by indirect evidence that lowering the cost of acquiring long-term care insurance also reduced informal care demand (Mommaerts 2025). By slight contrast, Coe, Goda, and Van Houtven (2023) also find that coresidence is declining in insurance coverage, though informal care itself does not. Among adults with dementia, Choi et al. (2021) likewise find that those with a spouse or adult child available, in particular a coresident adult child, receive more informal care, use less formal care, and are less likely to enter a nursing home, though the authors caution that the association is not causal. However, fully accounting for the endogeneity of prices in a formal GE model has yet to be undertaken in a rigorous quantitative framework which would more decisively pin down this relationship, particularly on aggregate.

We thus proceed by building a model of the household that allows for a flexible substitution margin between formal and informal care demand. Parameters associated with this margin will then be internally calibrated in our later quantitative exercises to ensure a joint set of moments characterizing markets for formal and informal care are best matched alongside the aggregate, sectoral data points we outlined in Section 2.

3.3 Model of the Household

In any given period t there exists J living cohorts of overlapping generations. Consumers/households are heterogeneous by age, $j \in \{1, \ldots, J\}$.[22] Households can be either of working age, $j < J_R$, or

[21] One important caveat should be mentioned: a large share of the gap is plausibly methodological and compositional rather than substantive. ATUS is a one-day diary (so episodic care that occurs on non-diary days is missed), it excludes "on call" or supervisory time unless recorded as an activity, and our moments are conditional on selection into the ATUS elder-care module but average over a broad caregiver population (age 25–64) rather than restricting attention to primary caregivers or to caregivers of disabled elders. By contrast, retrospective caregiver surveys (and caregiver supplements attached to disability samples) are designed to identify active and often high-intensity caregiving situations and elicit "typical" weekly hours using recall, which can both raise measured hours and be more sensitive to a survey's question verbiage and reference periods (Miller and Haughton 2021).

[22] We use the terms "household" and "consumer" synonymously.

retired, $j \geq J_R$. Households die with certainty after age J, though some fraction die accidentally each period. Each working-age cohort is characterized by a representative household. However, for each $j \geq J_R$ there are two types of living households — 1) those where all residents are healthy and residing independently at home; 2) those that require some degree of living assistance. Thus, at any given age households can be characterized by two possible types — 1) independent; 2) dependent — though, all households of age $j < J_R$ are of the independent type. We will index the living states of existence such that i denotes independent and d denotes dependent.

3.3.1 Population Dynamics

Let M_{jt} be the population of age-j households alive in period t.[23] Let $M^d_{jt} < M_{jt}$ be the mass of age-j consumers that are living-dependent, and let M^i_{jt} be the mass who are independent. When $j < J_R$, $M^i_{jt} = M_{jt}$, however when $j \geq J_R$, $M^d_{jt} + M^i_{jt} = M_{jt}$. Let ψ_{jt} be the fraction of age-j agents who are dependent, such that $\psi_{jt} = M^d_{jt}/M_{jt}$.[24] Being dependent is assumed to be a conditionally absorbing state (i.e., as long as you continue to be alive, you will continue to be dependent, though you could still transition to death): consumers may transition from independence to dependence but not the other way around. Being dependent also affects one's probability of survival. Let s^d_{jt} be the probability that an age-j dependent in period t survives to be an age-$j+1$ agent in period $t+1$. Such an agent will, naturally, continue to be dependent if they survive. Let s^i_{jt} be the probability that an age-j independent in period t survives to be an age-$j+1$ agent in period $t+1$. For the independent agent of age $j \geq J_R$, s^i_{jt} is simply the survival probability but provides no information about whether or not the agent transitions to dependency in $t+1$. Rather, let $\varpi_{j+1,t+1}$ ("varpi") be the incidence of dependency: the probability that an age-j independent agent in period t who survives is dependent at age $j+1$ in $t+1$. The probability that an age-j independent agent survives and becomes dependent in $t+1$ is then $s^i_{jt}\varpi_{j+1,t+1}$. The incidence $\varpi_{j+1,t+1}$ is distinct from the share $\psi_{j+1,t+1}$, which also counts surviving agents who were already dependent at age j.

There are thus three possible states of existence — 1) living and independent, 2) living and dependent, 3) dead. Depending on the age of the agent the probability one transitions from independence to dependence may be zero. Death is, of course, an absorbing state. Let Φ_{jt} be the transition matrix for an age-j agent in period t. This object is

$$\Phi_{jt} = \begin{pmatrix} s^i_{jt}(1-\varpi_{j+1,t+1}) & s^i_{jt}\varpi_{j+1,t+1} & 1-s^i_{jt} \\ 0 & s^d_{jt} & 1-s^d_{jt} \\ 0 & 0 & 1 \end{pmatrix}, \tag{10}$$

where $\varpi_{j+1,t+1} = 0$ for all $j < J_R - 1$. Further, for agents age $j < J_R$ we will typically write $s_{jt} \equiv s^i_{jt}$, since all working-age adults are assumed to be independent. (10) is a fully exogenous

[23] Recall, M_t (no j subscript) is the total mass (population) of consumers alive in period t.
[24] Obviously, for $j < J_R$ this value is zero.

object. Let $\overline{\Phi}_{jt}$ be the sub-matrix of (10), having removed the last row and column (i.e., it only describes how the population of surviving agents evolves). Further, use bold notation, $\boldsymbol{M}_{jt}$, to denote the stacked vector $(M^i_{jt}, M^d_{jt})^\top$. We take Φ_{jt} to have rows indexed by the current state and columns by next period's state; since $\boldsymbol{M}_{jt}$ is a column vector of masses, the law of motion uses $\overline{\Phi}^\top_{jt}$. The population evolves according to

$$\boldsymbol{M}_{j+1,t+1} = \overline{\Phi}^\top_{jt} \cdot \boldsymbol{M}_{jt}. \tag{11}$$

$\overline{\Phi}_{jt}$ will of course be sparse for young agents. Dividing the dependent row of (11) by the mass of survivors gives the law of motion for the dependent share,

$$\psi_{j+1,t+1} = \frac{s^d_{jt}\psi_{jt} + s^i_{jt}(1-\psi_{jt})\,\varpi_{j+1,t+1}}{s^d_{jt}\psi_{jt} + s^i_{jt}(1-\psi_{jt})}, \qquad \psi_{J_R-1,t} = 0, \tag{12}$$

so that $\psi_{J_R,t} = \varpi_{J_R,t}$, while at older ages $\psi_{j+1,t+1} > \varpi_{j+1,t+1}$ whenever $\psi_{jt} > 0$ and $\varpi_{j+1,t+1} < 1$. Thus, because dependency is absorbing, the share accumulates past incidence as long as some dependents survive past the onset of dependency.

Finally, note that the ratio of dependents to working-age care providers in any given period will be a key driver of how care demand evolves over time, as we will show. Aging thus hits care demand through several different channels at once: 1) dependency incidence, ϖ, may change over time due to changes in medical practices or exogenous technological advances reducing the prevalence of diseases (e.g., dementia and Alzheimer's) which lead to dependence; 2) dependents themselves may survive longer due to exogenous changes in the conditional survival rate, s^d, leading to protracted episodes of dependence; 3) at the same time, if less new workers enter the economy due to declining fertility/natality this will have an impact on how much informal care can be supplied. These exogenous demographic forces will compete against each other in determining supply and demand for informal care. We will discuss this in more detail as the model presentation evolves and especially in our quantitative exercises later in Sections 4 and 5.

3.3.2 Preferences

Agents age $j \geq J_R$ differ by type in terms of how their flow utility is generated. Let $\widetilde{n}_{jt}$ be a composite good that describes a dependent agent's demand for both formal and informal nursing-care services. Independent agents have no demand for such services. This object is a composite of formal nursing-home care purchased on the market, n_{jt}, and informal care provided by altruistic loved ones, q_{jt}. Further, let ζ_{jt} be the efficiency by which q_{jt} received from loved ones may be converted to care services. We assume this depends on the age of the recipient of care, so it is like a consumption-efficiency shifter. The care composite is

$$\widetilde{n}_{jt} \equiv \mathcal{G}(n_{jt}, q_{jt}; \zeta_{jt}) := \left(\nu_n n_{jt}^{\rho_n} + (1-\nu_n)(\zeta_{jt} q_{jt})^{\rho_n}\right)^{\frac{1}{\rho_n}}, \tag{13}$$

where $\mathcal{G}$ is the function describing how formal and informal care are combined to generate the composite. The CES composite allows for ρ_n to be flexible, enabling us to pin-down the macro elasticity of substitution for formal and informal care; when $\rho_n < 0$ the two are gross complements, while when $\rho_n \in (0, 1)$ they are gross substitutes. Like Koreshkova and Lee (2024) we carefully distinguish between formal nursing care and informal care in order to provide an avenue to estimate their aggregate substitutability, whereas much of the literature on late-life saving treats nursing-home care as an exogenous expense (DeNardi, French, and Jones 2010; Kopecky and Koreshkova 2014) or folds it into a single medical good (De Nardi, French, and Jones 2016).

Retired dependent agents have preferences over consumption, c_{jt}, and the nursing-care composite, $\widetilde{n}_{jt}$, according to the isoelastic function:

$$u_j^d(c_{jt}, \widetilde{n}_{jt}) := \frac{\left(\nu_d(\xi c_{jt})^{\rho_d} + (1-\nu_d)\widetilde{n}_{jt}^{\rho_d}\right)^{\frac{1-\sigma}{\rho_d}}}{1-\sigma}. \tag{14}$$

The parameter ρ_d controls the within-period substitutability between non-nursing consumption c_{jt} and the care composite $\widetilde{n}_{jt}$ (elasticity of substitution $1/(1-\rho_d)$). As $\rho_d \to 0$, the CES aggregator inside (14) converges to the Cobb–Douglas index $(\xi c_{jt})^{\nu_d}\widetilde{n}_{jt}^{1-\nu_d}$. Separately, when $\sigma = 1$ (log utility), (14) is interpreted as the log of the CES aggregate (i.e., $u_j^d = \frac{1}{\rho_d}\log\left(\nu_d(\xi c_{jt})^{\rho_d} + (1-\nu_d)\widetilde{n}_{jt}^{\rho_d}\right)$); combining $\sigma = 1$ with $\rho_d \to 0$ yields $u_j^d = \nu_d \log(\xi c_{jt}) + (1-\nu_d)\log\widetilde{n}_{jt}$. The parameter ν_d governs the weight on non-nursing consumption relative to care. The parameter ξ effectively discounts the dependent agent's utility from consumption, reflecting the fact that such agents have less autonomy, which we interpret as less enjoyment of consumption, than their independent peers. We index u_j^d by j for purposes which will become apparent when we describe the process of reverse altruism below.

Note that the policy functions describing a household's optimal choice of c_{jt} will vary depending on whether or not the household is dependent or independent. Since this is a model of households, an independent agent is an independent household. It could be a two-person household where both adults are healthy and independent. Any dependent household is a household that requires care time, even if it is only one adult requiring it. Still, this is an aggregate model and there are only three types of agents by cohort, so we can think of preferences as being representative (and thus characterizing average behavior) for cohort-type pairs.

Following Kitao (2014) and Cooley and Henriksen (2018), working-age and independent retired agents have additively separable isoelastic preferences over consumption and leisure. Working-age adults will also be alruistic, internalizing their dependent elders' utility. An independent retiree, meanwhile, only cares about consumption and leisure,

$$u^i(c_{jt}, \ell_{jt}) := \frac{c_{jt}^{1-\sigma}}{1-\sigma} + \chi\frac{\ell_{jt}^{1-\gamma}}{1-\gamma}, \tag{15}$$

where $1/\sigma$ is the intertemporal elasticity of substitution for consumption, γ is the curvature of the

utility function for leisure, and χ is the weight on leisure relative to consumption.[25] The effective amount of leisure available to a working-age agent depends on the age-specific cost of labor-market participation. Following Kitao (2014), we model this fixed cost as a function of age:

$$\phi_j = \kappa_1 + \kappa_2\, j^{\kappa_3}, \tag{16}$$

where j indexes the agent's age. The participation cost ϕ_j is measured in units of time, representing the disutility associated with labor market participation (e.g., commuting, job preparation) that increases with age. When an agent works positive hours, effective leisure for working-age agents is given by the time constraint

$$\ell_{jt} = 1 - h_{jt} - q^s_{jt} - \phi_j, \qquad h_{jt},\, q^s_{jt} \geq 0, \tag{17}$$

where h_{jt} is market work and q^s_{jt} is informal care supply. Retirees neither work nor supply care to elders, so that $\ell_{jt} = 1$ for $j \geq J_R$.

Let $\bar{j}$ be the age difference between a care receiver and a care provider.[26] Thus, an age- $j + \bar{j}$ agent is eligible for informal care by an age-j agent but only so long as $j + \bar{j} \geq J_R$ and only if they are of the dependent type. An age-j agent with $j < J_R$ has additively separable preferences over consumption, leisure, and the utility of dependent elders:

$$u(c_{jt}, \ell_{jt}; u^d_{j+\bar{j}}) := \frac{c_{jt}^{1-\sigma}}{1-\sigma} + \chi \frac{\ell_{jt}^{1-\gamma}}{1-\gamma} + \underline{\beta}\, u^d_{j+\bar{j}}\big(c_{j+\bar{j},t}, \mathcal{G}(n_{j+\bar{j},t}, q_{j+\bar{j},t}(q^s_{jt}); \zeta_{j+\bar{j},t})\big). \tag{18}$$

Agents that are young enough may not have any dependent elders and so would face the same utility function as that in (15) but dropping the i superscript. The parameter $\underline{\beta} \in (0,1)$ describes the degree to which young agents are care about their dependent elders (i.e., altruism). Altruism is directed at the representative dependent of age $j + \bar{j}$: the young do not condition on how long that elder has required care, so $u^d_{j+\bar{j}}$ is evaluated at the pool's average state (more below).[27]

To distinguish between demand for care time and its supply, let q^s_{jt} be the supply of help time working-age agents provide to those aged $j + \bar{j}$. Help received is a function of the supply of help provided by working-age adults, $q_{j+\bar{j},t}(q^s_{jt})$. The actual level of help received by age-$j + \bar{j}$ dependent agents will vary in the cohort population levels, specifically the ratio of age-$j + \bar{j}$ dependents to working-age adults of age j, $M^d_{j+\bar{j},t}/M_{jt}$. In the context of our exercise, we will often refer to this object the "dependency ratio," and it will be a key driver of much of the variation in

[25] For retirees $\ell_{jt} = 1$, so the leisure term in (15) reduces to the constant $\chi/(1-\gamma)$: it shifts the level of $\mathcal{V}^i_{j,t}$ but enters no marginal condition, so it affects no allocation or price, only welfare comparisons across types, which we do not make.

[26] One can think of this as the difference between a parent and a child.

[27] Bequests in the model are accidental and shared equally, so caregiving does not buy an inheritance. For cohort-representative households this is innocuous: most people die with little wealth and inheritances are concentrated at the top (Poterba, Venti, and Wise 2012; Feiveson and Sabelhaus 2018), Medicaid pays for formal care once wealth is spent down (Sloan and Shayne 1993; Brown and Finkelstein 2008), and care provided by children to elders is not guided by a strategic bequest motive (Sloan, Picone, and Hoerger 1997; Perozek 1998).

the formal/informal care tradeoff margin, as we discuss in further detail throughout the paper. Note that it can fluctuate over time for several reasons: changes to dependency incidence, ϖ_{jt}, due to changes in care practices or health patterns may cause $M^d_{j+\bar{j},t}/M_{jt}$ to fall, while relatively slow growth in the working-age population coupled with population aging may cause the ratio to rise. The extent to which these two forces are in tension will determine how care practices, on aggregate, evolve.

Finally, for notational completeness assume $u^d_{j+\bar{j}} = 0$ when $j + \bar{j} < J_R$ or $j + \bar{j} > J$. Further, we must assume that $\bar{j}$ is large enough that $J - \bar{j} \leq J_R - 1$ (i.e., care-giving adults are all of working age, not retired).

3.3.3 Time Use, Budget Constraints, & Asset Accumulation

All agents are endowed with one unit of time. Assuming $\bar{j}$ is sufficiently large enough, retired agents do not work but only care about leisure, thus setting $\ell_{jt} = 1$. Meanwhile, working-age agents work and supply time caring for elders if $j + \bar{j} \geq J_R$.[28] Households are assumed to supply labor equi-proportionally to both sectors and receive the labor-share-weighted average wage $\overline{w}_t = w_t(L_{ct} + \omega_t L_{nt})/L_t$, so that aggregate labor income $\overline{w}_t L_t$ equals firms' total labor cost $w_t L_{ct} + \omega_t w_t L_{nt}$.[29]

Working-age agents earn labor income elastically at efficiency-wage rate $\overline{w}_t \eta_j$ and remit taxes τ_t to finance the social transfer system (Social Security plus Medicare and Medicaid) for retired agents; η_j is an age-specific efficiency scalar. They also save in capital assets, $a_{j+1,t+1}$, which are subject to a gross rate of return, $R_t = r_t + 1 - \delta$, where δ is the rate of capital depreciation. The budget constraint of an age $j < J_R$ agent is

$$p_{ct} c_{jt} + a_{j+1,t+1} \leq \overline{w}_t \eta_j (1 - \tau_t)(1 - \phi_j - \ell_{jt} - q^s_{jt}) + R_t(a_{jt} + b_t) + \pi_{jt}, \qquad \forall j < J_R, \tag{19}$$

where π_{jt} are cohort-specific dividends received from firms' profits. Accidental bequests, b_t, are received by agents from a stock left by those who accidentally die in the transition from $t-1$ to t; we assume all agents of all ages receive an equal share of aggregate bequests. Retired adults get income from their assets as well as via transfer payments from the government which depend on their dependency status, where $T^d_t > T^i_t$, generally. The budget constraint of a dependent agent is

$$p_{ct} c^d_{jt} + p_{nt} n^d_{jt} + a^d_{j+1,t+1} \leq T^d_t + R_t(\overline{a}^d_{jt} + b_t) + \pi^d_{jt}. \tag{20}$$

The value $\overline{a}^d_{jt}$ is the dependent pool's mass-weighted average of all assets carried forward by

[28] Households here divide their time among market work, leisure and care. On how time use shapes structural change and welfare more broadly, Bednar and Pretnar (2026) study structural change when all types of consumption take time, Boerma and Karabarbounis (2020) infer that the productivity of leisure time has risen, and Boerma and Karabarbounis (2021) find that accounting for home production widens inequality in living standards.

[29] Households do not choose which sector to work in since they are representative (though cohort-specific).

surviving (continuing) dependents plus those brought in by the newly dependent.[30] This object evolves according to

$$\overline{a}^d_{j+1,t+1} = \frac{s^d_{jt} M^d_{jt}\, a^d_{j+1,t+1} + s^i_{jt} \varpi_{j+1,t+1} M^i_{jt}\, a^i_{j+1,t+1}}{s^d_{jt} M^d_{jt} + s^i_{jt} \varpi_{j+1,t+1} M^i_{jt}}, \tag{21}$$

where $a^d_{j+1,t+1}$ and $a^i_{j+1,t+1}$ denote the asset levels chosen at age j by dependent and independent households respectively. The pooling works like an unanticipated, actuarially fair long-term-care risk pool: newly dependent households, who arrive with more assets, top up those who have been drawing down against care costs for longer, a within-cohort reallocation that neither creates nor destroys capital.

For an independent and retired agent the budget constraint is

$$p_{ct} c^i_{jt} + a^i_{j+1,t+1} \leq T^i_t + R_t(a^i_{jt} + b_t) + \pi^i_{jt}. \tag{22}$$

Note that, for retirees, the asset holdings and consumption levels clearly depend on their dependency type. Entering retirement, of course, $a^d_{J_R,t} = a^i_{J_R,t}$, since investment decisions were made prior to realizing the idiosyncratic dependency shock. Dividends, π_{jt}, are received from firm profits. We assume these are distributed in proportion to agents' period-t asset holdings, $\pi_{jt} = (a_{jt} + b_t)\Pi_t / K_t$.[31] Households take dividends as given (i.e., as lump-sum income): they do not internalize that investment may affect next period's dividend, so the private return on saving is R_{t+1}, even though the return on capital including profits is $R_{t+1} + \Pi_{t+1}/K_{t+1}$. Finally, note that all age $j = 1$ agents enter with $a_{1t} = 0$, though they receive bequests, $b_t > 0$. Further, a_{jt} need only satisfy $a_{jt} \geq 0$, and new investment, $\iota_{jt} \equiv a_{j+1,t+1} - (1-\delta)(a_{jt} + b_t)$ (with $\overline{a}^d_{jt}$ in place of a_{jt} for dependents), satisfies $\iota_{jt} > -\underline{\iota}$, where $\underline{\iota} > 0$ is some incremental borrowing limit. This allows for elderly people to spend down accumulated assets, and it is safe to assume that this dis-investment constraint never binds.

3.3.4 Recursive Optimization Problems

Households face only idiosyncratic uncertainty over two possible shocks: 1) whether or not they will survive; 2) conditional upon surviving past retirement, whether or not they will be dependent. Households are assumed to know how Z_{ct}, Z_{nt}, μ_{ct}, μ_{nt}, $\{s_{jt}\}_j$, $\{s^i_{jt}\}_j$, $\{s^d_{jt}\}_j$, and $\{\varpi_{jt}\}_j$ (and hence, through (12), $\{\psi_{jt}\}_j$) evolve. Finally, because households know how the population distribution will evolve, they also need not forecast the evolution of b_t, so that the only endogenous state variable is a_{jt}. These assumptions are consistent with those in Krueger and Ludwig (2007), Cooley and Henriksen (2018), and Pretnar and Feldman (2026) and will allow us to readily simu-

[30] We track one asset position per type and age rather than one per history. A single policy function per type suffices, since the date at which dependency struck enters neither preferences, the budget constraint, nor (10): a dependent household's problem depends only on its assets, its age, and the period.

[31] It should be apparent that $\pi^d_{jt} = (\overline{a}^d_{jt} + b_t)\Pi_t / K_t$, for example.

late transition and growth paths in order to feasibly isolate the forces driving long-term structural transformation amidst population aging.

Assume at $j = J$ agents cease saving and divide their remaining resources amongst c_{Jt} and n_{Jt}, depending on their type. Retirees solve the following recursive optimization problems, depending on their current dependency status:

$$\mathcal{V}^d_{j,t}(a_{jt}) = \max_{c_{jt}, n_{jt}, a_{j+1,t+1}} \left\{ u^d_j\big(c_{jt}, \mathcal{G}(n_{jt}, q_{jt}; \zeta_{jt})\big) + \beta s^d_{jt} \mathcal{V}^d_{j+1,t+1}(a_{j+1,t+1}) \right\} \tag{23}$$

$$\begin{aligned} \text{and} \quad \mathcal{V}^i_{j,t}(a_{jt}) = \max_{c_{jt}, a_{j+1,t+1}} \Big\{ & u^i(c_{jt}, 1) \\ & + \beta s^i_{jt} \Big[(1 - \varpi_{j+1,t+1}) \mathcal{V}^i_{j+1,t+1}(a_{j+1,t+1}) + \varpi_{j+1,t+1} \mathcal{V}^d_{j+1,t+1}(a_{j+1,t+1}) \Big] \Big\}. \end{aligned} \tag{24}$$

The parameter β characterizes an agent's own time preferences and is different from $\underline{\beta}$ — the degree of altruism. Notice that in (23), dependent agents do not directly choose how much help, q_{jt}, they will receive. Also in (23), since dependency is an absorbing state (conditional upon survival), the expected future value of living is simpler than in (24), where we must account for the possibility that an independent agent in period t becomes dependent in period $t + 1$. Both problems are solved subject to the budget constraints (20) and (22), respectively, with the dependent's assets entering at the pooled position (21) and the care composite given by (13) with q_{jt} taken as given, together with $a_{j+1,t+1} \geq 0$, the dis-investment limit on ι_{jt}, and the terminal condition $a_{J+1,t+1} = 0$.

For young agents age $j < J_R - 1$ the recursive optimization problem is

$$\mathcal{V}_{j,t}(a_{jt}) = \max_{c_{jt}, \ell_{jt}, q^s_{jt}, a_{j+1,t+1}} \left\{ u\big(c_{jt}, \ell_{jt}; u^d_{j+\bar{j}}\big) + \beta s_{jt} \mathcal{V}_{j+1,t+1}(a_{j+1,t+1}) \right\}. \tag{25}$$

When young agents are one year away from retirement, they must take into consideration that they may (possibly) immediately transition into the dependent state. Agents age $j = J_R - 1$ have the following value function:

$$\begin{aligned} \mathcal{V}_{j,t}(a_{jt}) = \max_{c_{jt}, \ell_{jt}, q^s_{jt}, a_{j+1,t+1}} \Big\{ & u\big(c_{jt}, \ell_{jt}; u^d_{j+\bar{j}}\big) \\ & + \beta s_{jt} \Big[(1 - \varpi_{j+1,t+1}) \mathcal{V}^i_{j+1,t+1}(a_{j+1,t+1}) + \varpi_{j+1,t+1} \mathcal{V}^d_{j+1,t+1}(a_{j+1,t+1}) \Big] \Big\}. \end{aligned} \tag{26}$$

Again, both (25) and (26) are solved subject to the budget constraint (19), the time constraint (17) with ϕ_j from (16), $a_{j+1,t+1} \geq 0$ and the dis-investment limit on ι_{jt}, and the informal care market clearing condition (30), through which q^s_{jt} enters $u^d_{j+\bar{j}}$.

3.4 Optimal Choices

Young agents age $j < J_R - 1$ (i.e., at least two periods away from retirement) make consumption/savings decisions to satisfy the following, standard Euler equation:

$$\begin{aligned} \frac{\partial u}{\partial c_{jt}} &= R_{t+1}\beta s_{jt}\left(\frac{p_{ct}}{p_{c,t+1}}\right)\frac{\partial u}{\partial c_{j+1,t+1}}, \\ \text{where} \qquad \frac{\partial u}{\partial c_{jt}} &= c_{jt}^{-\sigma}. \end{aligned} \tag{27}$$

The discount factor β governs patience, as usual, while the curvature parameter σ controls the life-cycle consumption profile, as usual. The survival probability s_{jt} introduces a mortality discount that reduces the effective incentive to save at advanced ages.

Because consumption and leisure enter the working-age agent's utility additively in (18), the intratemporal first-order condition equates the marginal utility of leisure with the marginal value of work time:

$$\chi \ell_{jt}^{-\gamma} = \frac{c_{jt}^{-\sigma}}{p_{ct}} \overline{w}_t \eta_j (1 - \tau_t). \tag{28}$$

Supposing working-age adults provide some positive level of help, $q_{jt}^s > 0$, the leisure–care-supply margin equates the marginal utility of leisure with the discounted marginal increase to utility their dependent elder would get from the supplied informal care, weighted by the ratio of care suppliers to care receivers:

$$\chi \ell_{jt}^{-\gamma} = \underline{\beta} \, \frac{\partial u^d_{j+\bar{j}}}{\partial \widetilde{n}_{j+\bar{j},t}} \frac{\partial \mathcal{G}}{\partial q_{j+\bar{j},t}} \left(\frac{M_{jt}}{M^d_{j+\bar{j},t}}\right). \tag{29}$$

Note that, because $\mathcal{G}$ depends on $n_{j+\bar{j},t}$ and $q_{j+\bar{j},t}(q^s_{jt})$ non-separably, the marginal product of $q_{j+\bar{j},t}$ also depends on the level of formal care provided. Thus, young agents respond to their elder's choice of $n_{j+\bar{j},t}$ when setting q^s_{jt}.

Since cohorts are representative and q^s_{jt} is a cohort-wide average, this average informal care supply per dependent will itself depend on the number of dependents requiring such care for every available working-age provider. This is where the term $M_{jt}/M^d_{j+\bar{j},t}$ in (29) comes from — the restriction that the informal market for help time provided by those of age j to those age $j + \bar{j}$ must clear:[32]

$$M_{jt}\, q^s_{jt} = M^d_{j+\bar{j},t}\, q_{j+\bar{j},t}. \tag{30}$$

Helpers thus take the dependent's consumption and formal-care choices as given, so no term for the dependent's response to the care received enters (29). Note that higher $\underline{\beta}$ (more altruistic

[32] The object $\frac{\partial q_{j+\bar{j},t}}{\partial q^s_{jt}}$ is then just the ratio $M_{jt}/M^d_{j+\bar{j},t}$.

young people) increases q^s_{jt}, which through the care composite raises $\widetilde{n}_{j+\bar{j},t}$. Holding the dependent's consumption and prices fixed, demand for market-purchased nursing services is lower at higher levels of altruism when $\rho_d < \rho_n$: whether more informal care displaces formal care at the dependent's margin thus depends on $\rho_d - \rho_n$, not on the sign of ρ_n. Non-nursing/nursing complementarities can thus drive increasing non-nursing-consumption at ever-higher levels of altruism ($\underline{\beta} \to_- 1$), so that the more altruistic young people are *and* the stronger the non-nursing/nursing complementarities, the more old people divert resources from formal nursing expenditure to non-nursing consumption. Further, non-separabilities in $\mathcal{G}$ and $u^d_{j+\bar{j}}$ ensure that young agents internalize the dependents' tradeoff between formal care and non-nursing consumption when providing informal care.

An independent retiree's choice of consumption and continued asset holdings satisfy the Euler equation:

$$\frac{\partial u^i}{\partial c_{jt}} = R_{t+1}\beta s^i_{jt}\left(\frac{p_{ct}}{p_{c,t+1}}\right)\left[(1-\varpi_{j+1,t+1})\frac{\partial u^i}{\partial c_{j+1,t+1}} + \varpi_{j+1,t+1}\frac{\partial u^d_{j+1}}{\partial c_{j+1,t+1}}\right]. \tag{31}$$

Note that (31) characterizes dynamic consumption/savings decision making for all agents age $j \geq J_R - 1$, since an age $J_R - 1$ agent could enter retirement as a dependent. The non-zero probability one becomes dependent next period, $\varpi_{j+1,t+1}$, creates a consumption-insurance motive: independent retirees anticipate potentially becoming dependent and adjust their savings accordingly. Whether this raises savings as a precautionary buffer against costly nursing-care needs depends on whether the dependent state's marginal utility of consumption next period exceeds the independent state's. The larger transfers dependents receive ($T^d_t > T^i_t$), the weight ν_d, and the scalar ξ can each push the other way, so the sign is a quantitative question. While $\frac{\partial u^i}{\partial c_{jt}}$ is straightforward, complementarities between nursing-care and non-nursing-care consumption in u^d_{j+1} are embedded in the expression for a dependent's marginal utility of consumption. These objects are

$$\frac{\partial u^i}{\partial c_{jt}} = c_{jt}^{-\sigma} \qquad \text{and} \qquad \frac{\partial u^d_j}{\partial c_{jt}} = \nu_d \xi^{\rho_d} c_{jt}^{\rho_d - 1}\left(\nu_d(\xi c_{jt})^{\rho_d} + (1-\nu_d)\widetilde{n}^{\rho_d}_{jt}\right)^{(1-\rho_d-\sigma)/\rho_d}. \tag{32}$$

The parameter ρ_d, together with ρ_n, governs the price elasticity of nursing demand, which disciplines the nursing expenditure share (see Section 4.2 for further discussion). Absent informal care, $\rho_d > 0$ (gross substitutes) means dependents substitute away from care services as nursing prices rise. The weight ν_d determines the relative importance of non-nursing consumption versus care in the dependent's utility. The scalar ξ discounts the dependent's valuation of consumption. A lower ξ lowers the level of dependent utility, but with $\rho_d < 0$ and $\sigma > 1$ the dependent's marginal utility of consumption is decreasing in ξ, so through (31) a relatively lower ξ would result in relatively higher savings in anticipation of dependency.

Markets in this economy are incomplete in two ways. Within a cohort, retirees hold a single asset and cannot buy annuities or long-term-care insurance, so they bear survival and dependency risk themselves. They self-insure by saving, which the dependency probability in (31) turns into

the consumption-insurance motive described above. The state-contingent transfers $T_t^d > T_t^i$ and the informal care of altruistic working-age households also provide partial insurance. The missing private market matches the small U.S. long-term-care insurance market and evidence that family care crowds it out (Pauly 1990; Brown and Finkelstein 2008; Braun, Kopecky, and Koreshkova 2019; Ko 2022; Mommaerts 2025).

The dependent's Euler equation simply embeds the nursing-care and non-nursing-care consumption tradeoff:

$$\frac{\partial u_j^d}{\partial c_{jt}} = R_{t+1}\beta s_{jt}^d \left(\frac{p_{ct}}{p_{c,t+1}}\right) \frac{\partial u_{j+1}^d}{\partial c_{j+1,t+1}}. \tag{33}$$

The dependent retired agent also selects formal nursing care to satisfy the marginal rate of substitution condition with non-nursing-care consumption (of course, taking what the altruistic younger agents supplies as given):

$$\frac{p_{nt}}{p_{ct}} \frac{\partial u_j^d}{\partial c_{jt}} = \frac{\partial u_j^d}{\partial \widetilde{n}_{jt}} \frac{\partial \mathcal{G}}{\partial n_{jt}}, \tag{34}$$

where

$$\frac{\partial u_j^d}{\partial \widetilde{n}_{jt}} = (1-\nu_d)\widetilde{n}_{jt}^{\rho_d - 1}\left(\nu_d(\xi c_{jt})^{\rho_d} + (1-\nu_d)\widetilde{n}_{jt}^{\rho_d}\right)^{(1-\rho_d-\sigma)/\rho_d} \qquad \text{and} \qquad \frac{\partial \mathcal{G}}{\partial n_{jt}} = \nu_n n_{jt}^{\rho_n - 1}\widetilde{n}_{jt}^{1-\rho_n}, \tag{35}$$

and so it is thus clear that the elder chooses n_{jt} in response to the supply of informal care, $q^s_{j-\bar{j},t}$, which enters $\widetilde{n}_{jt}$ due to the non-separability of formal and informal care in $\mathcal{G}$. The intratemporal split between non-care consumption and formal care depends on ρ_d, ν_d, ξ, ν_n, ρ_n, relative prices, and (through $\widetilde{n}_{jt}$) the informal care received:

$$\frac{p_{nt}}{p_{ct}} \nu_d \xi^{\rho_d} c_{jt}^{\rho_d - 1} = (1-\nu_d)\nu_n n_{jt}^{\rho_n - 1}\widetilde{n}_{jt}^{\rho_d - \rho_n}. \tag{36}$$

And, thus, by taking $q^s_{j-\bar{j},t}$ as given, the dependent agent can adjust the allocation of resources devoted to both c_{jt} and n_{jt}, where the margin of adjustment is governed by the relative values of ρ_d (the care/non-care elasticity) and ρ_n (the formal/informal elasticity).

No single agent in our model optimizes over both n_{jt} and q_{jt}. Because the altruistic concern of the young, $\underline{\beta}\, u^d_{j+\bar{j}}$, is separable from the utility they get from decisions that affect their household directly (consumption and leisure), we can combine the dependent's condition for n_{jt} with the helper's condition for $q^s_{j-\bar{j},t}$ (i.e., (29) together with (28)). The dependent's marginal utility of care cancels, but the two households' marginal utilities of consumption do not, since they belong to

different agents. This gives the marginal rate of substitution between formal and informal care:

$$\frac{\partial \mathcal{G}/\partial n_{jt}}{\partial \mathcal{G}/\partial q_{jt}} = \frac{p_{nt}}{\overline{w}_t \eta_{j-\bar{j}}(1-\tau_t)} \left(\frac{M_{j-\bar{j},t}}{M^d_{jt}}\right) \frac{\underline{\beta}\, \partial u^d_j/\partial c_{jt}}{c^{-\sigma}_{j-\bar{j},t}} \qquad \text{where} \qquad \frac{\partial \mathcal{G}}{\partial q_{jt}} = (1-\nu_n)\zeta^{\rho_n}_{jt} q^{\rho_n-1}_{jt} \widetilde{n}^{1-\rho_n}_{jt}. \tag{37}$$

After solving out for the ratio of formal to informal care, using $q_{jt} = (M_{j-\bar{j},t}/M^d_{jt})\, q^s_{j-\bar{j},t}$ from the informal-care clearing condition, we get

$$\frac{n_{jt}}{q^s_{j-\bar{j},t}} = \left[\frac{(1-\nu_n)\, p_{nt}\, \underline{\beta}\, \partial u^d_j/\partial c_{jt}}{\nu_n \overline{w}_t \eta_{j-\bar{j}}(1-\tau_t)\, c^{-\sigma}_{j-\bar{j},t}}\right]^{\frac{1}{\rho_n-1}} \left(\frac{\zeta_{jt} M_{j-\bar{j},t}}{M^d_{jt}}\right)^{\frac{\rho_n}{\rho_n-1}}. \tag{38}$$

Thus, the supply of informal care from the young depends not only on the young agent's opportunity cost of time, summarized by $\overline{w}_t \eta_{j-\bar{j}}(1-\tau_t)$, and on how much formal elder care their elders can afford, but also on the dependent's marginal utility of consumption relative to the helper's, scaled by $\underline{\beta}$, where the quality parameter ζ_{jt} amplifies informal care productivity.

3.5 The Allocation of Care Inputs

Two elasticities shape how care moves between the market and the home — the inner elasticity between formal and informal care, $1/(1-\rho_n)$, in (13), and the outer elasticity between care and non-care consumption, $1/(1-\rho_d)$, in (14). Neither object by itself determines the sign of the response of one kind of care to the other, but rather the relative elasticity, $\rho_d - \rho_n$, does, and it enters on both sides of the informal-care relationship.

On the dependent's side informal care enters (36) only through $\widetilde{n}^{\rho_d-\rho_n}_{jt}$. Log-differentiating (36) at fixed c_{jt} and prices gives

$$\frac{d\ln n_{jt}}{d\ln q_{jt}} = \frac{(\rho_d-\rho_n)\left[1-\nu_n(n_{jt}/\widetilde{n}_{jt})^{\rho_n}\right]}{1-\rho_n-(\rho_d-\rho_n)\,\nu_n(n_{jt}/\widetilde{n}_{jt})^{\rho_n}}, \tag{39}$$

where $\nu_n(n_{jt}/\widetilde{n}_{jt})^{\rho_n}$ is formal care's share of the composite. The denominator is positive for any $\rho_n, \rho_d < 1$ (i.e., finite elasticity of substitution), so more informal care lowers the demand for formal care exactly when $\rho_d < \rho_n$. This holds whether or not formal and informal care are themselves complements.

On the helper's side substituting (13) and (14) into (29) gives

$$\chi \ell^{-\gamma}_{jt} = \underline{\beta}\,(1-\nu_d)(1-\nu_n)\,\zeta^{\rho_n}_{j+\bar{j},t}\, q^{\rho_n-1}_{j+\bar{j},t}\, \widetilde{n}^{\rho_d-\rho_n}_{j+\bar{j},t} \left(\nu_d(\xi c_{j+\bar{j},t})^{\rho_d} + (1-\nu_d)\widetilde{n}^{\rho_d}_{j+\bar{j},t}\right)^{\frac{1-\rho_d-\sigma}{\rho_d}} \left(\frac{M_{jt}}{M^d_{j+\bar{j},t}}\right). \tag{40}$$

Holding $q_{j+\bar{j},t}$ and $c_{j+\bar{j},t}$ fixed, the elasticity of the right side of (40) with respect to formal care is

$$\nu_n \left(\frac{n_{j+\bar{j},t}}{\widetilde{n}_{j+\bar{j},t}} \right)^{\rho_n} \left[(\rho_d - \rho_n) + (1 - \rho_d - \sigma) \frac{(1-\nu_d)\widetilde{n}^{\rho_d}_{j+\bar{j},t}}{\nu_d (\xi c_{j+\bar{j},t})^{\rho_d} + (1-\nu_d)\widetilde{n}^{\rho_d}_{j+\bar{j},t}} \right]. \tag{41}$$

When this elasticity is negative, a cut in formal care raises the value of an informal hour to the helper. The first term in brackets is the relative elasticity, $\rho_d - \rho_n$. The second term reflects that less care also shrinks the dependent's consumption–care aggregate, which works against the first term when $1 - \rho_d - \sigma > 0$, with a weight equal to care's share of that aggregate. When $\rho_d < \rho_n$ and $1 - \rho_d - \sigma > 0$, the elasticity is negative as long as care's share of total consumption is below $(\rho_n - \rho_d)/(1 - \rho_d - \sigma)$; meanwhile, when $1 - \rho_d - \sigma \leq 0$, $\rho_d < \rho_n$ suffices for elasticity negativity.

Together, (39) and (41) show how cost disease moves care toward the home. A rise in p_{nt} leads dependents to purchase less formal care. With $\rho_d < \rho_n$ that cut raises the value of informal care to helpers, who supply more of it, and more informal care lowers formal demand further. If instead $\rho_d > \rho_n$, the bracketed term in (41) exceeds care's share of the total consumption multiplied by $1 - \sigma - \rho_n$, so that whenever $\sigma \leq 1 - \rho_n$ informal care falls along with formal care. Both are direct effects, holding the dependent's consumption and the provider ratio $M_{jt}/M^d_{j+\bar{j},t}$ fixed. Section 4.2 discusses the elasticity values the calibration selects.

3.6 Aggregation and Equilibrium

Aggregating the household problem and closing the model is mechanical, and we state it in full in Technical Appendix C. Three features are worth flagging here because they are used later. First, capital and labor are supplied by households and are mobile across sectors, so both sectors face the common rental rate r_t while nursing firms pay the wedge-adjusted wage $\omega_t w_t$; the sectoral input shares K_{nt}/K_t and L_{nt}/L_t that the calibration targets follow from cost minimization at those prices together with the CES cost shares in (9). Second, the market for informal care clears bilaterally rather than through a price: the hours supplied by helpers of age j must equal the hours received by dependents of age $j + \bar{j}$, which is what converts per-helper time into per-recipient care. Third, the government runs a balanced payroll-tax budget each period, and transfers are split between dependent and independent retirees by the fixed rule $T^i_t = \varphi_t T^d_t$, so φ_t enters as an exogenous policy series rather than as an equilibrium object. Accidental bequests are redistributed lump-sum and equally, and firm profits are returned to households in proportion to their asset holdings.

4 Calibration

We now seek to quantify the full GE model, putting together the production side as described by the two-sector model in Section 2.2 and the demand side outlined in Section 3.3. We desire a plausible calibration that yields simulations which best mimic the qualitative properties of the data. In Section 5 we will then use the fully calibrated model to study the causes and consequences

of nursing-sector structural transformation through our model's lens. We will also quantify the total social cost (formal plus informal) of elder care and engage in forward-looking projections to assess how the future of the U.S. economy may evolve given our findings regarding technological change, market structure (sectoral markups), and demographics, while also considering several policy counterfactuals.

First, though, we need to calibrate a full GE model to match time series of both aggregate and dis-aggregated outcomes along a dynamic transition path. Our calibration is designed to target several time series we previously analyzed in Section 2, as well as the unconditional informal care profile (see Technical Appendix A.4). In addition we seek to ensure that the canonical macro accounting ratios hold. For 5-year intervals beginning in 1990 and continuing through 2020, we target the following time series:[33] 1) the relative nursing price, $p_t = p_{nt}/p_{ct}$;[34] 2) the nursing market expenditure share, $\theta_{Nt} = p_t N_t / X_t$; 3) nursing-sector capital input shares, $\theta^K_{nt} = K_{nt}/K_t$; and 4) nursing-sector efficiency labor input shares, $\theta^L_{nt} = L_{nt}/L_t$. We also target the following aggregates as averages over the entire transition path: 1) capital/output ratio, K_t/Y_t; 2) total consumption/output ratio, $(p_{nt}N_t + p_{ct}C_t)/Y_t$; 3) average net five-year growth rate of p_t; 4) average net five-year growth rate of θ_{Nt}.[35] From the micro data we target two separate objects: 1) unconditional average informal care hours;[36] 2) the age profile of elder care provision, where we take within-age-group averages over the whole time series.[37]

The computational model starts with an initial steady state assumed in 1980, allowing two periods (1980 and 1985) of transition from steady state to commence the dynamics; transitions then continue out to a far-off steady state in 2150, though all exogenous growth variables stop changing in 2100 to allow for convergence to stationarity. We follow the solution procedure for OLG model transition paths outlined in Krueger and Ludwig (2007) and deployed as a modified version in Pretnar and Feldman (2026) who fully endogenize all prices.

In the remainder of this section we proceed piece-wise by explaining how we ex-ante set certain parameters to match various aggregates from data while also describing the internal calibration undertaken to tune the remaining parameters. We conclude this section by discussing the calibration fitness and its implications before moving on to our quantitative exercises in Section 5.

4.1 Period Length, Demographics, and Exogenous Preference Parameters

Let j and t each by 5-year periods, so that j encompasses 5-year age intervals, and $t = 1, 2, 3, \ldots$, where $t = 1$ is the initial steady state (e.g., 1980). Suppose all agents enter the economy at age 20 and live to be a maximum of 99 years old, so that $J = 16$; retirement occurs at age 65 with $J_R = 10$. Further, we set $\bar{j} = 7$ so that an adult at age 60-64 provides elder care to a dependent at age 95-99.

[33] Note that we start our calibration in 1990 due to data limitations on the sectoral labor share as can be seen in Figure A.2a of Technical Appendix A.2.

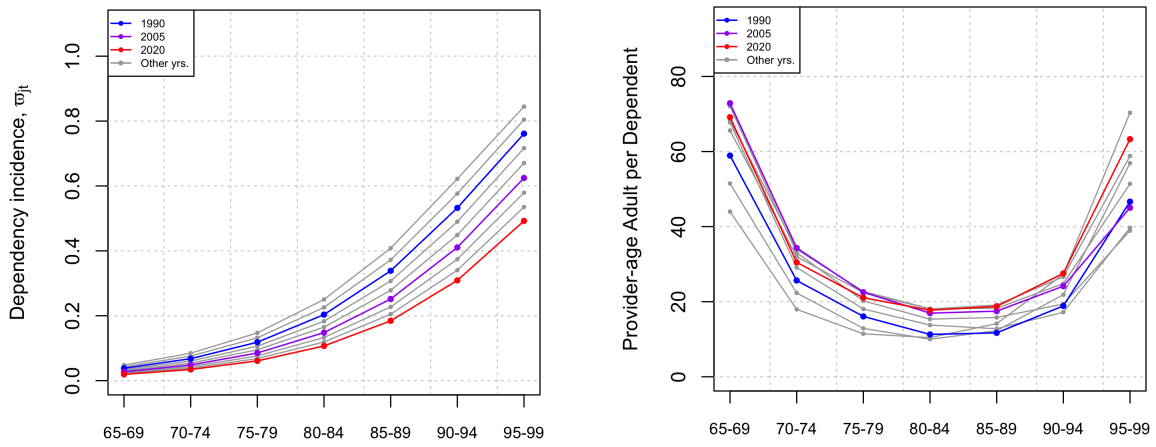
[34] For the sake of comparison purposes within the calibration only, we normalize relative prices to unity in 1990.

[35] We compute the latter two as geometric averages over the 1990–2020 sample.

[36] Given this time series is flat, we simply target the average.

[37] Again, we use the unconditional series by age, here, since the model lacks a formal extensive margin: see Figure A.5a of Technical Appendix A.4.

Both survival rates and population levels inputs are taken from U.N. Dept. of Economic & Social Affairs (2024).[38] We use two sources (studies of dementia and corresponding dependencies) to arrive at values for the dependency incidence ϖ_{jt} — Corrada et al. (2010) and Langa et al. (2017) — and obtain the dependent shares ψ_{jt} from (12). Recall from (38) that the aggregate number of dependents per provider dictates the amount of formal care per informal care hour demanded. As it will turn out, the evolution of this object is critical toward driving some of the second-order demand-side effects of structural transformation on the informal care market. We thus plot this object here along with the dependency rate, ϖ_{jt}, in Figure 3 as computed under the procedure in Technical Appendix B.2.1 using U.N. Dept. of Economic & Social Affairs (2024) data and findings from Corrada et al. (2010), Langa et al. (2017), and Garre-Olmo et al. (2019).



(a) $\{\varpi_{jt}\}_{j,t}$ (b) $\{M_{j-\bar{j},t}/M^d_{jt}\}_{j,t}$

Figure 3: Panel (a) shows the dependency incidence profiles ϖ_{jt} by the age bin a household enters, 65–69 through 95–99 (incidence is zero at younger ages). Panel (b) shows providers per dependent, $M_{j-\bar{j},t}/M^d_{jt}$, by recipient age. Both panels show every model period from 1980 to 2020, with 1990 in blue, 2005 in purple, 2020 in red and the remaining years in gray.

Our period length implies a capital depreciation rate of $\delta = 1-(1-0.096)^5 \approx 0.396$. We set the 5-year discount rate at $\beta = 0.983^5 \approx 0.9178$. We set the curvature parameter, $\sigma = 1.1$, corresponding to an intertemporal elasticity of substitution of $1/\sigma \approx 0.909$.

The values of χ (leisure weight in utility) and γ (labor/leisure and intertemporal income elasticity) are taken directly from Cooley and Henriksen (2018). The participation parameters — κ_1, κ_2, and κ_3 — are taken from Kitao (2014) and Cooley and Henriksen (2018), and the labor-productivity age-profile, $\{\eta_j\}_j$, is taken directly from Hansen (1993) and Pretnar and Feldman (2026).[39]

Transfer taxes are set to $\tau_t = \tau = 0.153$: the transfer tax is the combined employer and employee contribution to Social Security (OASDI) and Medicare (HI) under the Federal Insurance

[38] Technical Appendix B.2.1 gives more demographic detail — how we construct population levels, survival rates, dependency incidence and dependent mortality, and plots of each series.

[39] We plot both parameter sets in Technical Appendix B.2.2.

Contributions Act, and 0.153 is the statutory combined rate of 12.4% for OASDI and 2.9% for HI that has been in effect since 1990 (Social Security Administration (2024), Table 2.A3). Because the model's wage is the marginal product of labor, the full employer-plus-employee rate is the appropriate wedge on labor income. The transfer ratio $\varphi_t = T_t^i/T_t^d$ is constructed from program data rather than assumed. Both retiree types receive Social Security and Medicare, so T_t^i is the annualized average retired-worker benefit (Social Security Administration (2024), Table 5.B) plus Medicare outlays per enrollee (Medicare Payment Advisory Commission 2024; Centers for Medicare & Medicaid Services 2024). Dependents additionally qualify for Medicaid coverage of institutional long-term care, which is the dominant payer for nursing-home services, so T_t^d adds Medicaid nursing-home spending by source of funds (Centers for Medicare & Medicaid Services 2024) divided by the resident count (National Center for Health Statistics 2004; Harris-Kojetin et al. 2019). The resulting series is stable over the sample, ranging from 0.356 in 2000 to 0.430 in 2020 and averaging 0.397: an independent retiree receives roughly two-fifths of the transfer a dependent receives, the gap being almost entirely Medicaid's coverage of nursing-home care.[40]

4.2 Demand for Care and Altruism

Together, (13) and (14) comprise a nested CES function with different parametric hierarchies, each of which govern different layers of tradeoff margins. The inner nest governs the margin between purchased and informally provided care. The weight ν_n sets the level of that mix: raising it shifts dependents toward market care, which lifts the nursing market expenditure share θ_{Nt} and lowers average elder care hours. The curvature ρ_n instead sets how far the mix moves when care becomes expensive, and so is identified from the co-movement of elder care hours with the relative nursing price rather than from the level of either series. The efficiency shifter ζ_{jt} enters only as $(\zeta_{jt}q_{jt})^{\rho_n}$. Were it constant, it would be absorbed into ν_n, ν_d (or ξ), and $\underline{\beta}$, up to a level shift in u^d that affects saving through (31) when $\sigma \neq 1$. We therefore let it vary with the age of the care recipient but fix it over time, and restrict it to the two-parameter form

$$\zeta_j = \underline{\zeta}_1 \exp\{\underline{\zeta}_2(j - J_R)\}, \qquad J_R \leq j \leq J-1, \tag{42}$$

where $\underline{\zeta}_1$ is the efficiency of informal care received by the youngest dependents and $\underline{\zeta}_2$ its proportional change per five-year age bin of the recipient. To note, we also do not believe that fixing this series over time is at all problematic for two reasons: 1) the ATUS elder-care bins are too small to ascertain whether average care hours provided *by age* actually move over time; 2) we observe in Figure 2b (extensive margin conditional) and its unconditional counterpart in Figure A.5b of Technical Appendix A.4 that average care hours per day provided per person age 25-64 does not change over the sample in a statistically significant way. Note that we depart from (42) at the single oldest recipient bin, ages 95–99, whose efficiency ζ_J we calibrate freely to best match the age profile. Further, there is a behavioral reason that ζ_J may be magnitudes *lower* than ζ_j with $j < J$:

[40] See Technical Appendix B.2.4 for a plot of φ_t over time.

since ζ_J governs the efficiency of those providing care to age-J agents, we may expect that older agents, especially those near the end of life, may be considerably more difficult to care for than those who are less old.

The outer nest governs the margin between ordinary consumption and care in total. The weight ν_d sets how much of a dependent's resources go to care of either kind: raising it tilts the dependent toward ordinary consumption and shrinks the care composite as a whole, so it lowers the nursing market expenditure share and average elder care hours *together*. That is what separates it from ν_n, which moves the same two series in opposite directions by substituting purchased for informal care; targeting the expenditure share and elder care hours jointly therefore identifies the two weights separately. The curvature ρ_d, jointly with ρ_n, determines the sign of the expenditure share's response to price. Because informal care enters the composite without being purchased, the relevant condition is not simply that $1/(1-\rho_d)$ lies below unity. Holding a dependent's total spending fixed, a rising relative nursing price *raises* formal care's share of that spending if and only if $\rho_n + (\rho_d - \rho_n)\,\nu_n (n_{jt}/\widetilde{n}_{jt})^{\rho_n} < 0$, which reduces to $\rho_d < 0$ only when informal care is absent. A rising θ_{Nt} alongside a rising p_t is the pattern we observe; given ρ_n, which the co-movement of elder care hours with the relative price pins down, it restricts ρ_d. The calibrated values select the case of Section 3.5. With ρ_d far below ρ_n (Table 1), more informal care lowers the demand for formal care, and care's share of the dependent's consumption–care aggregate stays below $(\rho_n - \rho_d)/(1 - \rho_d - \sigma)$ at every dependent age and period of the calibrated path, so a cut in formal care raises the value of an informal hour to helpers. Note that the within-period marginal rate of substitution between c_{jt} and $\widetilde{n}_{jt}$ depends on ν_d and ξ only through the single combination $\nu_d \xi^{\rho_d}/(1-\nu_d)$, so the observed consumption–care split cannot separate them. We thus fix this value.

The altruism weight $\underline{\beta}$ sits outside both nests, entering only the leisure/care supply margin (29) as the shadow price of caregiving time, where it pins the *level* of average elder care hours that ρ_n leaves free. Its reach is wider than that single moment: because effective leisure nets out care time, $\ell_{jt} = 1 - h_{jt} - q^s_{jt} - \phi_j$, caregiving competes directly with market work, so $\underline{\beta}$ also moves aggregate labor supply and the capital-output ratio, while by raising $\widetilde{n}_{j+\bar{j},t}$ it depresses demand for market-purchased care, given $\rho_d < \rho_n$ (see Section 3.5). Meanwhile, $\underline{\beta}$ and ν_n are nearly indistinguishable in terms of identification unless both elder care hours and market labor supply are targeted jointly. Note finally that $\underline{\beta}$ is not scale-free: it multiplies $\partial u^d/\partial \widetilde{n}$ and so inherits whatever ν_d, ρ_d, and ξ imply, meaning small values need not indicate weak altruism and $\underline{\beta}$ cannot be re-tuned independently of (14).

4.3 Markups, Productivities, and Production Technology Parameters

We face several choices when it comes to setting the time-varying markups and productivities, as well as the static production technology elasticities, either exogenously versus varying some of them within an internal calibration routine to achieve quality model fitness. In this section we discuss our quantitative reasoning with regards to how we choose these parameters.

We start by explaining which objects we exogenously fix, aligning our choices with those from the growth accounting exercises in Section 2 whenever possible. We fix $\omega_t = \overline{\omega} = 0.4313$ — average nursing to economy-wide wage ratio (BLS 2025a; BEA 2026b; BEA 2026f). We set the net 5-year growth rate of Z_{ct} to be constant according to estimates from Feenstra, Inklaar, and Timmer (2015) — $g_{Z_c} = 0.0323$;[41] we normalize $Z_{c1} = 1$ in the initial 1980 steady state.

The non-nursing capital intensity, α_c, will dictate the degree to which markup *levels* in that sector bite. Because that sector is Cobb-Douglas, its labor cost share is exactly $1 - \alpha_c$ irrespective of factor prices, so labor's share of non-nursing *revenue* is $(1 - \alpha_c)/[1 + (\mu_{ct} - 1)\theta_{C_c,t}]$ by (3), where $\theta_{C_c,t}$ is the share of sector-c output going to consumption (recall sector-c contains consumption *and* investment, I_t). Requiring the markup to exceed marginal cost, $\mu_{ct} \geq 1$, therefore bounds the capital intensity by the measured labor share, and since the restriction must hold period by period and α_c is constant, the binding constraint must accommodate the largest value the labor share takes, $\alpha_c \leq 1 - \max_t \theta_{L_c,t}$. Taking $\theta_{L_c,t}$ from the Penn World Tables aggregate labor share (Feenstra, Inklaar, and Timmer 2015), which peaks at 0.640 in 2001, the restriction caps α_c at 0.360 over 1990–2019.[42] While the growth accounting exercise in Section 2 tells us what kinds of relative-markup and relative-TFP *growth rates* can rationalize the data, the full GE calibration requires specifying sectoral markup *levels*. This is fairly straightforward for μ_{ct}: we choose $\alpha_c = 0.33$ and then use the aggregate labor share and the consumption share $\theta_{C_c,t}$ to compute the time series

$$\mu_{ct} \;=\; 1 + \frac{(1 - \alpha_c)/\theta_{L_c,t} - 1}{\theta_{C_c,t}}, \tag{43}$$

whose least-squares quadratic trend across the seven sample points we feed into the model.[43]

We internally calibrate the time-varying technology and markup levels associated with the nursing sector — Z_{n1}, g_{Z_n}, μ_{n1}, and g_{μ_n} — as well as the fixed production elasticities, α_n and ϵ. The nursing productivity level and its growth rate, Z_{n1} and g_{Z_n}, primarily target the relative price of nursing-home care and its growth. The nursing markup level and its growth rate, μ_{n1} and g_{μ_n}, primarily target the shares of aggregate capital and labor allocated to the nursing sector, along with α_n and ϵ. Also, nursing productivity is not restricted to a single growth rate, as we let Z_{nt} grow at g_{Z_n} through 2005 and at g'_{Z_n} thereafter, with the level continuous at the break. We do this because a constant g_{Z_n} forces $\{p_t\}_t$ onto a path whose curvature the data reject: the relative price rises fastest early in the sample and decelerates after the mid-2000s, and a one-rate path cannot reproduce that bend.

[41] We use g_y to denote the net growth rate; recall $\widetilde{g}_y = \ln(1 + g_y)$.

[42] We assume that the aggregate share and the non-nursing sectoral shares are identical.

[43] We measure $\theta_{C_c,t}$ from NIPA Table 1.5.5, with I_t fixed investment and C_t (personal consumption expenditure net of nursing homes), both in real terms. Technical Appendix B.2.3 plots the sampled μ_{ct}, the values of which run between 1.065 (2000) and 1.174 (2010).

Table 1: Calibrated Parameters

Externally Calibrated Parameters

Parameter	Value	Description	Source/Justification
$\bar{j}$	7	Dependency Age Gap	35-year Age Gap
J	16	Maximum Age	Die by Age 99
J_R	10	Retirement Age	Retire by Age 65
$\{M_{jt}\}_{j,t}$	Tech. App. B.2.1	Population	U.N. Dept. of Economic & Social Affairs (2024)
$\{s_{jt}\}_{j,t}$	Tech. App. B.2.1	Survival Rates	U.N. Dept. of Economic & Social Affairs (2024)
$\{\varpi_{jt}\}_{j,t}$	Tech. App. B.2.1	Dependency Incidence	Corrada et al. (2010) and Langa et al. (2017)
$\{\lambda_j\}_j$	Tech. App. B.2.1	Dependent Mortality	Garre-Olmo et al. (2019)
$\{\psi_{jt}\}_{j,t}$	Tech. App. B.2.1	Dependency Fraction	Computed, (12)
δ	0.396	Depreciation	9.6% Annual
β	0.9178	Discounting	0.983 Annual Rate
σ	1.1	Consumption Curvature	Intertemporal Elast. 0.909
γ	4.0	Leisure Curvature	Cooley and Henriksen (2018)
χ	0.4123	Leisure Weight	Cooley and Henriksen (2018)
κ_1	0.0531	Participation Avg.	Kitao (2014) and Cooley and Henriksen (2018)
κ_2	0.00149	Participation Age Weight	Kitao (2014) and Cooley and Henriksen (2018)
κ_3	1.4178	Participation Curvature	Kitao (2014) and Cooley and Henriksen (2018)
$\{\phi_j\}_j$	Tech. App. B.2.2	Participation Profile	Computed
$\{\eta_j\}_j$	Tech. App. B.2.2	Labor Efficiency	Hansen (1993)
τ	0.153	Transfer Tax Rate	Social Security Administration (2024), Table 2.A3
$\{\varphi_t\}_t$	Tech. App. B.2.4	Transfer Ratio	Computed
$\overline{\omega}$	0.4313	Avg. Sectoral Wage Ratio	(BLS 2025a; BEA 2026b; BEA 2026f)
g_{Z_c}	0.0323	Z_{ct} 5-yr. Growth Rate	Feenstra, Inklaar, and Timmer (2015)
Z_{c1}	1	Initial S.S. Z_{ct}	Normalization
α_c	0.33	Non-nursing Capital Share	Assumed
$\{\mu_{ct}\}_t$	Tech. App. B.2.3	Sector-c Markup	Computed with Quadratic Trend

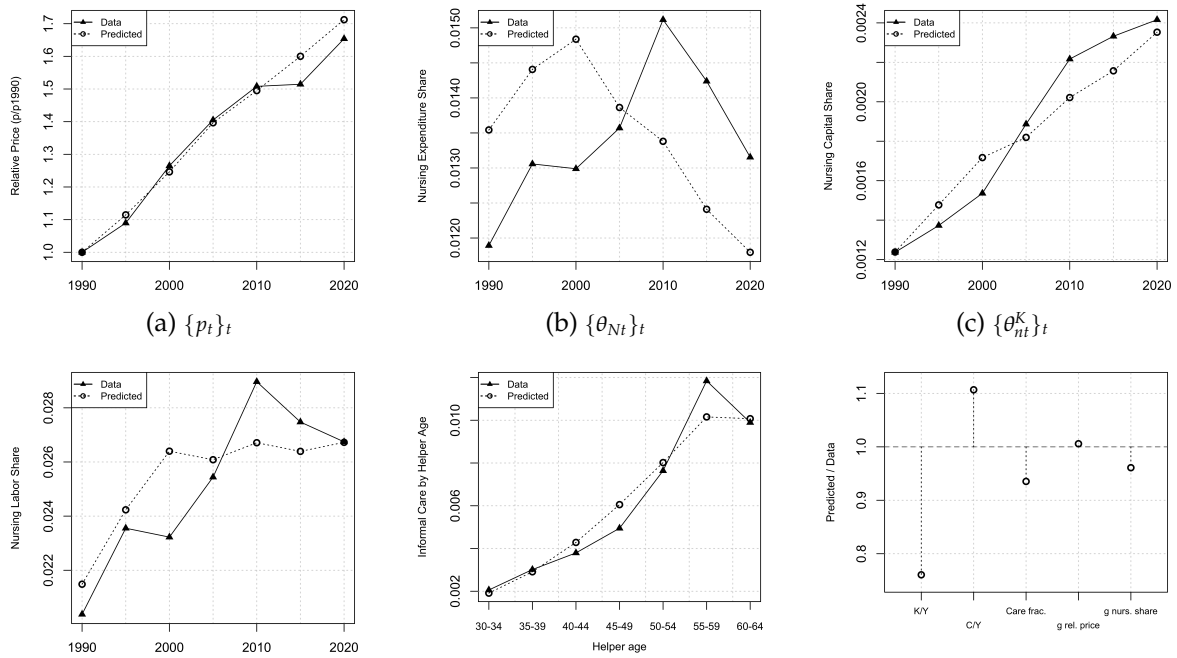
Internally Calibrated Parameters

Parameter	Value	Description	Primary Moment(s)
ν_n	0.8480	Formal Care Weight	Normalization (see text)
ρ_n	-0.3614	Formal/Informal Elast.	$\{p_t\}_t$; Elder Care Hrs.
ν_d	0.9813	Non-nursing Care Weight	$\{\theta_{Nt}\}_t$; Elder Care Hrs.
ρ_d	-3.7783	Cons./Care Elast.	Co-movement of $\{p_t, \theta_{Nt}\}_t$
$\underline{\zeta}_1$	37.9169	Care Efficiency, Level (recipient ages 65–94)	Age Profile of Care (level)
$\underline{\zeta}_2$	-0.8517	Care Efficiency, Age Gradient (recipient ages 65–94)	Age Profile of Care (shape)
ζ_J	0.2925	Care Efficiency, Recipient Ages 95–99	Age Profile of Care (95–99)
ξ	1.0000	Consumption Utility Discounting	Normalization (see text)
$\underline{\beta}$	0.4763	Altruism	Held fixed (see text)
g_{Z_n}	-0.1239	Z_{nt} 5-yr. Growth Rate, through 2005	$\{p_t, \theta_{Nt}, K_{nt}/K_t, L_{nt}/L_t\}_t$
g'_{Z_n}	-0.0800	Z_{nt} 5-yr. Growth Rate, after 2005	$\{p_t\}_t$ (curvature)
Z_{n1}	1.0000	Initial S.S. Z_{nt}	Normalization (see text)
g_{μ_n}	-0.0552	μ_{nt} 5-yr. Growth Rate	$\{K_{nt}/K_t, L_{nt}/L_t\}_t$
μ_{n1}	1.8872	Initial S.S. μ_{nt}	$\{K_{nt}/K_t, L_{nt}/L_t\}_t$
ϵ	2.1487	Sector-n Elasticity	$\{K_{nt}/K_t, L_{nt}/L_t\}_t$
α_n	0.3331	Sector-n Weight	$\{K_{nt}/K_t, L_{nt}/L_t\}_t$

4.4 Discussion and Model Fitness

Table 1 shows the externally and internally calibrated parameter values, their descriptions, their sources/justification, and for the internally calibrated ones, the moment(s) they target. Figure 4 shows time series of fitted model moments relative to the data. Figures 4a through 4d show the aggregate moments' time series along the transition path as predicted by the model (dotted lines and circles) against data (solid lines and triangles). Figure 4e shows the age profile of care hours provided by informal caregivers. Figure 4f shows the scalar moments' deviation from their data (or assumed) values.

The nursing sector block carries nearly all of the leverage over the relative price and the sectoral input shares, so that feedback effects from household-side substitutability of formal and informal care have little impact on prices at this particular parameterization. The household block, in turn, governs how formal and informal care respond to those prices rather than the prices themselves. Z_{n1} and μ_{n1} set where the time series begin, and g_{Z_n} and g_{μ_n} determine how fast they move. Meanwhile, α_n and ϵ divide the input shares between capital and labor. They enter the cost shares in (9) jointly, but the time variation in $r_t/(\omega_t w_t)$ along the transition path separates them in principle: ϵ governs how the cost shares move with that ratio and α_n governs their level. Rising wages push nursing sector output toward using more capital, and the high ϵ is what lets nursing's capital share grow faster than its labor share.



(a) $\{p_t\}_t$ (b) $\{\theta_{Nt}\}_t$ (c) $\{\theta^K_{nt}\}_t$

(d) $\{\theta^L_{nt}\}_t$ (e) Avg. Help by Caregiver Age (f) Scalar Moments

Figure 4: In this figure we plot predicted (dotted lines and circles) time series of model moments along our targeted transition paths — panels (a) through (d) — against data (solid lines and triangles), along with the average informal care provision profile in (e) and scalar moment departures from data in (f).

The household preference parameters exert leverage of a different kind. To some degree both ν_n and $\underline{\beta}$ dictate the formal/informal substitution margin, so they move the nursing expenditure share and elder care hours in opposite directions, whereas benchmark tests revealed that ν_d moves them in the same direction. Targeting allocations of both formal and informal care jointly is what separately identifies the ν_d and ν_n weights. The curvature ρ_d is disciplined not by any level but by the co-movement of θ_{Nt} with p_t. The ζ_j's act almost entirely on the age profile of care with little reach into the aggregates, so that the age profile of care provision almost entirely identifies $\underline{\zeta}_1$, $\underline{\zeta}_2$, and ζ_J. We match this profile well in Figure 4e.

Turning to the production block, the calibration selects $\alpha_n = 0.3331$ and $\epsilon = 2.1487$, placing nursing capital and labor in the gross-substitutes region.[44] Our calibration a;so suggests that the implied mean capital cost share of sector n is 0.073, roughly 28% of the 0.26 that Donahoe (2000) reports for the broader health sector. This is to be expected if residential long-term care is substantially more labor-intensive than health care in aggregate as hospitals, imaging, and diagnostic services carry the kinds of equipment that nursing facilities largely do not.

Looking at the demand side, note that the empirical literature has generally found formal (marketized) and informal (non-market) care to be substitutes: Mommaerts (2018) finds coresidence and nursing-home utilization to be substitutes, and Mommaerts (2025) finds that the availability of informal caregivers lowers both long-term-care insurance purchase and Medicaid spending on formal care. Our model is the first to study the substitution between formal and informal care in a full GE setting with *all* of capital rental rates, wages, and sectoral output prices being endogenous. We separate two important elasticities — that between formal and informal care and that between care and non-care consumption. We find both margins to be gross complements, but the *relative elasticity* is what matters, and it delivers the formal/informal substitution effects the micro studies find. Within the care composite, $\mathcal{G}$, the calibration selects $\rho_n = -0.36$, an elasticity of substitution between purchased nursing services and informally provided care of $1/(1-\rho_n) = 0.73$. In the outer nest encompassing the non-nursing and care consumption tradeoffs, the calibration selects $\rho_d = -3.78$, an elasticity between ordinary consumption and care of 0.21. Because the nursing production block pins down the path of p_t, the role of ρ_d, together with ρ_n, is not to generate the rise in the relative price but to govern how the nursing expenditure share *responds* to it: with both curvatures negative, θ_{Nt} *rises* with p_t in Figures 4a and 4b, which our benchmarking tests suggest no substitutable specification could generate. Turning to our estimate of $\rho_n = -0.36$, formal and informal care are gross complements within the composite. This does not by itself mean that the two move together: rather, the relative magnitude of ρ_d and ρ_n determines this. Because the two elasticities differ markedly, care substitutes far more readily with itself (i.e., formal for informal and vice versa) than with ordinary consumption.

Our aggregate estimates are consistent with the substitution effects from the micro literature (Van Houtven, Coe, and Skira 2013; Mommaerts 2018; Mommaerts 2025). Because $\rho_d < \rho_n$, more

[44] Note that the growth accounting in Technical Appendix B.1 can only restrict ϵ and α_n to a locus of pairs given an assumed capital income share, though it agrees with the calibration on the sign and ordering of the two price forces—differential productivity dominates and relative markups fall.

informal care lowers the demand for formal care in (39) even though formal and informal care are gross complements within $\mathcal{G}$: this is the same response the micro studies document. Our estimates are *aggregate* elasticities: they also account for the margin between care and non-care consumption and are disciplined with prices determined in GE, although at our parameterization household-side feedback effects turn out to be small.

4.5 Predicted Output

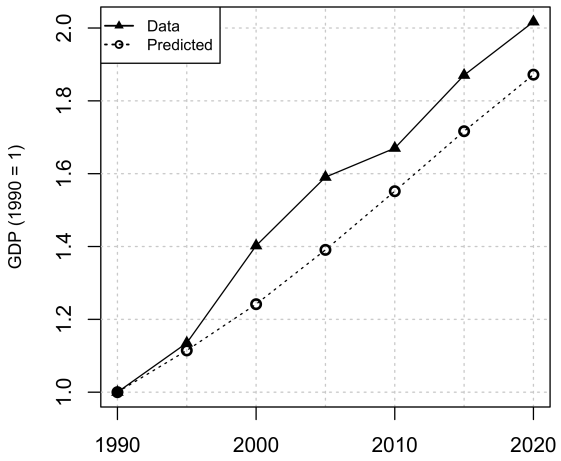


Figure 5: This figure shows model-predicted GDP ($Y_t = I_t + p_{ct}C_t + p_{nt}N_t$) against non-targeted data from BEA (2026e), normalized to unity in 1990.

Note that we do not target output growth in our calibration. GDP in our model is $Y_t = I_t + p_{ct}C_t + p_{nt}N_t$ with the investment good as numeraire. In Section 5 when we investigate the aggregate implications of population aging alongside the demand for both informal and formal, one thing we seek to understand is how the various channels impact long-run growth. Thus, to assess whether our model can credibly predict output growth, Figure 5 shows model-predicted GDP against U.S. data (BEA 2026e), normalized to unity in 1990. Note that the model smoothly predicts the non-targeted evolution of GDP from 1990-2020, absent the late 1990s tech-induced jump, which is likely due to our assumption that g_{Z_c} (growth rate of non-nursing-sector TFP) is constant. Nonetheless, given this is a model about long-run changes, we are satisfied with our model's performance in predicting this non-targeted time series.

5 Quantitative Exercises

We now use the quantitative model to engage in several exercises seeking to understand both which forces are primarily responsible for the current state of the aggregate economy through our model's lens (how we got here) and what might happen moving forward (projecting where we are going). One goal is to use the model to price out the value of informal care relative to both

aggregate output, aggregate expenditure, and formal care itself, in order to understand how such derived valuations have changed over time or may evolve in the future.

We thus begin our quantitative exercises by pricing informal care under two conventions and using those prices to value it against formal care, output and total expenditure (Section 5.1). We then proceed to counterfactually decompose the fundamental drivers of formal, nursing-sector structural change (Section 5.2). Finally, we examine several model-implied projections as to how population aging and continued technological and sectoral changes may impact the evolution of the U.S. economy through 2100 (Section 5.3).

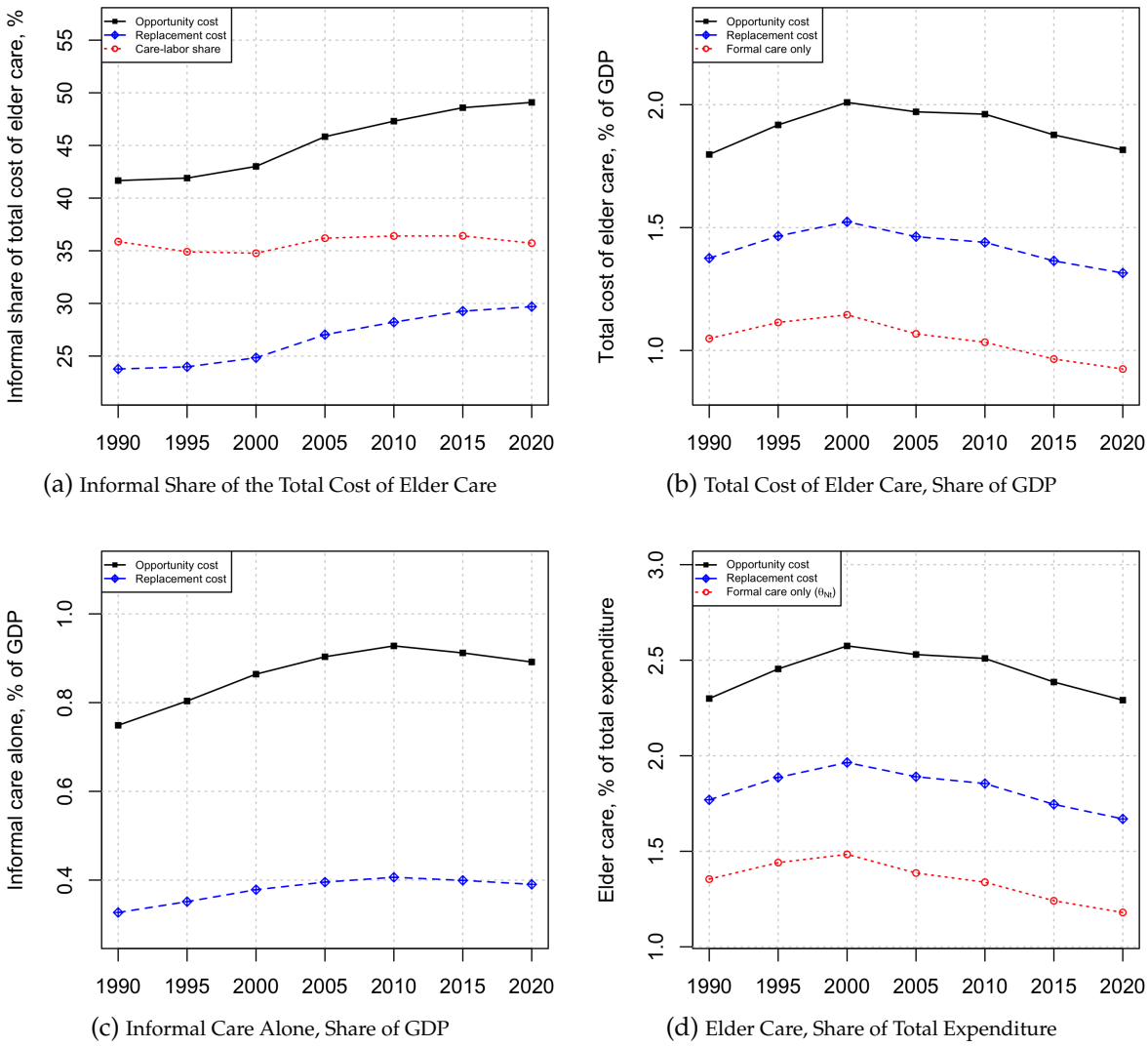
5.1 Aggregate Average Price of Informal Care

While the formal half of elder care's cost is priced by the market and its total value is $p_{nt}N_t$, the informal component is a quantity of time, $Q_t = \sum_{j=1}^{J_R-1} M_{jt}q_{jt}^s = \sum_{j=J_R}^{J} M_{jt}^d q_{jt}$, and converting it into resources requires a price that no market reports. We consider two aggregate average pricing schema, readily attainable from our model environment, and each supported by the literature — 1) the average opportunity cost (*oc*) of the caregiver's time and 2) the average replacement cost (*rc*) of hiring a market substitute.[45] Note that we refer to these objects as "averages" or "aggregate averages" because they do not *exactly* correspond to the marginal cost of care incurred directly by an informal care provider. There are, of course, two perspectives on which convention is appropriate. The literature on caregiving prices an hour at what the caregiver gives up, following the Becker (1965) treatment of time allocation, because it asks how caregiving depresses employment and earnings (Muurinen 1986; Van Houtven, Coe, and Skira 2013). The national accounts for household production instead price an hour at the wage of a hired household worker, because they value what is produced at home (Bridgman et al. 2012; Bridgman 2016a; Bridgman, Craig, and Kanal 2022; Bridgman 2016b; Bridgman, Duernecker, and Herrendorf 2018; Bridgman 2023; Bridgman 2025; Bridgman 2026).[46] The two conventions thus answer different questions: the opportunity cost measures what caregivers give up, which is the cost our caregivers weigh when they choose q_{jt}^s, and the replacement cost measures what the same care would cost to buy.

The *oc* measure values an hour of informal care at what the caregiver gives up by providing it. In the model an hour of q_{jt}^s enters (19) at $\overline{w}_t\eta_j(1-\tau_t)$ net of tax, and is subtracted from the

[45] Both often offer different valuations of the same allocation. van den Berg, Brouwer, and Koopmanschap (2004) and Koopmanschap et al. (2008) survey the available conventions, and van den Berg et al. (2006) show that the opportunity cost and replacement cost methods each attain materially different valuations. Hurd et al. (2013) make the same point for U.S. dementia costs, reporting the informal component under both a replacement-cost and a forgone-wage convention and obtaining different totals. The choice of convention is therefore not innocuous.

[46] These authors reject the opportunity cost argument because it ties productivity at home to productivity in the market, so that a doctor would clean a house more productively than a paid housekeeper (Bridgman, Duernecker, and Herrendorf 2018), and they attribute the gap between market wages and household-worker wages to low returns to skill at home and a higher disutility of market work (Bridgman 2025).



(a) Informal Share of the Total Cost of Elder Care

(b) Total Cost of Elder Care, Share of GDP

(c) Informal Care Alone, Share of GDP

(d) Elder Care, Share of Total Expenditure

Figure 6: This figure shows different measures as to how informal care and the total value of elder care relate to other aggregate outcomes. Panel (a) shows the informal share of the total cost of elder care under the *oc* and *rc* conventions, together with the care-labor share, $L_t^q/(L_t^q + L_{nt})$. Panel (b) shows the total cost of elder care as a share of GDP for both *oc* and *rc*, with formal care alone, $p_{nt}N_t/Y_t$, in red. Panel (c) shows the value of informal care alone as a share of GDP for both *oc* and *rc*. Panel (d) shows elder care's share of total expenditure with informal care counted in both numerator and denominator, against the nursing expenditure share $\theta_{Nt} = p_{nt}N_t/X_t$, in red.

effective labor supply, $L_t = \sum_{j=1}^{J_R-1} M_{jt}\eta_j(1 - \phi_j - \ell_{jt} - q_{jt}^s)$. Let

$$L_t^q \equiv \sum_{j=1}^{J_R-1} M_{jt}\,\eta_j\,q_{jt}^s \tag{44}$$

denote the efficiency labor withdrawn from the market by caregiving; this object is distinct from Q_t in that it embeds the age-specific labor-productivity profile, $\{\eta_j\}_j$. The opportunity-cost valuation

of informal care is

$$V_t^{\text{oc}} = \overline{w}_t L_t^q, \tag{45}$$

where we use V (distinct from a consumer's equilibrium continuation value, $\mathcal{V}$) to denote the period-specific (static) "value" of informal care (i.e., informal outlay).[47] The replacement-cost measure values an hour of informal care at the cost of buying the same hour from the market at the market wage rate for formal nursing-care labor, $\omega_t w_t$; this measure assumes the hired worker is as efficient as the caregiver being replaced.[48] The replacement-cost aggregate value is

$$V_t^{\text{rc}} = \omega_t w_t L_t^q. \tag{46}$$

With a price in hand, the total social value of elder care, E_t^l, and the informal share of it, θ_{qt}^l are:

$$E_t^l = p_{nt} N_t + V_t^l \qquad \text{and} \qquad \theta_{qt}^l = \frac{V_t^l}{E_t^l}, \quad \forall l \in \{oc, rc\}. \tag{47}$$

Define the average cost (price) of an hour of informal care under convention l as

$$p_{qt}^l := \frac{V_t^l}{Q_t}, \qquad \forall l \in \{oc, rc\}, \tag{48}$$

so that $V_t^l = p_{qt}^l Q_t$. Because V_t^l and Q_t both sum over every provider age, p_{qt}^l is an economy-wide weighted average: the cost of an hour supplied at each provider age, weighted by that age's share of all informal care hours:

$$p_{qt}^{oc} = \sum_{j=1}^{J_R-1} \frac{M_{jt}\, q_{jt}^s}{Q_t}\, \overline{w}_t \eta_j, \qquad p_{qt}^{rc} = \sum_{j=1}^{J_R-1} \frac{M_{jt}\, q_{jt}^s}{Q_t}\, \omega_t w_t \eta_j. \tag{49}$$

This is *not* the marginal cost of an hour supplied by a provider of any particular age.[49]

Finally, note that both formal and informal care are ultimately labor, and the model expresses both in the same efficiency units: L_{nt} is the labor employed by the nursing sector and L_t^q in (44) is the labor absorbed by informal caregiving. The informal share of nursing labor is thus $L_t^q/(L_t^q + L_{nt})$, which we also analyze for comparative purposes.

Figure 6 demonstrates how both the informal value of elder care and the total value of elder care under various pricing schema have evolved along the calibrated transition path. In Figure 6a we see that informal care accounts for between 24% and 49% of the total resource cost of elder care, depending on how an hour of it is priced — 24% to 30% under *rc* and 42% to 49% under *oc*, with the

[47] Recall that the household wage is a function of the allocation of sectoral labor and the relative sectoral wage — $\overline{w}_t = w_t(L_{ct} + \omega_t L_{nt})/L_t$.

[48] This is the basis for other imputations of the aggregate value of informal caregiving in the U.S. (see, e.g., National Alliance for Caregiving and AARP (2004), National Alliance for Caregiving and AARP (2009), and Chari et al. (2015)).

[49] Technical Appendix D.1 works at the margin instead: it compares, age by age, the dependent's willingness to pay for one more hour of informal care with that hour's after-tax cost to the provider who supplies it.

price-free care-labor share running between 35% and 36%. Counting only the value of market care as in the NIPA data therefore captures between 51% (*oc*) and 70% (*rc*) of what elder care *actually* cost the economy in 2020. Figure 6b shows that model-predicted total outlay on elder care went from between 1.38% and 1.80% (*rc* and *oc* respectively) of total output in 1990 to between 1.32% and 1.82% in 2020. Comparing the black and blue lines in Figure 6b to the red line demonstrates that including informal care time roughly doubles the fraction of total resources devoted to caring for the old under *oc* and raises it by about two fifths under *rc*;[50] similar magnitudes are attained via comparisons in Figure 6d examining the elder care share of total *expenditure*. Finally, notice that counting informal care tends to shift elder care's share in levels at magnitudes that depend on the pricing schema, but leaves trends over time practically unchanged.

5.2 Contributors to Structural Transformation

In this section we fix different exogenous time series one at a time to understand how the various drivers of the model economy's evolution contributed to nursing-sector structural transformation from 1990 to 2020. We leave some of the more detailed breakdowns of the results of this exercise for Technical Appendix D.2, focusing here on two constellations of counterfactuals as follows.

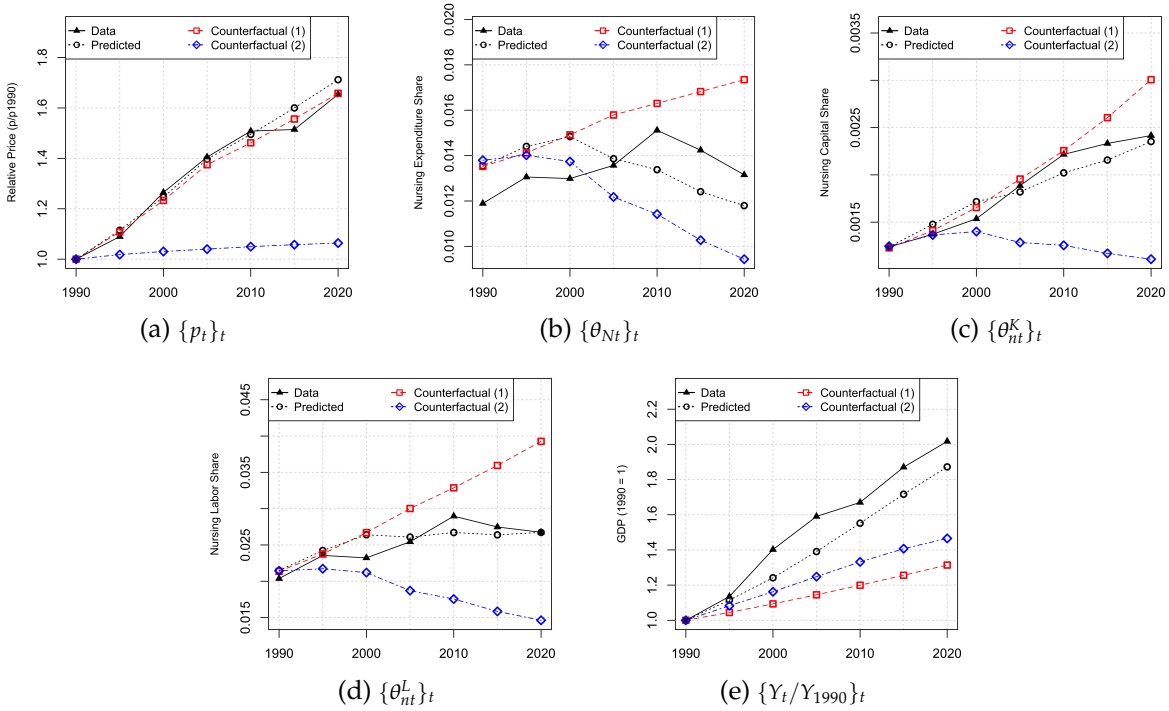
First, in counterfactual (1) we turn off population growth and population aging, freezing $\{M_{jt}, s_{jt}, \varpi_{jt}, \varphi_t\}_{j,t}$ at its 1990 value, while allowing all other exogenous objects to evolve: this counterfactual thus simulates the sensitivity of the model results to traditional drivers of structural change (i.e., productivities and markups). Second, we do the opposite: in counterfactual (2) we prevent $\{Z_{ct}, \mu_{ct}, Z_{nt}, \mu_{nt}\}_t$ from growing after 1990, while allowing the demographic variables to evolve, thus demonstrating the role of demographic changes in driving transformations. Under each of these scenarios we examine how the key aggregates would have evolved.[51]

Figure 7 shows the counterfactual (1) and (2) simulations in red and blue respectively against the predicted baseline (dotted lines and circles) and data (solid lines and triangles).[52] Several things stand out. First, it is clear that cost disease is entirely responsible for rising relative prices. Still, demographic variation does appear to have some impact on nursing-sector aggregates: looking at Figure 7b, had demographic change *not* occurred (red line), cost disease would have helped continue to drive the nursing expenditure share up into the 2020s. This is primarily due to dependency incidence, ϖ_{jt}, falling over the sample, so holding it at its 1990 profile keeps more of an aging population dependent and raises the nursing expenditure share above the baseline. By contrast holding population and survival, M_{jt} and s_{jt}, at 1990 instead removes aging and lowers the share. Falling incidence is the stronger demographic force in the demography counterfactual, though. Not surprisingly, absent this particularly positive aspect of demographic change, relatively more resources (capital and labor) would have to go to the slow-growing sector (nursing) plagued by cost disease (Figures 7c and 7d).

[50] This is also reflected in the informal shares of GDP in Figure 6c.

[51] In Technical Appendix D.2 we also show the results of counterfactual simulations, turning off one driver at a time to assess more detailed sensitivities.

[52] Counterfactual care series are in Figures D.2 and D.3 in Technical Appendix D.2.1.



(a) $\{p_t\}_t$ (b) $\{\theta_{Nt}\}_t$ (c) $\{\theta^K_{nt}\}_t$

(d) $\{\theta^L_{nt}\}_t$ (e) $\{Y_t/Y_{1990}\}_t$

Figure 7: This figure is analogous to Figures 4a through 4d and Figure 5 but includes the simulated series of counterfactuals (1) in red and (2) in blue. Panels (a) through (d) feature the relative price, nursing expenditure share, nursing share of capital, and nursing share of labor, while panel (e) features 1990-normalized GDP.

Meanwhile, within counterfactual (2) — frozen production-side series — turning off declining Z_{nt}, and to a lesser extent rising Z_{ct}, leads to a 48% reduction in relative prices by 2020, demonstrating that the slow growth in the nursing sector has the strongest impact on relative price growth. If nursing-sector markups had not fallen, relative prices would have risen by 41% from 1990 to 2020, still a fraction of the 71% predicted from the baseline. The technology and markup series have little effect on the shape of the nursing expenditure share profile, though they do have implications for GDP growth, which we now discuss.

Consequences of both demographic and technological change for output are small in spite of what they may seem to be at first glance. In counterfactual (1) GDP is 29.7% lower relative to the baseline, but that gap is almost entirely one of population *size* due to turning off population growth: holding population and survival at 1990 accounts for nearly all of it, while holding incidence at 1990 barely moves output. When looking at GDP per capita in counterfactual (1), it is only 1.2% lower. Freezing population and survival removes three decades of cohort growth and improving survival, which shrinks output and population alike. Most of the small loss instead comes from holding incidence at 1990, which keeps more households dependent and shifts labor and capital toward the nursing sector. Meanwhile, in counterfactual (2) GDP growth is dampened because the dominant sector (non-nursing-consumption) does not see its substantive TFP growth. Thus, under this scenario while the populace may be spending relatively less on formal nursing

services, their overall income falls due to relatively slow growth that is driven *only* by population increases (blue line) as opposed to meaningful improvements to productivity (i.e., increasing Z_{ct}). In counterfactual (2) we also further turn off markup growth (or degrowth in the case of sector n) and sector-n productivity growth. Foremost, markup growth almost exclusively affects prices and not real outcomes. Second, regarding sector-n, its contribution to aggregate output is small ($< 1.4\%$ in 1990). Thus, *not* letting cost disease happen (e.g., declining TFP) in such a small sector does not improve outcomes much on aggregate.

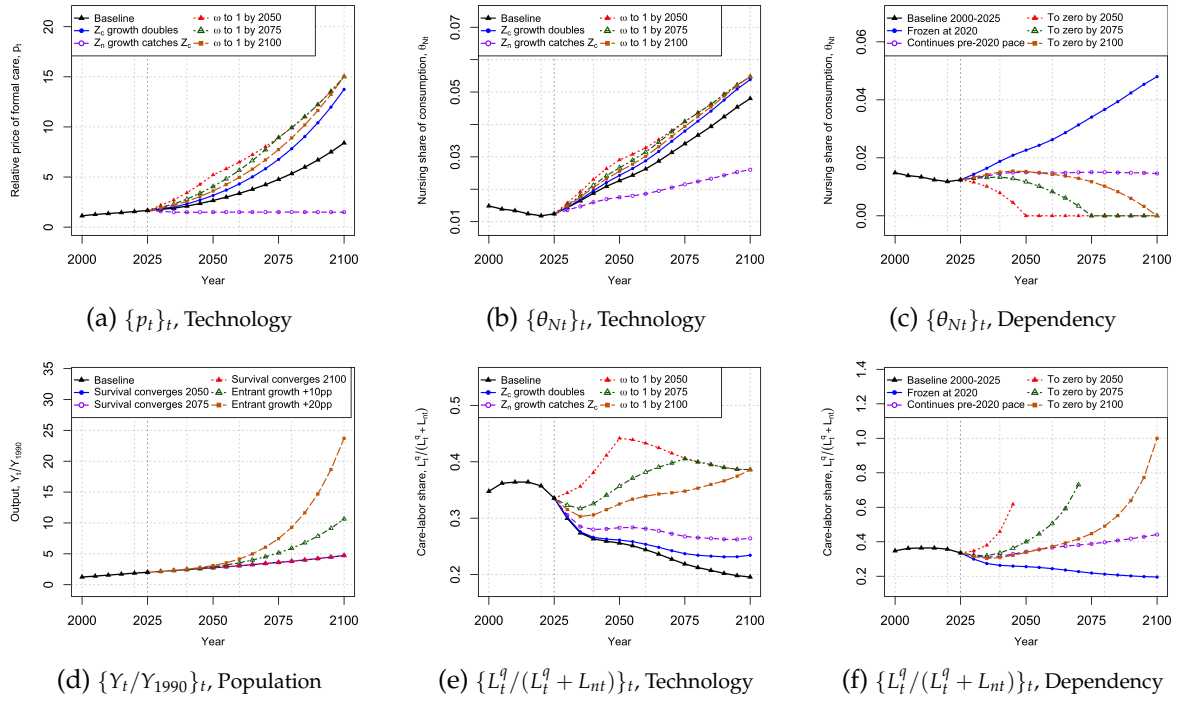
5.3 Forward Projections

How might the economy evolve into the future, and how might different scenarios of demographic change associated with population aging and dependency incidence affect its evolution? What about different rates of technical change and changes to market concentration? In this section we consider four different constellations of future projections, simulating the economy forward to year 2100 in each, in order to understand how economic activity (both formal and informal) may evolve under different scenarios.

The four sets of future projections take the following flavors. First, we consider different paths for the evolution of dependency incidence, ϖ_{jt}, letting all other exogenous time series evolve according to either their U.N. Dept. of Economic & Social Affairs (2024) projections or the growth paths we estimate. Second, we hold ϖ_{jt} after 2020 fixed and perturb the rates of technological change after 2020, including ω_t, in different ways. Third, we do the same for market concentration, holding all else fixed. Finally, fourth, we consider several changes to the exogenous evolution of the population, independent of variation in ϖ_{jt}, including i) convergence in survival rates by type and ii) changes to the number of entrants (i.e., increases to natality, wherein the projections suggest a continued decline). In all scenarios the households are assumed to anticipate that the economy evolves according to our baseline evolution, and each perturbation begins in 2030 and is unanticipated by households alive in 2025, who internalize the new paths from then on. We include discussions of the first, second, and fourth sets of projections here in the main text, relegating the relatively mundane (and obvious) findings from the third set (markups) to Technical Appendix D.3.

We commence by consulting the model's predictions as to the evolution of aggregates. Figure 8 shows the projected series that carry the main dynamics, and Technical Appendix D.3.1 shows the remainder. Each series starts with the calibrated baseline from 2000 to 2025 and continues out as projections thereafter all the way to 2100.[53] Many of the implications ascertained in Section 5.2 carry forward here: 1) demographics do not affect prices (Figures D.5a and D.5b in Technical Appendix D.3.1), though dependency rates can affect expenditure shares, driving down demand for formal care given $M^d_{jt}/M_{j-\bar{j},t}$ rises thus driving up demand for informal care (Figure 8c); 2)

[53] We start the new forward simulations *after* 2025 and pin the 1980 to 2025 baseline calibrated path to such simulations. The idea is not to disturb the equilibrium pre-2025 wherein expectations regarding 2025 outcomes are baked into 2020 decisions.



(a) $\{p_t\}_t$, Technology (b) $\{\theta_{Nt}\}_t$, Technology (c) $\{\theta_{Nt}\}_t$, Dependency

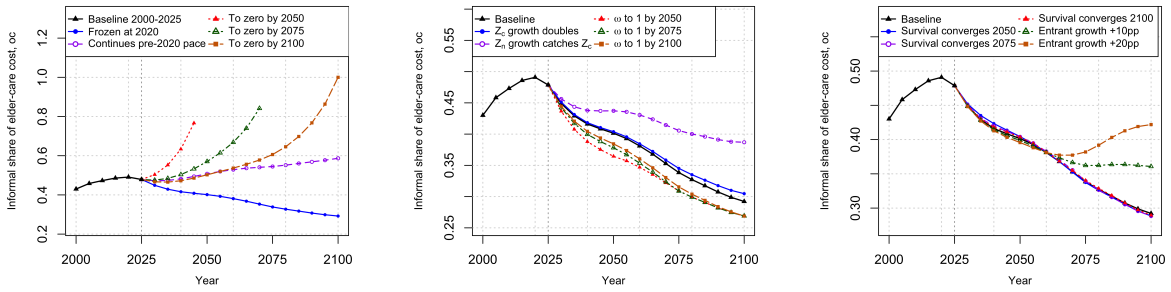
(d) $\{Y_t/Y_{1990}\}_t$, Population (e) $\{L_t^q/(L_t^q + L_{nt})\}_t$, Technology (f) $\{L_t^q/(L_t^q + L_{nt})\}_t$, Dependency

Figure 8: Projections of the series that carry the main dynamics, 2000–2100, under the dependency-rate (Dependency), technology and sectoral wage catch-up (Technology), and survival and natality (Population) scenarios. The black line is the baseline, and each colored line is one scenario, solved as an unanticipated shock after 2025. In the Dependency panels the black line ends at the 2025 branch and the blue line, with ϖ_{jt} frozen at 2020, continues it. Technical Appendix D.3.1 shows the remaining series.

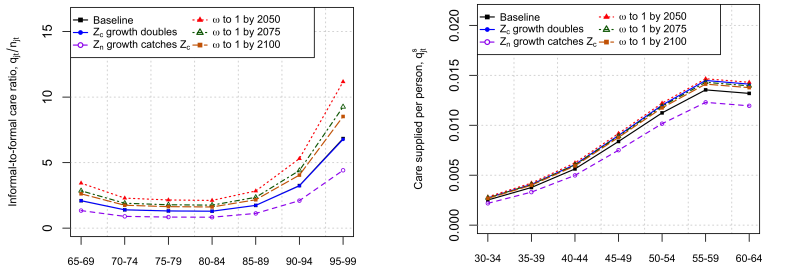
technological change and the associated sectoral labor-margin (increasing ω_t) affect the allocation of inputs and prices, as well as the expenditure share in predictable ways, and only faster nursing productivity growth halts the rise in p_t, while nursing wages catching up to economy-wide wages raises it the most (Figures 8a and 8b); 3) the demographic margin that actually affects aggregate output the most is increasing natality (Figure 8d), though increasing dominant-sector technological change (c-sector) also would increase growth significantly (Figure D.5e in Technical Appendix D.3.1); 4) absent either increases to nursing-sector technological change, significant declines in dependency incidence, and/or significant increases in natality, the nursing sector will grow to become $\approx 5\%$ of all consumption by 2100; 5) as either dependency declines or nursing-sector labor becomes more expensive (increasing ω_t), the informal care-labor share will rise, demonstrating how relatively weak complementarities ($\rho_d < \rho_n$) in the production of $\widetilde{n}_{jt}$ lead to substitution away from formal care in the face of increasing formal production input costs driving up market prices (Figures 8f and 8e).

The latter implications show that capital/labor input substitutability in the formal nursing sector and care-margin substitutability for the formal/informal tradeoff are interactive forces in driving both the allocation of liquidity to formal expenditure and off-market time by care providers

to informal outlay. This is especially salient if ω_t (i.e., nursing-sector wage relative markdown and/or nursing-sector labor productivity) is changing (see Figures 8b and 8e, and Figures D.4b and D.4e in Technical Appendix D.3.1). When letting nursing-sector wages catch up at different rates ($\omega_t \underset{_}{\to} 1$), the formal sector substitutes capital for labor yet still sees its overall share of consumption expenditure grow: because ω_t prices nursing labor rather than raising its efficiency, a rising ω_t is a cost shock, and the relative price rises by more than nursing output N_t falls, so that N_t ends the century well below its fixed-ω path. Informal labor's share of all care labor then rises (Figure 8e) because the nursing sector employs far less labor than on the fixed-ω path with θ^L_{nt} growing much more slowly. This brings us to a more nuanced discussion as to how assumptions regarding the projection of exogenous drivers of demographic and technical change affect inferences pertaining to a more detailed breakdown of the formal/informal care margin.



(a) Informal Share, *oc*, Dependency (b) Informal Share, *oc*, Technology (c) Informal Share, *oc*, Population



(d) q_{jt}/n_{jt}, 2050, Technology (e) q^s_{jt} by Helper Age, 2050, Technology

Figure 9: We show projected informal care statistics under *oc* pricing for 2000–2100, under the different dependency-rate (Dependency), technology and sectoral wage catch-up (Technology), and survival and natality (Population) scenarios. Panela (a) through (c) show the informal share of total elder-care costs under *oc*. Panels (d) and (e) show the care profiles by age under the 2050 technology scenarios. Technical Appendix D.3.2 shows informal care by age under the other scenarios.

Figure 9 shows how the informal share of elder care carries forward under each set of different projections, and Figure D.7 in Technical Appendix D.3.2 shows the care margins by age. When needing a price for informal care, we only use the Becker (1965) *oc* measure. Although we do not plot it here, providers per dependent, $M_{j-\bar{j},t}/M^d_{jt}$, is exogenous and changes only as we exogenously perturb and vary any of the demographic variables — ϖ_{jt} for the dependency-rate

scenarios and s_{jt} and M_{1t} (new entrants) for the survival and natality scenarios.

Along the baseline path, providers per dependent fall at every recipient age after 2050 while formal care keeps getting more expensive due to cost disease, so each provider supplies more care and informal care still rises relative to formal care at every age (the blue lines in Figures D.7b and D.7c in Technical Appendix D.3.2). Yet along this baseline, which holds ϖ_{jt} at its 2020 profile (the blue lines in Figures 9a and 8f), the informal share of elder care falls through 2100, as cost disease pulls spending and labor into the nursing sector. Falling dependency risk, ϖ, reverses this, because it raises providers per dependent and so the amount of informal care relative to formal care is higher at every age relative to the baseline (the purple, red, green and orange lines in Figure 9a; by age, Figure D.7a in Technical Appendix D.3.2).

The technology-projection scenarios leave providers per dependent untouched and act instead through the price of formal care. Letting nursing wages catch up (the red, green and orange lines) raises informal care relative to formal care at every age and draws more care from each informal provider, while letting nursing productivity catch up (the purple line) has the opposite effect (Figures 9d and 9e). Still, though, the informal share of the cost of elder care moves opposite to the informal-to-formal care ratio and to care supplied per provider due to the formal price increase. Notice, the informal share is lower than in the baseline when nursing wages catch up and higher when nursing productivity catches up (in Figure 9b, the red, green and orange lines fall below the black baseline and the purple line stays above it). A nursing-wage catch-up raises spending on formal care even as dependents buy less of it, while a nursing-productivity catch-up holds that spending down. Meanwhile, faster entrant growth (the green and orange lines in Figure 9c) raises providers per dependent only for the youngest recipients by 2050 and lifts the informal share only late in the century, while converging survival rates (the blue, purple and red lines, which lie on the black baseline in Figure 9c) barely moves the informal share.

6 Conclusion

The care sector does not exhibit the usual marketization patterns associated with service-sector structural change. Negative nursing productivity growth raises the relative price of formal care, and, given that care is a much stronger complement to other consumption than formal care is to informal care, cost disease raises informal care relative to formal care, together with more providers per dependent. That gap in complementarities also explains why micro data find formal and informal care to be substitutes (Orsini 2010; Mommaerts 2018; Mommaerts 2025), since, as we show, the relative substitution elasticity is what matters. Population aging, meanwhile, barely contributes to relative price movements, and falling age-specific dependency nearly offsets aging leading to rising dependency levels. Informal care comprises from 29.7% to 49.1% of the total cost of elder care in 2020, depending on how an hour is priced. Looking forward, among the scenarios we consider, only faster nursing productivity growth halts the rise in the relative market price, and long-run output per person depends mainly on non-nursing productivity, though total out-

put also grows with the size of entering cohorts. Elder care's claim on output roughly triples by 2100 at fixed age-specific dependency incidence, which is the assumption behind standard cost projections (Brookmeyer, Gray, and Kawas 1998; Alzheimer's Association 2026), and rises far less if incidence keeps declining.

The model's mechanics carry direct implications for policy. The relative price of formal care depends on relative productivities, relative markups, and the nursing wage, so policies acting on those objects move the price, while policies acting on demand mainly move quantities. Nursing wages rising relative to economy-wide wages would raise the price of formal care and formal care expenditure, while shifting the quantity of care toward informal caregivers. Meanwhile, faster nursing productivity growth lowers that price and lowers informal care relative to formal care. Medical advances that reduce dependency at given survival rates lower the nursing expenditure share and raise informal care's share of elder care. Larger entering cohorts raise output the most among demographic margins, mostly by raising population levels of working-age adults.

Aging and Structural Change: Cost Disease in the Care Sector

Technical Appendix
(for online publication only)

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November 6, 2026

A Data Appendix

A.1 Primary Data Sources and Series Identifiers

Unless otherwise noted, national accounts and fixed-assets series are from the Bureau of Economic Analysis (BEA) and labor-market series are from the Bureau of Labor Statistics (BLS). For transparency and replication, the main series used in Section 2 are:

- **National accounts (BEA NIPA).** GDP: Table 1.5.5 (BEA 2026e). PCE price index (deflator): Table 1.5.4 (BEA 2026e). Expenditure shares and category-level prices: Tables 2.5.5 and 2.5.4 (BEA 2026e). Compensation: Table 1.12 (BEA 2026d). Hours worked: series B4701C0A222NBEA (BEA 2026c).
- **Capital stocks (BEA Fixed Assets).** Current-cost net stocks by industry: Table 3.1ESI (BEA 2026a). Chain-type quantity indexes by industry: Table 3.2ESI (BEA 2026a).

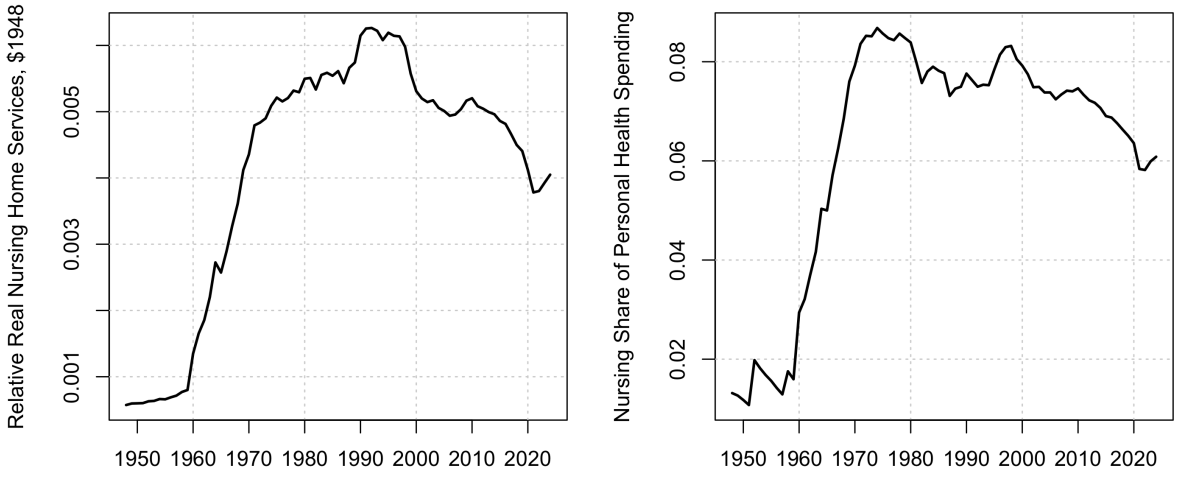
- **Employment and wages (BLS CES).** Nursing and Residential Care Facilities (NAICS 623) employment: CEU6562300001 (BLS 2025a). Total private employment: USPRIV (BLS 2025b). Nursing wages: CEU6562300003 (BLS 2026b).
- **Productivity / labor share.** Economy-wide TFP and labor share: Penn World Tables (Feenstra, Inklaar, and Timmer 2015).

Demographics. Demographic profiles and projections (population by age, survival rates, and the working-age-to-retired ratio) are constructed from the United Nations World Population Prospects (U.N. Dept. of Economic & Social Affairs 2024). Our dependency-rate profiles use dementia prevalence as a proxy for dependency (Langa et al. 2017), and we adjust mortality conditional on dependency using dementia mortality rate ratios from Garre-Olmo et al. (2019).

Micro data. Technical Appendix A.4 uses the American Time Use Survey (ATUS) to measure informal elder-care time inputs (BLS 2026a).

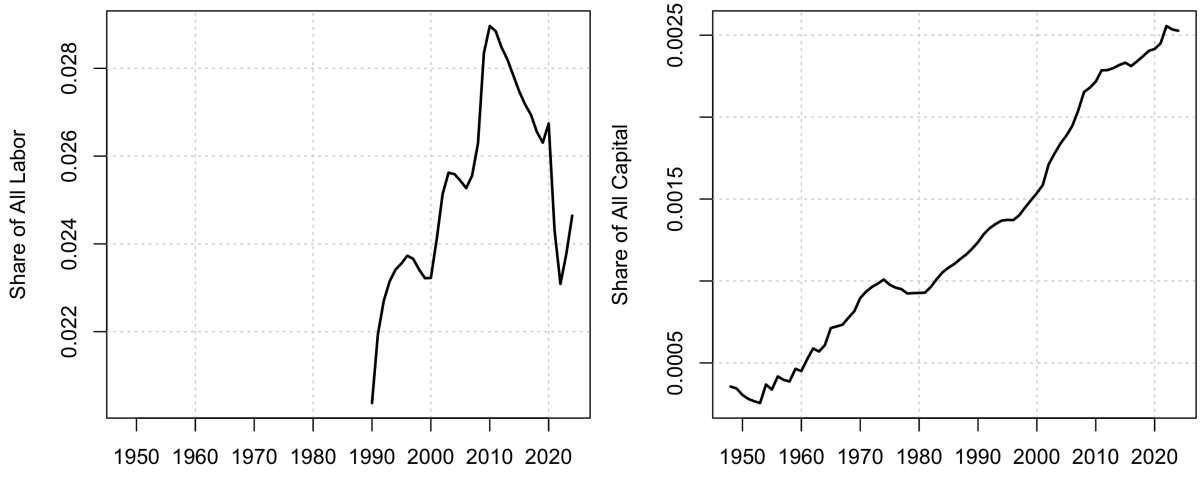
A.2 Sectoral Input/Output Data

This appendix supplements the sectoral aggregate data exploration in Section 2.1. Figure A.1a shows the real quantity of nursing-home consumption relative to all other consumption in \$1948. Figure A.1b shows the nominal share of all health-services expenditure attributable to nursing-home expenditure. Both figures feature data from the BEA (2026e). Figure A.2 supplements the structural change decompositions in Section 2.1 by showing the shares of all labor and capital that go toward nursing-home services. Figure A.3 shows nursing sector versus full economy wages; note that data are available only from 2007 to 2022, though the ratio is fairly constant, oscillating between 0.41 and 0.44. We arrive at the value of the series $\{\omega_t\}_t$ by dividing the points in the blue line of Figure A.3 by the same figure's red line.



(a) Real Quantity Ratio (b) Nursing Share of Health Services

Figure A.1: Panel (a) shows the real quantity ratio in $1948 of nursing-home services consumption to non-nursing-home consumption. Panel (b) shows the share of health-services spending that is driven by nursing-home expenditure.



(a) Nursing Share of Labor (b) Nursing Share of Capital

Figure A.2: Panel (a) shows the share of all private sector labor in the nursing-home sector, where the denominator is "All Employees, Total Private" (BLS 2025b). Panel (b) shows the nominal share of all private capital in the nursing-home sector.

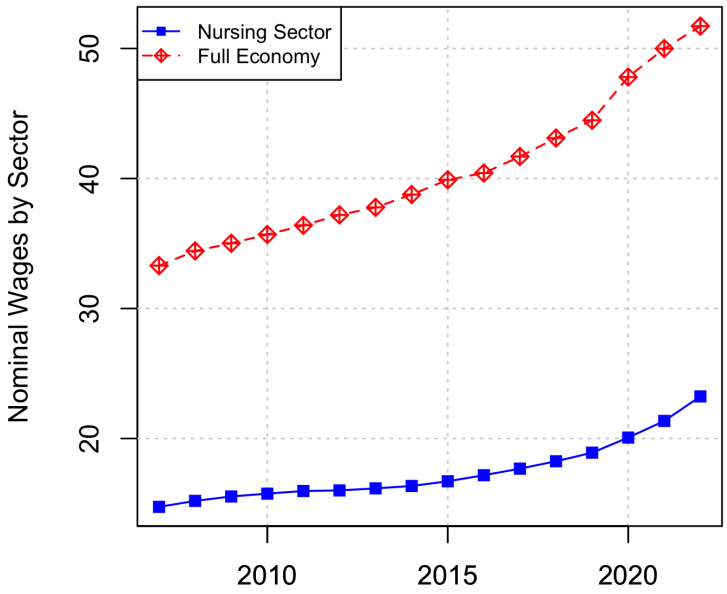


Figure A.3: Nursing-sector wages are computed from BLS (2026b). Full-economy wages are computed from BEA (2026b) and BEA (2026f). Data are available from 2007 to 2022 and the nursing/economy-wide ratio is relatively constant between 0.41 and 0.44.

A.3 Demographic Data

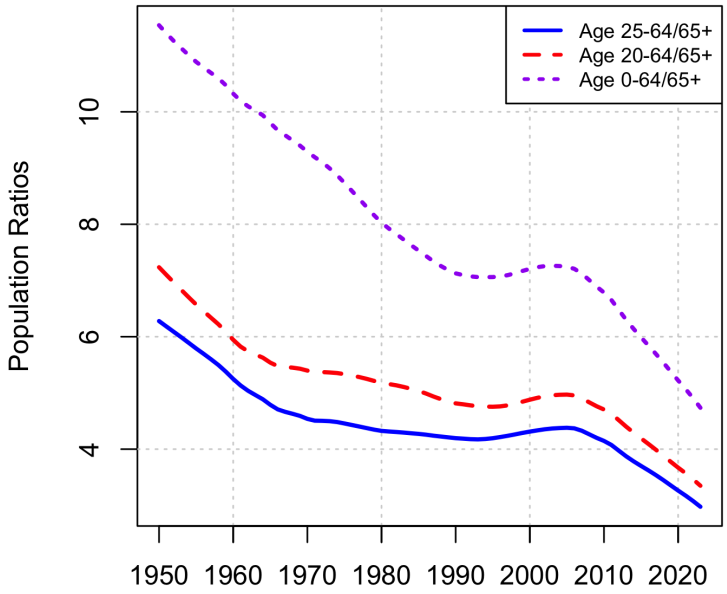
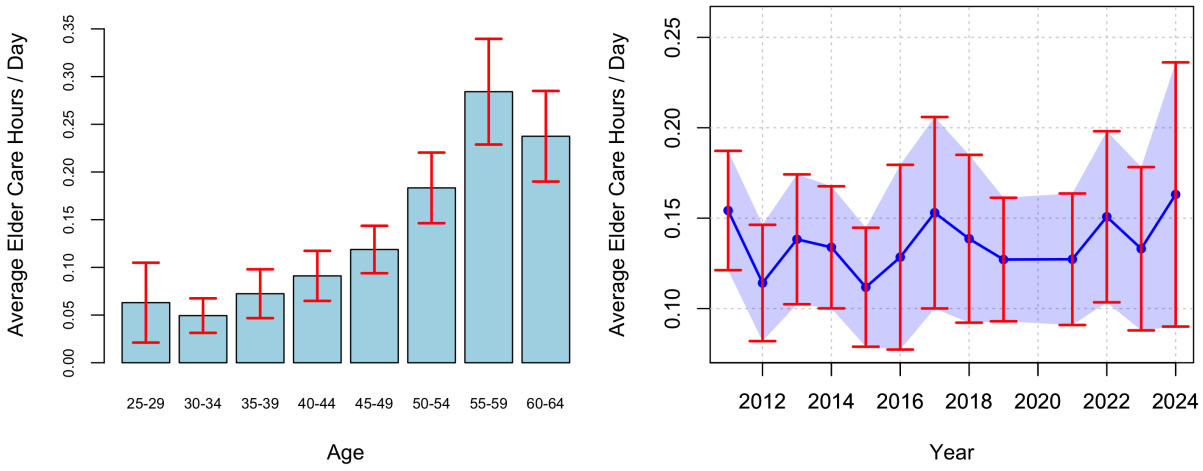


Figure A.4: We plot the time series of U.S. population ratios for years 1950-2023 (U.N. Dept. of Economic & Social Affairs 2024).

A.4 Time Use Micro Data



(a) Care Time Provided by Age

(b) Care Time Provided over Time

Figure A.5: This figure is analogous to Figure 2 in the main text, except we do not condition on those who provide positive care. Rather, this figure presents the average amount of care provided by everyone in the economy, conditional upon age in panel (a) and over time in panel (b).

B Estimation and Calibration Appendix

B.1 Growth Accounting Identification Estimates

B.1.1 Identifying Relative Markup and TFP Growth Rates

Log-differencing the relative price (6) yields the generalized decomposition:

$$\widetilde{g}_{p,t} = \widetilde{g}_{\mu,t} + \widetilde{g}_{Z_c,t} - \widetilde{g}_{Z_n,t} - \Big[\alpha_c \widetilde{g}_{r,t} + (1-\alpha_c)\widetilde{g}_{w,t}\Big] + \Big[\theta_{K_n,t}\widetilde{g}_{r,t} + \theta_{L_n,t}\big(\widetilde{g}_{w,t} + \widetilde{g}_{\omega,t}\big)\Big], \tag{B.1}$$

where $\theta_{K_n,t}$ and $\theta_{L_n,t}$ are the CES within-sector factor cost shares defined in (9), which generally vary with factor prices. These are stated with the production structure in Section 2. The cost shares are what make the CES case differ from Cobb-Douglas here.[54] The key implication of $\epsilon \neq 1$ for the decomposition is that the cost shares $\theta_{K_n,t}$ and $\theta_{L_n,t}$, which weight the pass-through of factor-price changes to relative prices, vary over time, whereas these objects are constant in the Cobb-Douglas case. In the latter case, (B.1) simplifies to

$$\widetilde{g}_{p,t} = \widetilde{g}_{\mu,t} + \widetilde{g}_{Z_c,t} - \widetilde{g}_{Z_n,t} - \Big[\alpha_c \widetilde{g}_{r,t} + (1-\alpha_c)\widetilde{g}_{w,t}\Big] + \Big[\alpha_n \widetilde{g}_{r,t} + (1-\alpha_n)\big(\widetilde{g}_{w,t} + \widetilde{g}_{\omega,t}\big)\Big], \tag{B.2}$$

which is simply (B.1) evaluated at the constant cost shares $\theta_{K_n} = \alpha_n$ and $\theta_{L_n} = 1-\alpha_n$, and collects to the compact form $\widetilde{g}_{p,t} = \widetilde{g}_{\mu,t} + \widetilde{g}_{Z_c,t} - \widetilde{g}_{Z_n,t} + (\alpha_n - \alpha_c)\big(\widetilde{g}_{r,t} - \widetilde{g}_{w,t}\big) + (1-\alpha_n)\widetilde{g}_{\omega,t}$ (i.e. the Pretnar and Feldman (2026) decomposition with an added wedge term in $\widetilde{g}_{\omega,t}$). We state (B.2) in factor prices rather than in nursing factor quantities: the two are equivalent only if the nursing capital–labor first-order condition holds exactly in the data, and writing it in prices keeps the Cobb-Douglas case a literal special case of (B.1). Under CES the GE effect in (B.1) appears as the difference between two unit-cost terms: a consumption-sector component, $-\big[\alpha_c \widetilde{g}_{r,t} + (1-\alpha_c)\widetilde{g}_{w,t}\big]$, with the constant Cobb-Douglas cost shares, and a nursing-sector component, $\theta_{K_n,t}\widetilde{g}_{r,t} + \theta_{L_n,t}(\widetilde{g}_{w,t} + \widetilde{g}_{\omega,t})$, with *endogenous* cost shares. Relative price growth thus reflects the pass-through of common factor price changes (and of the nursing labor wedge) through cost shares that differ across the two sectors. When $\epsilon < 1$ (gross complements), a rise in $r_t/(\omega_t w_t)$ increases $\theta_{K_n,t}$. Conversely, when $\epsilon > 1$, the nursing sector can substitute toward the cheaper input, and the pass-through falls as r_t rises. Full CES thus provides additional flexibility to help explain how the capital and labor shares of the nursing sector have evolved.

For the markup identification step, the Pretnar and Feldman (2026) procedure generalizes naturally. Log-differentiating the nursing expenditure share $\theta_{Nt} = p_t N_t / X_t$,[55] where $X_t = C_t + p_t N_t$,

[54] When $\epsilon = 1$ the cost shares collapse to the constants $\theta_{K_n,t} = \alpha_n$ and $\theta_{L_n,t} = 1-\alpha_n$, and (B.1) collects to

$$\widetilde{g}_{p,t}\big|_{\epsilon=1} = \widetilde{g}_{\mu,t} + \widetilde{g}_{Z_c,t} - \widetilde{g}_{Z_n,t} + (\alpha_n - \alpha_c)\big(\widetilde{g}_{r,t} - \widetilde{g}_{w,t}\big) + (1-\alpha_n)\widetilde{g}_{\omega,t},$$

where the penultimate term captures the GE effect. Applying in addition the Cobb-Douglas nursing first-order condition $\widetilde{g}_{K_n,t} - \widetilde{g}_{L_n,t} = \widetilde{g}_{\omega,t} + \widetilde{g}_{w,t} - \widetilde{g}_{r,t}$ would re-express that term in nursing factor quantities as $(\alpha_n - \alpha_c)(\widetilde{g}_{L_n,t} - \widetilde{g}_{K_n,t})$ with wedge $(1-\alpha_c)\widetilde{g}_{\omega,t}$, which is the form in Pretnar and Feldman (2026). We do not make that substitution, because it holds only insofar as the nursing first-order condition is satisfied in the data.

[55] Expenditure shares are denoted with the sectoral subscript, N and C.

gives $\widetilde{g}_{\theta_N,t} = \widetilde{g}_{p,t} + \widetilde{g}_{N,t} - \widetilde{g}_{X,t}$. Substituting the CES price decomposition (B.1) for $\widetilde{g}_{p,t}$ and the CES production function growth rate $\widetilde{g}_{N,t} = \widetilde{g}_{Z_n,t} + \theta_{K_n,t}\widetilde{g}_{K_n,t} + \theta_{L_n,t}\widetilde{g}_{L_n,t}$, the nursing-sector TFP terms cancel and we obtain:

$$\widetilde{g}_{\theta_N,t} = \widetilde{g}_{\mu,t} + \widetilde{g}_{Z_c,t} - \Big[\alpha_c\widetilde{g}_{r,t} + (1-\alpha_c)\widetilde{g}_{w,t}\Big] + \theta_{K_n,t}(\widetilde{g}_{r,t} + \widetilde{g}_{K_n,t}) + \theta_{L_n,t}(\widetilde{g}_{w,t} + \widetilde{g}_{\omega,t} + \widetilde{g}_{L_n,t}) - \widetilde{g}_{X,t}. \tag{B.3}$$

The cancellation of $\widetilde{g}_{Z_n,t}$ means relative markup growth can be identified from (B.3) without knowledge of nursing-sector TFP, exactly as in the Cobb-Douglas case. Note, however, that (B.3) weights the growth of nursing inputs by the cost shares, while the output elasticities of (2) equal those cost shares only when the nursing capital–labor condition (8) holds. For $\epsilon \neq 1$, (B.3) therefore imposes that condition on the data, ensuring that whatever values for $\widetilde{g}_{\mu,t}$ are obtained from the decomposition are done so conditional upon (8) being satisfied. At $\epsilon = 1$ the output elasticities are α_n and $1-\alpha_n$ identically, and nothing is imposed. When $\epsilon = 1$, the cost shares become constants $\theta_{K_n,t} = \alpha_n$ and $\theta_{L_n,t} = 1-\alpha_n$, the Cobb-Douglas factor demand condition implies $\widetilde{g}_{r,t} + \widetilde{g}_{K_n,t} = \widetilde{g}_{w,t} + \widetilde{g}_{\omega,t} + \widetilde{g}_{L_n,t}$, and (B.3) collapses to:

$$\widetilde{g}_{\theta_N,t} = \widetilde{g}_{\mu,t} + \widetilde{g}_{Z_c,t} + \alpha_c\widetilde{g}_{K_n,t} + (1-\alpha_c)\widetilde{g}_{L_n,t} + (1-\alpha_c)\widetilde{g}_{\omega,t} - \widetilde{g}_{X,t}, \tag{B.4}$$

which is the Pretnar and Feldman (2026) result augmented to allow for growth in the nursing labor wedge ω_t.

Given an assumption that non-nursing-sector TFP grows at a known rate $\widetilde{g}_{Z_c,t}$ (e.g., from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015)), and taking the nursing labor wedge ω_t as identified ex-ante from the observed sectoral wage ratio in the data, relative markup growth $\widetilde{g}_{\mu,t}$ is identified directly from (B.3) under additional assumptions on two of $\theta_{K_n,t}$, α_n, and ϵ, where any two of these also pin down the third via (9). In the quantity-substituted Cobb-Douglas form (B.4), one would not even need a value for α_n to identify the relative markup growth rate, given data. An assumption on α_n is in any case needed to ascertain the growth rate of nursing-sector TFP. This is because, procedurally, we work backwards, first backing out $\widetilde{g}_{\mu,t}$ from (B.3) (with the Cobb-Douglas case as its $\epsilon \to 1$ limit), then using $\widetilde{g}_{\mu,t}$ along with various parametric assumptions on $\theta_{K_n,t}$, α_n, and/or ϵ to back out $\widetilde{g}_{Z_n,t}$ from either (B.1) (with factor shares from (9)) or (B.2), respectively.

B.1.2 Implementation with Data

We quantify relative markup and nursing-sector TFP growth under alternative assumptions on nursing technology. We set the non-nursing Cobb–Douglas capital share to $\alpha_c = 0.33$, motivated by the long-run aggregate labor share once proprietors' income is imputed to labor (Gollin 2002) and by feasibility of implied non-nursing markups.[56] For the nursing sector we assume an average

[56] We discuss the feasibility issues when calibrating the full model in Section 4.3.

capital cost share $\overline{\theta}_{K_n} = 0.26$ (Donahoe 2000) and consider a Cobb–Douglas benchmark and CES specifications that vary (α_n, ϵ).

We measure the real wage as compensation per hour worked and recover the real rental rate from the non-nursing first-order condition, $r_t = \alpha_c(Y_t - p_{nt}N_t)/(K_t - K_{nt})$, with all nominal series deflated by the aggregate PCE price index.[57] The nursing effective wage is $\omega_t w_t$, where ω_t is the nursing/economy wage ratio; in our baseline we fix $\omega_t = \overline{\omega} = 0.4313$ at its 2007–2022 mean (See Figure A.3 in Technical Appendix A.2). We take expenditure shares and price indexes from BEA NIPA tables and split sectoral labor using BLS CES employment series; economy-wide TFP is from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015).

We identify several different relative markup and nursing-sector TFP growth rates conditional on several different sets of assumptions, including seven models with the full CES specification for the nursing sector and one model that assumes the nursing sector is Cobb-Douglas. For the nursing-sector general CES models we pre-impose several values for α_n ex-ante — $\alpha_n \in \{0.05, 0.10, 0.15, 0.20, 0.26, 0.35, 0.50\}$. With the mean of the nursing-sector capital share in hand, r_t computed from the non-nursing-sector first-order condition, real-valued data for w_t (\$2017 = 1) computed from BEA (2026d) and BEA (2026c), along with $\overline{\omega} = 0.4313$ from the observed sectoral wage ratio, we can invert a logged version of the capital factor share expression in (9) to back out estimates for the substitution elasticity, ϵ_t, conditional on the α_n-assumption from the following:

$$\ln\left(\frac{\overline{\theta}_{K_n}}{1-\overline{\theta}_{K_n}}\right) = \epsilon_t \ln\left(\frac{\alpha_n}{1-\alpha_n}\right) + (1-\epsilon_t)\ln\left(\frac{r_t}{\overline{\omega} w_t}\right). \tag{B.5}$$

For each assumed value of α_n, we solve (B.5) for the implied ϵ_t at each date and report the arithmetic mean $\widehat{\epsilon} = T_{obs}^{-1}\sum_t \epsilon_t$, where the average runs over the $T_{obs} = 30$ annual observations from 1990 to 2019 for which all of r_t, w_t, and the sectoral input series are jointly available. For each $(\alpha_n, \widehat{\epsilon})$ pair, we run the full two-stage decomposition by first backing out relative markups $\widetilde{g}_{\mu,t}$ from the expenditure share identity (B.3) (with the Cobb-Douglas case as its $\epsilon \to 1$ limit), then recovering relative TFP residually from the price decomposition in either (B.1) or (B.2).[58] The key advantage of this approach is that every $(\alpha_n, \widehat{\epsilon})$ pair under the CES structure generates a self-consistent set of markups and TFP paths, so that no parameter is assumed in one stage and contradicted in another.

The implied elasticity estimates, which we plot against assumed values of α_n in Figure B.1, decrease monotonically: at $\alpha_n = 0.05$, $\widehat{\epsilon} \approx 2.06$ (gross substitutes); at $\alpha_n = 0.26$, $\widehat{\epsilon} \approx 1.0$ (essentially Cobb-Douglas); and at $\alpha_n = 0.35$, $\widehat{\epsilon} \approx 0.90$ (slight complements). Note that the estimate is invariant to α_c exactly at $\alpha_n = \overline{\theta}_{K_n}$, where (B.5) returns $\widehat{\epsilon} = 1$ for any factor price ratio. Thus, sensitivity to α_c is confined to values of α_n away from 0.26. The Donahoe (2000) benchmark of

[57] Capital stocks are from the BEA Fixed Assets Accounts: we use current-cost net stocks (Table 3.1ESI) for the levels entering r_t and chain-type quantity indexes (Table 3.2ESI) for capital *quantities* in the growth decomposition. Complete series identifiers and additional source details are reported in Technical Appendix A.1.

[58] Note that when running the decomposition on the Cobb-Douglas assumption, we set both $\epsilon = 1$ and $\alpha_n = 0.26$ ex-ante.

$\alpha_n = 0.26$ approximately corresponds to the Cobb-Douglas special case, rendering any *assumed* Cobb-Douglas case, where we ex-ante fix *both* α_n and ϵ, redundant.

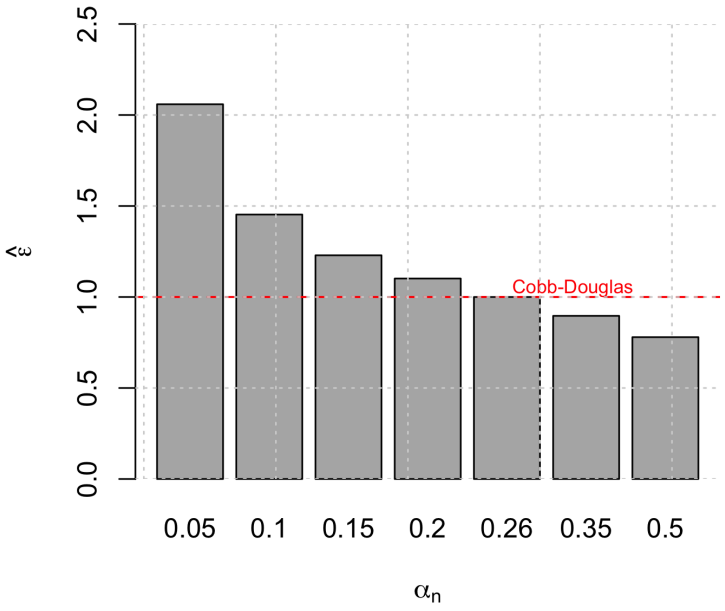
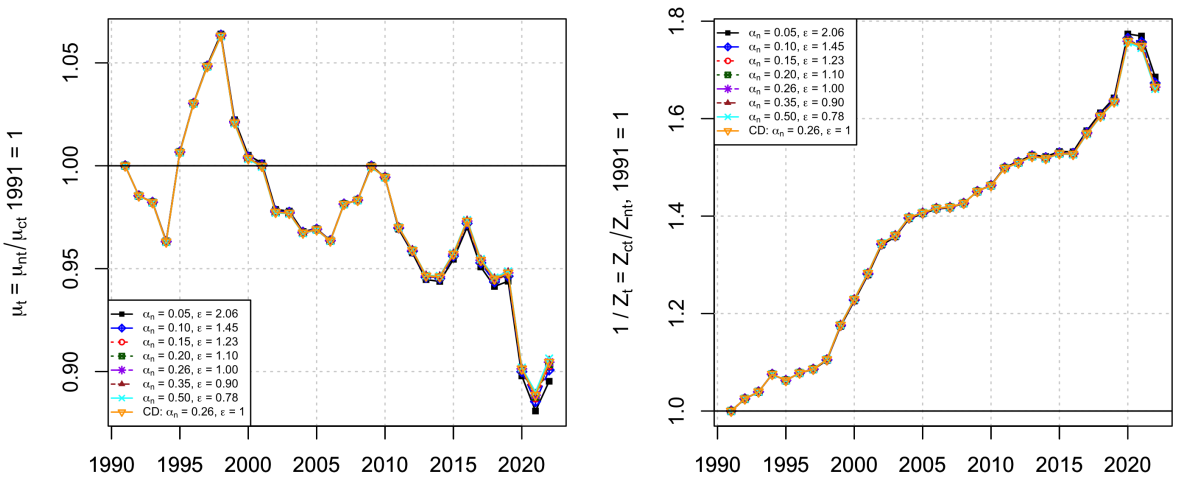


Figure B.1: This is the implied $\widehat{\epsilon}$ estimated under each different assumed α_n from equation (B.5). Each bar is one $(\alpha_n, \widehat{\epsilon})$ pairing consistent with the same average nursing capital cost share $\overline{\theta}_{K_n} = 0.26$, not a competing estimate of ϵ. The locus crosses the Cobb–Douglas line exactly at $\alpha_n = 0.26$. All estimates are computed with $\alpha_c = 0.33$ assumed.



(a) Relative Markups (b) Inverse Relative TFP's

Figure B.2: Panel (a) features relative markups, μ_t, normalized to unity in 1991, under the different specifications of the nursing-sector production function. Panel (b) shows the inversed relative TFP's, $Z_t^{-1} = Z_{ct}/Z_{nt}$, also normalized to unity in 1991.

Figure B.2a shows relative markups backed out from (B.3), with the Cobb-Douglas case as its $\epsilon \to 1$ limit, while Figure B.2b shows the implied inverse relative productivities ($Z_t^{-1} = Z_{ct}/Z_{nt}$) backed out in the second stage from (B.1) and (B.2). Estimated relative markups and relative TFP's vary little across parametric specifications, suggesting their paths are almost entirely driven by the underlying data rather than by the assumed nursing technology. Two features stand out. First, the relative markup μ_t rises through the late 1990s, peaking around 1998, and then declines steadily thereafter, ending some 15% to 16% below that peak and 9.4% to 10.5% below its 1991 level by 2022. Since $\mu_t = \mu_{nt}/\mu_{ct}$, to the extent markups proxy for concentration, this says that market power in the nursing-home sector has not risen as fast as in the economy at large. Second, non-nursing-sector TFP grows some 66.1% to 68.5% more than nursing-sector TFP over the same window. With the markup term working *against* relative price growth and the productivity gap working strongly for it, the decomposition attributes the rise in the relative price of nursing care, and hence in the nursing expenditure share, overwhelmingly to Baumol's cost disease.

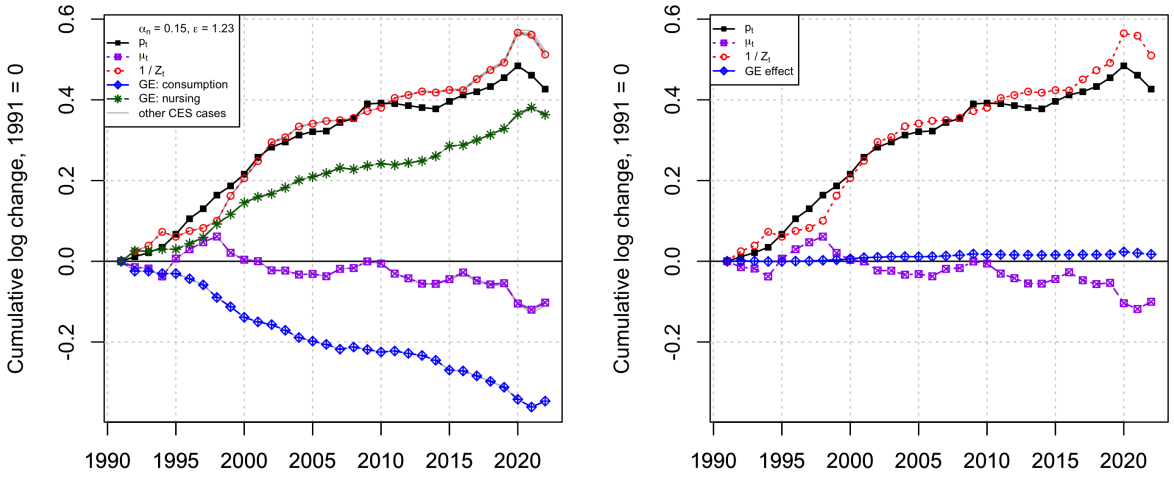


Figure B.3: This figure shows the cumulative log growth decomposition of the nursing-care relative price decomposition from (B.1) in panel (a) and (B.2) in panel (b). Panel (a) layers all of the separate CES compositions under different (α_n, ϵ) pairs in gray but highlights the $\alpha_n = 0.15$, $\epsilon = 1.23$ parameterization in colors. Note that variation across parameter pairs is almost imperceptible.

Figure B.3 shows the log-normalized relative price growth decompositions from (B.1) and (B.2). The full CES decomposition in (B.1) is accomplished in Figure B.3a, while the restricted Cobb-Douglas decomposition from (B.2) is in Figure B.3b. All colored lines sum to the value of the black line (the observed relative price). Purple lines denote the contribution of relative markups, while red lines denote the contribution of cost disease, as summarized by Z_t^{-1}. Green and blue lines in the full CES panel correspond to the two different components of the GE effect — the consumption-sector reallocation component, $-\left[\alpha_c \widetilde{g}_{r,t} + (1-\alpha_c)\widetilde{g}_{w,t}\right]$, is in blue and the nursing-sector component, $\theta_{K_n,t}\widetilde{g}_{r,t} + \theta_{L_n,t}(\widetilde{g}_{w,t} + \widetilde{g}_{\omega,t})$, is in green. In the Cobb-Douglas panel the blue line

is the GE effect. The CES cases in Figure B.3a, shown in gray behind the highlighted one, are nearly indistinguishable from one another, and by construction: since $\widehat{\epsilon}$ is fitted at each α_n so that (B.5) reproduces $\overline{\theta}_{K_n}$, every case implies almost the same average nursing cost share.

Examining Figure B.3, we see that GE effects are generally weak with sector-specific effects canceling each other out in the CES case. Not surprisingly, relative markups again also play little role in driving up prices, with their role generally less significant than GE effects in the Cobb-Douglas case. Differential sectoral TFP growth matters most.

B.2 Full GE Model Calibrated Parameter Values

In this appendix we present profiles of the Section 4 calibrated parameters not fully disclosed in the main text, both exogenously set and generated internally.

B.2.1 Demographic Parameter Values

Population levels M_{jt} are the population estimates and medium-variant projections by single year of age, which we sum into 5-year bins with $j = 1$ comprising ages 20–24 and $j = J$ the 95–99 bin into which we also fold the open-ended 100 and above group.[59] The population value for model-age j in period t is the bin total in the first year of the 5-year period.[60] We normalize the entire panel by the 40–44 bin in 1990.[61] Survival rates come from the life tables of U.N. Dept. of Economic & Social Affairs (2024), computed within a single year's table. As measured, the survival rate is a period-specific rather than a cohort-specific measure: it holds year-t mortality fixed across the five years the cohort actually lives through. Because M_{jt} and s_{jt} are drawn from different U.N. products, they do not satisfy $M_{j+1,t+1} = s_{jt} M_{jt}$ identically. The residual is net migration together with the cohort-versus-period timing noted above, and it runs at 0.9% to 2.5% of the receiving cohort per 5-year period at working ages, averaging 1.3%, while at retirement ages it averages 0.1% — the age pattern one would expect of U.S. net immigration. Figure B.4 plots M_{jt} and s_{jt}.

For $j \geq J_R$ we also must correct the survival rate for dependency type. Let λ_j be the factor by which a dependent's mortality hazard exceeds an independent's at the same age. We take data for λ_j from the matched cohort study of Garre-Olmo et al. (2019), averaging their male and female dementia mortality rate ratios; the resulting values fall from 2.75 at ages 65–74 to 1.15 at 95–99. Treating the binned rate as the independent survival probability, $s^i_{jt} = s_{jt}$, the dependent rate follows from proportional hazards, $s^d_{jt} = \left(s^i_{jt}\right)^{\lambda_j}$. We set $\lambda_j = 1$ for $j < J_R$, so the two types coincide before retirement. Figure B.7 plots λ_j.

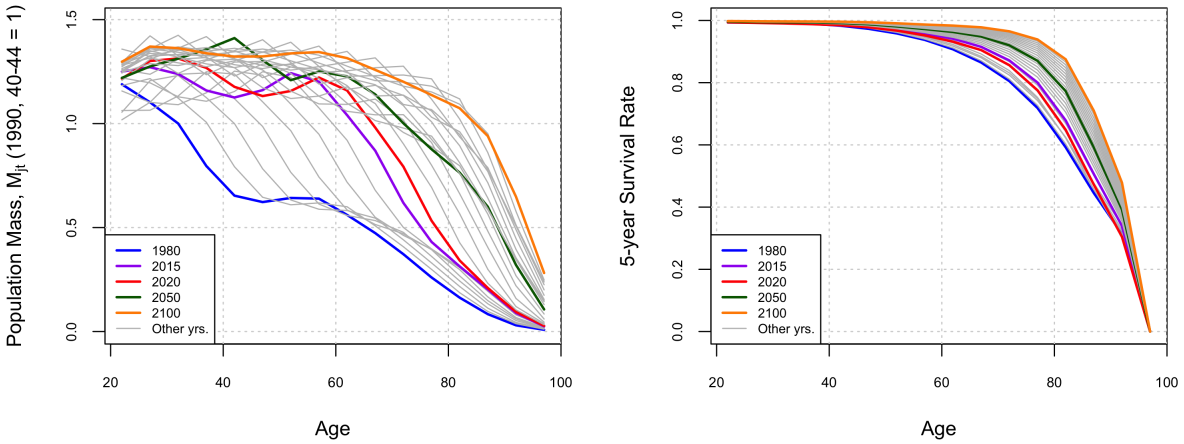
We use the onset of dementia as a proxy for the onset of dependency, admitting that this is likely a lower bound on the dependency rate. For the age profile we anchor on Corrada et al. (2010), who report 2005 annual dementia incidence rates of 12.7% at ages 90–94, 21.2% at 95–99, and 40.7% at 100 and above. These three rates imply a doubling time of 5.95 years. For the time profile we use Langa et al. (2017), who find that dementia prevalence fell from 11.6% of adults 65 and older in 2000 to 8.8% in 2012, a geometrically averaged annual decline of 2.276% per year; we assume that incidence declined at the same rate at every age and apply that decline to reach every other period. The resulting incidence profiles, ϖ_{jt}, are plotted in Figure 3a of the main text, and

[59] We ignore agents $<$ age 20 in our economic calculations.

[60] For example, to get M_{jt} for $j = 1$ in the period associated with the 5-year interval starting in 1990 and going through 1994, we add up the population of all 20-24 year-olds in 1990 and set this value for the population of age $j = 1$ agents in period t associated with the year 1990, so that the population is always that at the *beginning* of the 5-year period of interest.

[61] That normalization is immaterial to anything we report: every aggregate in the model is homogeneous of degree one in population, and every targeted moment is either a ratio or a per-capita quantity, so only the *shape* of the age profile and its evolution carry information.

the implied dependent shares, ψ_{jt}, in Figure B.5.



(a) $\{M_{jt}\}_{j,t}$ (b) $\{s_{jt}\}_{j,t}$

Figure B.4: Panel (a) shows population mass levels by age normalized to unity at the 40-44 cohort in 1990 (U.N. Dept. of Economic & Social Affairs 2024). Panel (b) shows 5-year survival rate profiles computed within each year's U.N. Dept. of Economic & Social Affairs (2024) life table.

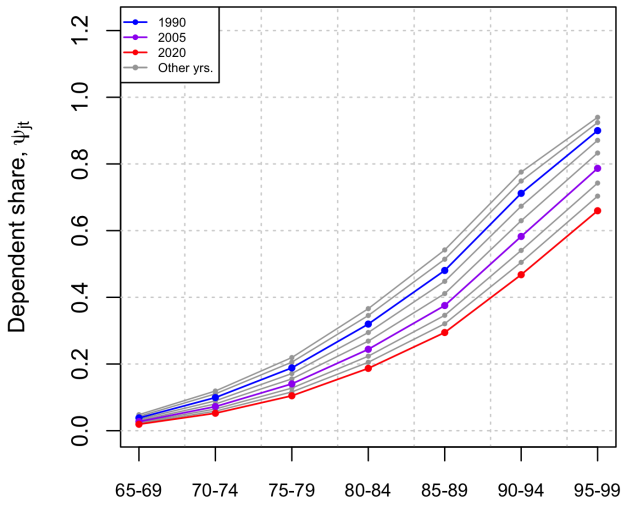


Figure B.5: This figure shows the dependent shares ψ_{jt} implied through (12) by the incidence profiles in Figure 3a.

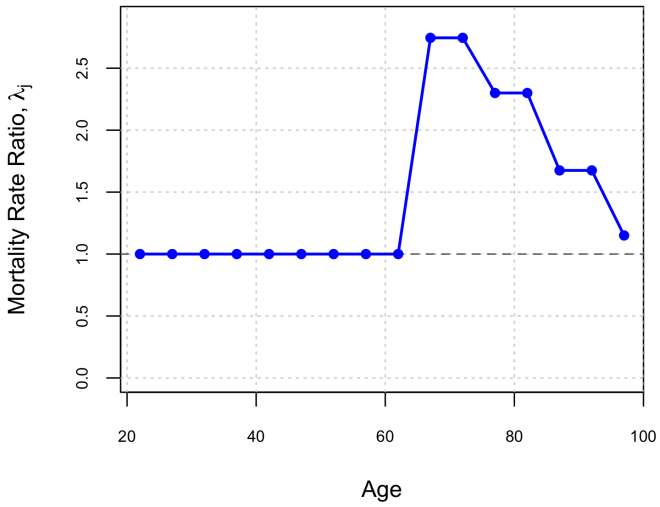
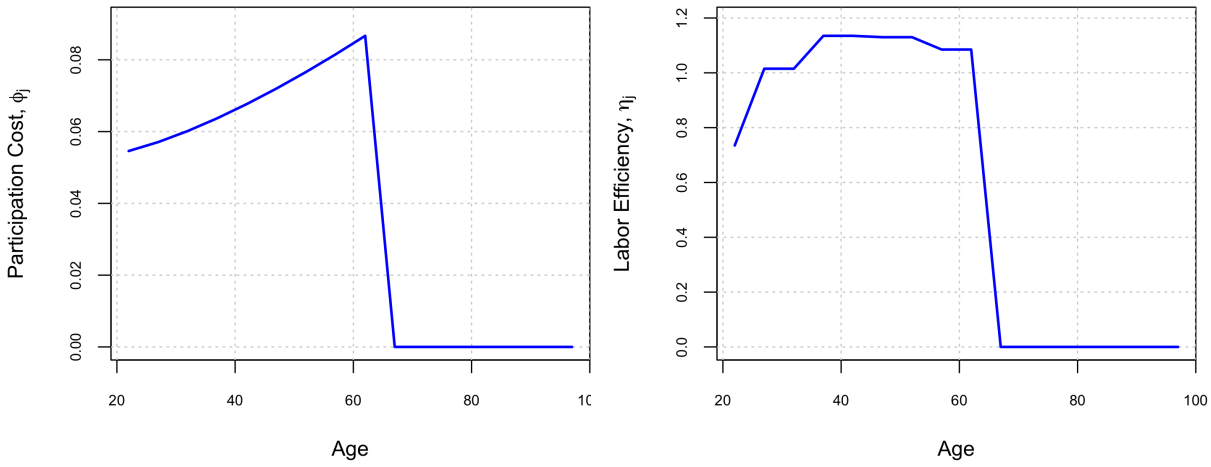


Figure B.6: $\{\lambda_j\}_j$

Figure B.7: This figure shows dementia mortality rate ratios λ_j, averaged over the male and female estimates of Garre-Olmo et al. (2019).

B.2.2 Time-use Parameter Values



(a) $\{\phi_j\}_j$ (b) $\{\eta_j\}_j$

Figure B.8: Panel (a) uses values from Kitao (2014) and Cooley and Henriksen (2018) to plot $\phi_j = \kappa_1 + \kappa_2 j^{\kappa_3}$. Panel (b) shows the values of η_j from Hansen (1993).

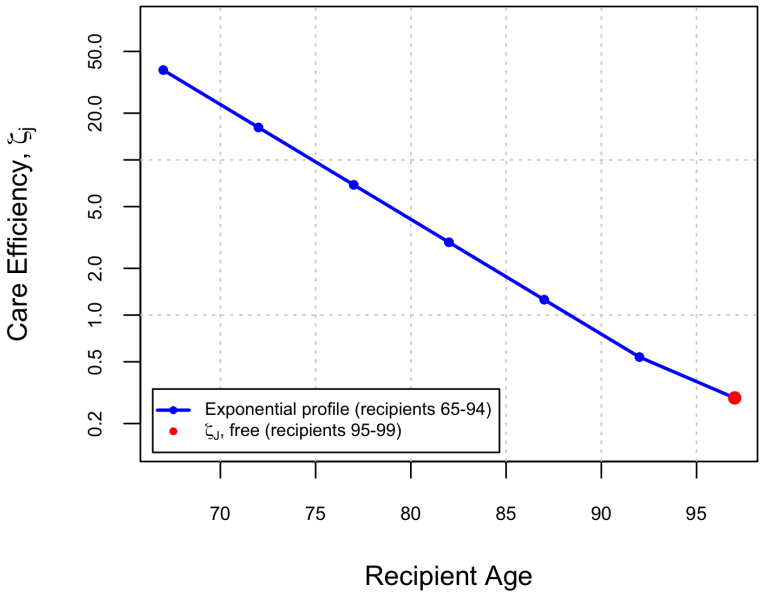


Figure B.9: This is the internally calibrated ζ_j profile. Recall that ζ_j represents the age-specific efficiency of care received. The vertical axis is logarithmic: the profile spans about five orders of magnitude, so on a linear axis every bin but the youngest would be indistinguishable from zero. Blue points are the exponential form (42) over recipient ages 65–94; the red point at 95–99 is the free terminal value ζ_J. Only dependent ages are shown, since $\zeta_j = 1$ below J_R is a normalization.

B.2.3 Production Parameter Values

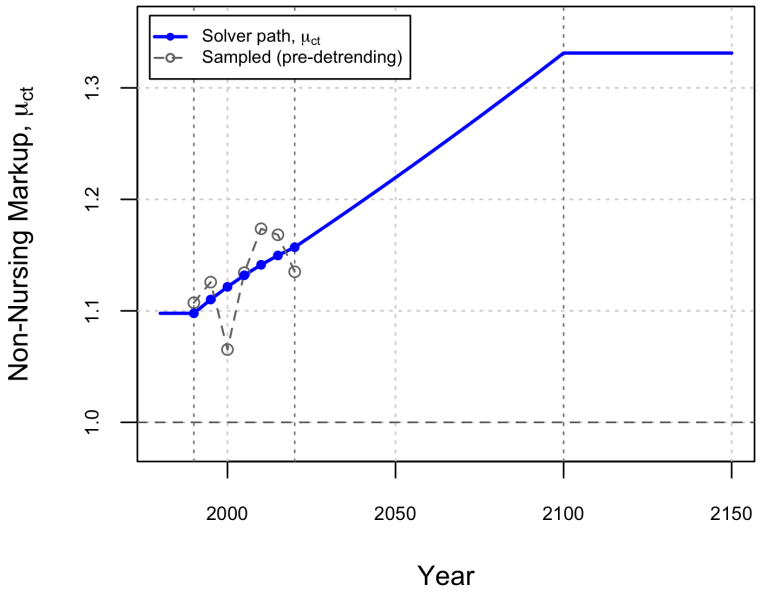


Figure B.10: This figure shows levels of μ_{ct} as estimated given aggregate labor share data from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015) and an assumed value of $\alpha_c = 0.33$. The gray circles and dashed line plot (43) evaluated at the seven sample points, before detrending; the dip at 2000 is the dot-com peak in the aggregate labor share, which the small markup excess amplifies. The blue line is the path the solver actually uses over the model's full horizon: the least-squares quadratic trend through those points over 1990–2020, backfilled at its 1990 value for the pre-sample periods, extrapolated beyond 2020 at the constant net rate $g_{\mu_c} = 0.0088$ implied by that trend, and held flat from 2100 to the terminal steady state.

B.2.4 Government Parameter Values

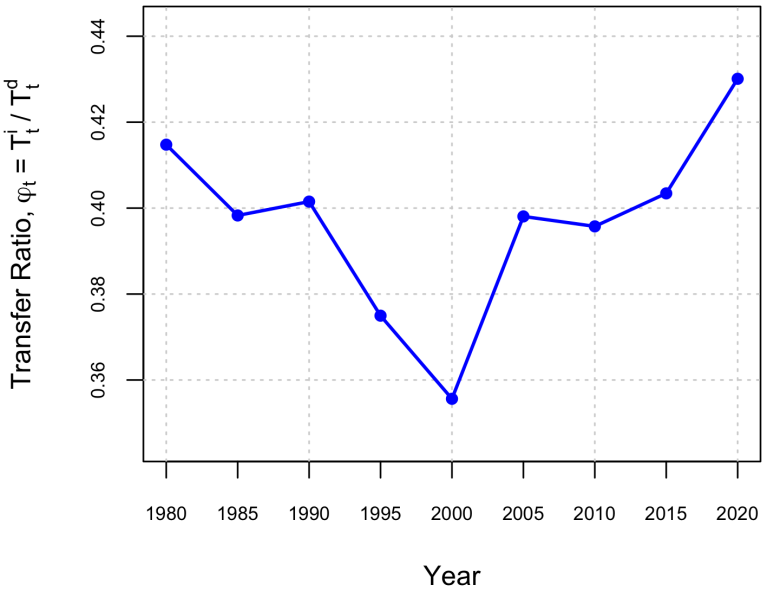


Figure B.11: The transfer ratio $\varphi_t = T_t^i / T_t^d$ over the model's nine five-year periods. T_t^i is the annualized average Social Security retired-worker benefit (Social Security Administration 2024) plus Medicare outlays per enrollee (Medicare Payment Advisory Commission 2024; Centers for Medicare & Medicaid Services 2024); T_t^d adds Medicaid nursing-home spending by source of funds (Centers for Medicare & Medicaid Services 2024) divided by the resident count (National Center for Health Statistics 2004; Harris-Kojetin et al. 2019).

C Aggregation and Equilibrium

Assume all household and firm decision variables satisfy natural interiority conditions. Capital and labor markets clear:

$$K_t = K_{ct} + K_{nt} = \sum_{j=1}^{J_R-1} M_{jt}(a_{jt} + b_t) + \sum_{j=J_R}^{J} \left[M^d_{jt}(\bar{a}^d_{jt} + b_t) + M^i_{jt}(a^i_{jt} + b_t) \right] \tag{C.1}$$

$$L_t = L_{ct} + L_{nt} = \sum_{j=1}^{J_R-1} M_{jt}\eta_j(1 - \phi_j - \ell_{jt} - q^s_{jt}). \tag{C.2}$$

Because capital and labor are mobile across sectors at the common rental rate r_t, with nursing firms paying $\omega_t w_t$ per unit of labor, cost minimization by nursing-sector firms yields factor demands

$$K_{nt} = \frac{\theta_{K_n,t} MC_{nt} N_t}{r_t}, \qquad L_{nt} = \frac{\theta_{L_n,t} MC_{nt} N_t}{\omega_t w_t}, \tag{C.3}$$

where MC_{nt} is the CES unit cost of nursing output defined beneath (5) and both $\theta_{K_n,t}$ and $\theta_{L_n,t} = 1 - \theta_{K_n,t}$ are the within-sector factor cost shares from (9). Total nursing-sector cost is $MC_{nt}N_t$, which the cost shares split between the two factors; dividing each factor's cost bill by its price recovers the quantity demanded. Note that labor cost is divided by $\omega_t w_t$, the nursing wage per efficiency unit of household labor. When $\epsilon = 1$ the shares collapse to the constants $\theta_{K_n,t} = \alpha_n$ and $\theta_{L_n,t} = 1 - \alpha_n$, recovering the Cobb-Douglas factor demands. The consumption sector absorbs the residual: $K_{ct} = K_t - K_{nt}$ and $L_{ct} = L_t - L_{nt}$. The nursing sector's shares of aggregate capital and labor are then

$$\theta^K_{nt} \equiv \frac{K_{nt}}{K_t} = \frac{\theta_{K_n,t} MC_{nt} N_t}{r_t K_t}, \qquad \theta^L_{nt} \equiv \frac{L_{nt}}{L_t} = \frac{\theta_{L_n,t} MC_{nt} N_t}{\omega_t w_t L_t}, \tag{C.4}$$

with the non-nursing sector taking the remainder, $\theta^K_{ct} = 1 - \theta^K_{nt}$ and $\theta^L_{ct} = 1 - \theta^L_{nt}$.

Consumption, investment, and formal nursing care services aggregate:

$$C_t = \sum_{j=1}^{J_R-1} M_{jt} c_{jt} + \sum_{j=J_R}^{J} \left[M^d_{jt} c^d_{jt} + M^i_{jt} c^i_{jt} \right] \tag{C.5}$$

$$I_t = \sum_{j=1}^{J_R-1} M_{jt} \iota_{jt} + \sum_{j=J_R}^{J} \left[M^d_{jt} \iota^d_{jt} + M^i_{jt} \iota^i_{jt} \right] \tag{C.6}$$

$$N_t = \sum_{j=J_R}^{J} M^d_{jt} n_{jt}. \tag{C.7}$$

The nursing sector's share of total consumption expenditure is

$$\theta_{Nt} = \frac{p_{nt} N_t}{p_{ct} C_t + p_{nt} N_t}. \tag{C.8}$$

Informal care aggregates by age j and $j + \bar{j}$ pair according to the clearing condition (30), with total informal care time

$$Q_t = \sum_{j=1}^{J_R-1} M_{jt} q^s_{jt} = \sum_{j=J_R}^{J} M^d_{jt} q_{jt}. \tag{C.9}$$

Total bequests made satisfy total bequests received. An agent of age j in $t-1$ who dies before t leaves the assets it chose in $t-1$, $a_{j+1,t}$, and agents of age J leave nothing:

$$\sum_{j=1}^{J_R-1} M_{j,t-1}(1-s_{j,t-1})a_{j+1,t} + \sum_{j=J_R}^{J-1}\left[M^d_{j,t-1}(1-s^d_{j,t-1})a^d_{j+1,t} + M^i_{j,t-1}(1-s^i_{j,t-1})a^i_{j+1,t}\right] = b_t M_t, \tag{C.10}$$

where $M_t = \sum_{j=1}^{J} M_{jt}$ and $a^d_{j+1,t}$ is the dying dependent's own saving, before pooling in (21). This identity, (10), and (21) imply that K_t in (C.1) equals aggregate saving in $t-1$, $\sum_j M_{j,t-1} a_{j+1,t}$, consistent with the law of motion for capital below. Profits satisfy

$$\Pi_t = \Pi_{ct} + \Pi_{nt} = \sum_{j=1}^{J_R-1} M_{jt}\pi_{jt} + \sum_{j=J_R}^{J}\left[M^d_{jt}\pi^d_{jt} + M^i_{jt}\pi^i_{jt}\right]. \tag{C.11}$$

Transfers satisfy

$$\tau_t \sum_{j=1}^{J_R-1} M_{jt}\overline{w}_t \eta_j (1-\phi_j - \ell_{jt} - q^s_{jt}) = \sum_{j=J_R}^{J}\left[M^d_{jt}T^d_t + M^i_{jt}T^i_t\right], \tag{C.12}$$

where we assume a fixed rule such that T^i_t is some fraction φ_t of the average dependent's transfers received, $T^i_t = \varphi_t T^d_t$. Like τ_t, the policy φ_t is assumed known to all agents.

C.1 Equilibrium

Given the set of sequences $\left\{\tau_t, \varphi_t, Z_{ct}, Z_{nt}, \mu_{ct}, \mu_{nt}, \omega_t, \{s_{jt}, s^i_{jt}, s^d_{jt}, \varpi_{jt}\}_j\right\}_t$, a two-sector monopolistically competitive equilibrium with transfers and reverse bequests consists of

i. Sequences of household policies for workers and retirees of both types, $\left\{c_{jt}, \ell_{jt}, q^s_{jt}, a_{j+1,t+1}\right\}_{j<J_R}$, $\left\{c^d_{jt}, n_{jt}, a^d_{j+1,t+1}\right\}_{j\geq J_R}$, and $\left\{c^i_{jt}, a^i_{j+1,t+1}\right\}_{j\geq J_R}$;

ii. Sequences of producer policies, $\left\{K_{ct}, L_{ct}, K_{nt}, L_{nt}, C_t, I_t, N_t\right\}_t$;

iii. Sequences of population distributions, $\left\{M_{jt}, M^d_{jt}, M^i_{jt}, \psi_{jt}\right\}_{j,t}$, and of dependent-pool asset positions, $\left\{\overline{a}^d_{jt}\right\}_{j,t}$;

iv. Sequences of prices, $\left\{p_{ct}, p_{nt}, p_{It}, w_t, \overline{w}_t, r_t\right\}_t$;

v. Sequences of bequests, $\left\{b_t\right\}_t$;

vi. Sequences of dividends, $\{\pi_{jt}, \pi^d_{jt}, \pi^i_{jt}\}_{j,t}$;

vii. Sequences of transfers, $\{T^d_t, T^i_t\}_t$;

such that

a. Taking prices, transfers, bequests, and dividends as given, working-age policies solve (25) (for $j < J_R - 1$) and (26) (for $j = J_R - 1$) subject to (19); dependent retirees' policies solve (23) subject to (20); and independent retirees' policies solve (24) subject to (22). Working-age agents take the informal care they supply as valued through the altruism term $\underline{\beta}\, u^d_{j+\bar{j}}$, taking the dependent's consumption and formal care as given, while dependent retirees take received care q_{jt} as given;

b. Firms in each sector minimize cost taking r_t and w_t as given. The nursing sector prices at a markup over marginal cost, $p_{nt} = \mu_{nt} MC_{nt}$, with factor demands given by (C.3). The non-nursing sector produces consumption and investment from the same technology and prices them differently: consumption sells at a markup, $p_{ct} = \mu_{ct} MC_{ct}$, while investment sells at marginal cost, $p_{It} = MC_{ct} = 1$, which is the numeraire. That sector absorbs residual factors, $K_{ct} = K_t - K_{nt}$ and $L_{ct} = L_t - L_{nt}$;

c. The capital market clears, $K_t = K_{ct} + K_{nt}$, with K_t given by (C.1);

d. The labor market clears, $L_t = L_{ct} + L_{nt}$, with L_t given by (C.2), and the wage households receive is the labor-share-weighted average $\overline{w}_t = w_t(L_{ct} + \omega_t L_{nt})/L_t$, so that aggregate labor income equals firms' total labor cost;

e. The market for formal nursing care clears, $N_t = \sum_{j=J_R}^{J} M^d_{jt} n_{jt}$, as in (C.7);

f. The market for non-nursing output clears, $C_t + I_t = Z_{ct} K_{ct}^{\alpha_c} L_{ct}^{1-\alpha_c}$, with C_t and I_t given by (C.5) and (C.6);

g. The informal care market clears age-pair by age-pair, $M_{jt} q^s_{jt} = M^d_{j+\bar{j},t} q_{j+\bar{j},t}$, as in (30);

h. Aggregate bequests left by those who die between $t-1$ and t equal aggregate bequests received, and are distributed equally across all living agents;

i. Firm profits are fully distributed as dividends, $\Pi_t = \Pi_{ct} + \Pi_{nt}$, in proportion to asset holdings, $\pi_{jt} = (a_{jt} + b_t)\Pi_t / K_t$;

j. The government budget balances period by period, as in (C.12), with $T^i_t = \varphi_t T^d_t$;

k. The population distribution evolves according to the survival probabilities s_{jt}, s^i_{jt}, s^d_{jt} and the dependency incidence ϖ_{jt}, as in (11) up to the migration wedge described in Technical Appendix B.2.1, with dependent shares ψ_{jt} following (12);

l. The aggregate capital stock evolves as $K_{t+1} = (1-\delta)K_t + I_t + \Xi_t$, where Ξ_t is the capital brought in by net migration. Migrants arrive with the average asset position of their age bin (Technical Appendix B.2.1), so

$$\Xi_t = \sum_{j=1}^{J-1} \left(\frac{M_{j+1,t+1}}{s^i_{jt} M^i_{jt} + s^d_{jt} M^d_{jt}} - 1 \right) \left(s^i_{jt} M^i_{jt}\, a^i_{j+1,t+1} + s^d_{jt} M^d_{jt}\, a^d_{j+1,t+1} \right),$$

with $M^i_{jt} = M_{jt}$, $M^d_{jt} = 0$, $s^i_{jt} = s_{jt}$, and $a^i_{j+1,t+1} = a_{j+1,t+1}$ for $j < J_R$. With no migration, $M_{j+1,t+1} = s^i_{jt} M^i_{jt} + s^d_{jt} M^d_{jt}$ by (11), so $\Xi_t = 0$;

m. The dependent pool's asset position evolves according to (21).

D Additional Quantitative Results

D.1 Marginal Benefits and Marginal Costs of Informal Care

In Section 5.1 we computed the aggregate average costs of informal care to arrive at adjusted measures of care's share of expenditure and GDP. We now set out a way to quantify the marginal benefits and marginal costs of informal elder care. Elder dependents benefit from receiving informal elder care in that it raises their nursing care consumption *and*, in the sense that informal care is only weakly complementary to formal care, frees up available liquidity for other purposes. On the margin we can value the benefits an elderly dependent receives from the informal care provision in terms of the value of the formal nursing consumption they would be willing to give up to receive one additional unit of informal care. Providers, meanwhile, give up after-tax labor income they could otherwise be earning. Both measures are age-specific, leading to an age-specific benefit-cost ratio profile.

Note that a dependent does not buy informal care, but they do buy formal care at p_{nt}, and they combine the two in the care composite $\mathcal{G}$ of (13). At their optimal mix one more hour of informal care is worth as much to them as $(\partial\mathcal{G}/\partial q_{jt})/(\partial\mathcal{G}/\partial n_{jt})$ units of formal care. Valuing those units at p_{nt} gives the most the dependent would give up for the hour of informal care (i.e., their willingness to pay, WTP_{jt}):

$$WTP_{jt} = p_{nt}\,\frac{\partial\mathcal{G}/\partial q_{jt}}{\partial\mathcal{G}/\partial n_{jt}} = p_{nt}\,\frac{(1-\nu_n)\zeta_j^{\rho_n}}{\nu_n}\left(\frac{q_{jt}}{n_{jt}}\right)^{\rho_n-1}. \tag{D.1}$$

Clearly, WTP_{jt} differs by recipient age. Meanwhile, the after-tax marginal cost of an hour of informal care supplied to a dependent of age j by a provider of age $j-\bar{j}$ is $\overline{w}_t\,\eta_{j-\bar{j}}\,(1-\tau_t)$. Note that this is a Becker (1965) *oc* valuation taken at the level of the individual rather than the aggregate of (45) in Section 5.1, the Becker (1965) formulation being the natural one for this model given the linearity of the provider's budget constraint, (19), in $q^s_{j-\bar{j},t}$. Both the *oc* value of care provided and (D.1) measure within-period tradeoffs.

Define the within-period benefit-cost ratio of an hour of informal care received at age j as

$$BC_{jt} = \frac{WTP_{jt}}{\overline{w}_t\,\eta_{j-\bar{j}}\,(1-\tau_t)}, \qquad j = J_R,\ldots,J. \tag{D.2}$$

Figure D.1 plots this profile. In our sample benefits to dependents have never exceeded 30% of costs to providers. Notice that costs relative to benefits have gone up over time, shifting BC_{jt} down at all ages from 1990 to 2020. Much of this level shift is attributable to informal caregivers per dependent, $M_{j-\bar{j},t}/M^d_{jt}$, rising (Figure 3b in Section 4.1), thus pushing down what the dependent is willing to pay for one more hour of informal care. Note that WTP_{jt} is $p_{nt}(q_{jt}/n_{jt})^{\rho_n-1}$ up to constants, and $\rho_n - 1 < 0$, so the rising informal-to-formal ratio lowers the value of an extra hour from the dependent's perspective. That roughly offsets the rise in the nursing price, so the benefit

stays flat while the cost (the provider's after-tax wage) grows with the economy. Equivalently, as providers per dependent rise, each provider sees their hour as doing more good and supplies care further down the dependent's marginal-value curve (see (29)). The cost side is also conservative. The provider's after-tax wage is a static measure of what an hour of care costs, and Skira (2015) estimates, in a dynamic model of caregiving and work with human-capital depreciation and job-offer frictions, a welfare cost of caregiving well above forgone wages. The provider's cost is therefore if anything understated, and BC_{jt} overstated.

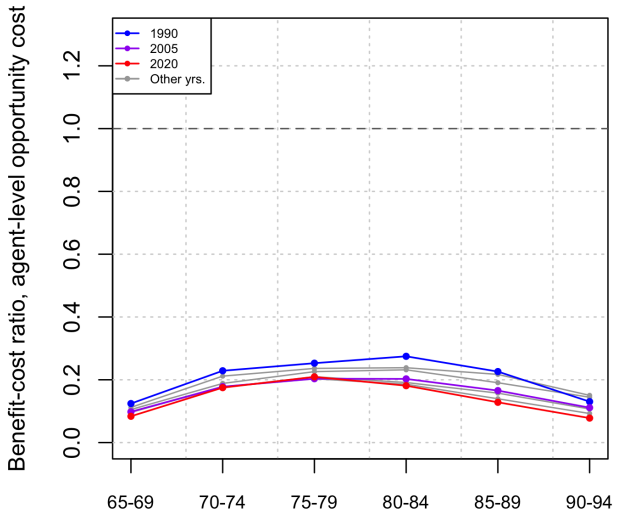


Figure D.1: This figure shows the benefit-cost ratio, BC_{jt}, profiles from (D.2) for all years in our targeted transition path, with 1990, 2005, and 2020 highlighted.

D.2 Structural Change Counterfactuals

Tables D.1 and D.2 report counterfactuals where we turn off each variable one-at-a-time to assess what is driving the broader outcomes in each constellation discussed in Section 5.2. The series held fixed in each counterfactual are named in the columns of each table; all other series are assumed to grow. The values in the tables represent either percent deviations relative to the 1990 base year (p_{nt}/p_{ct}), shares in levels, not deviations (θ_{Nt}, K_{nt}/K_t, L_{nt}/L_t), or percent deviations relative to the *simulated* baseline (Y_t, Y_t/M_t). The "all" counterfactuals are deviations or share levels when all of the (e.g.) demographic parameters are turned off in Table D.1 or all of the technology and markup series are turned off in Table D.2. The series-by-series breakdowns are discussed as supporting results for the broader "all" series presented as counterfactuals (1) and (2) in Section 5.2. Technical Appendix D.2.1 shows how the two constellations shaped informal care.

Table D.1: Demographic Counterfactuals

Moment	Year	Base	all	ϖ_{jt}	M_{jt}, s_{jt}
p_{nt}/p_{ct}	2000	1.246	-1.1%	0.1%	-1.1%
	2010	1.495	-2.2%	0.1%	-2.4%
	2020	1.712	-3.2%	0.2%	-3.4%
θ_{Nt}	2000	0.015	0.015	0.017	0.014
	2010	0.013	0.016	0.018	0.012
	2020	0.012	0.017	0.019	0.010
K_{nt}/K_t	2000	0.002	0.002	0.002	0.002
	2010	0.002	0.002	0.003	0.002
	2020	0.002	0.003	0.004	0.002
L_{nt}/L_t	2000	0.026	0.027	0.030	0.025
	2010	0.027	0.033	0.036	0.024
	2020	0.027	0.039	0.043	0.023
Y_t	2000	—	-11.8%	-0.1%	-11.7%
	2010	—	-22.6%	-0.4%	-22.3%
	2020	—	-29.7%	-1.0%	-29.2%
Y_t/M_t	2000	—	-1.9%	-0.1%	-1.8%
	2010	—	-2.7%	-0.4%	-2.4%
	2020	—	-1.2%	-1.0%	-0.4%

The series held fixed in each counterfactual name its column, and every such series is held at its 1990 value from 1990 onward. Share moments are reported as levels; all other entries are percent deviations from baseline.

Table D.2: Technology and Markup Counterfactuals

Moment	Year	Base	all	Z_{ct}	μ_{ct}	Z_{nt}	μ_{nt}
p_{nt}/p_{ct}	2000	1.246	−17.3%	−6.1%	2.6%	−23.3%	12.0%
	2010	1.495	−29.8%	−13.4%	4.5%	−38.2%	25.4%
	2020	1.712	−37.9%	−20.4%	6.1%	−47.7%	40.5%
θ_{Nt}	2000	0.015	0.014	0.014	0.015	0.013	0.016
	2010	0.013	0.011	0.012	0.014	0.010	0.015
	2020	0.012	0.009	0.010	0.012	0.008	0.014
K_{nt}/K_t	2000	0.002	0.001	0.002	0.002	0.001	0.002
	2010	0.002	0.001	0.002	0.002	0.002	0.002
	2020	0.002	0.001	0.002	0.002	0.002	0.002
L_{nt}/L_t	2000	0.026	0.021	0.026	0.027	0.023	0.025
	2010	0.027	0.018	0.025	0.027	0.021	0.024
	2020	0.027	0.015	0.024	0.026	0.019	0.022
Y_t	2000	—	−5.6%	−5.7%	−0.1%	0.2%	0.0%
	2010	—	−13.4%	−13.1%	−0.8%	0.4%	0.1%
	2020	—	−21.0%	−20.5%	−1.3%	0.5%	0.2%
Y_t/M_t	2000	—	−5.6%	−5.7%	−0.1%	0.2%	0.0%
	2010	—	−13.4%	−13.1%	−0.8%	0.4%	0.1%
	2020	—	−21.0%	−20.5%	−1.3%	0.5%	0.2%

The series held fixed in each counterfactual name its column, and every such series is held at its 1990 value from 1990 onward. Share moments are reported as levels; all other entries are percent deviations from baseline.

D.2.1 Informal Care under Counterfactuals (1) and (2)

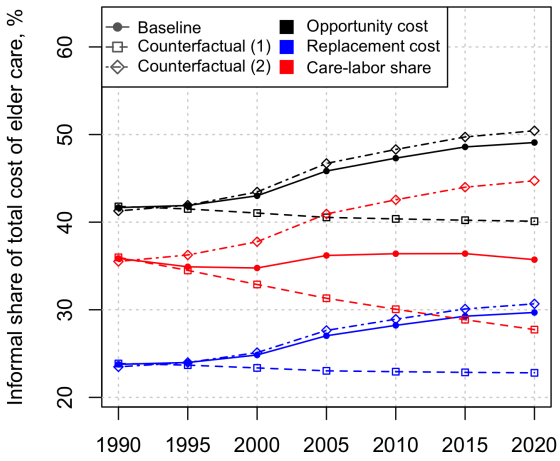
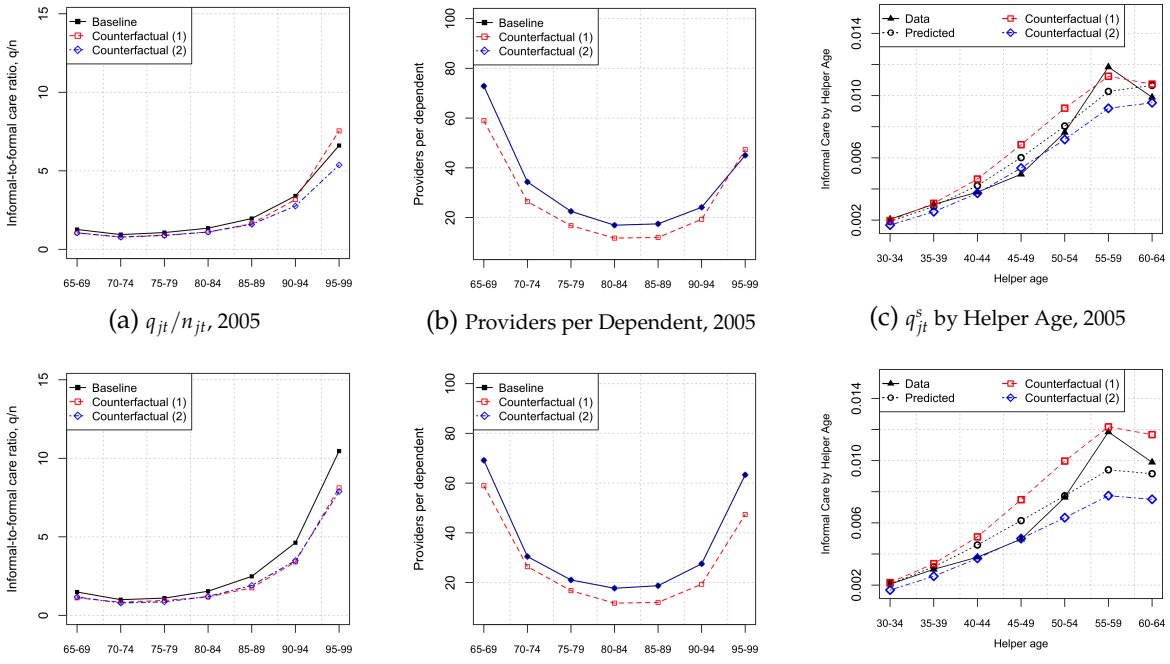


Figure D.2: Informal share of the total cost of elder care under counterfactuals (1) and (2) against the baseline, under *oc* (black) and *rc* (blue), with the care-labor share (red); line type and symbol identify the counterfactual.



(a) q_{jt}/n_{jt}, 2005 (b) Providers per Dependent, 2005 (c) q^s_{jt} by Helper Age, 2005

(d) q_{jt}/n_{jt}, 2020 (e) Providers per Dependent, 2020 (f) q^s_{jt} by Helper Age, 2020

Figure D.3: Care by age under counterfactuals (1) and (2) against the baseline. Panels (a)–(c) show 2005 and panels (d)–(f) show 2020: the informal-to-formal care ratio, q_{jt}/n_{jt}, and providers per dependent, $M_{j-\bar{j},t}/M^d_{jt}$, by recipient age, and care supplied per person by helper age, q^s_{jt}, as a fraction of a 24-hour day, against the targeted ATUS profile (Data).

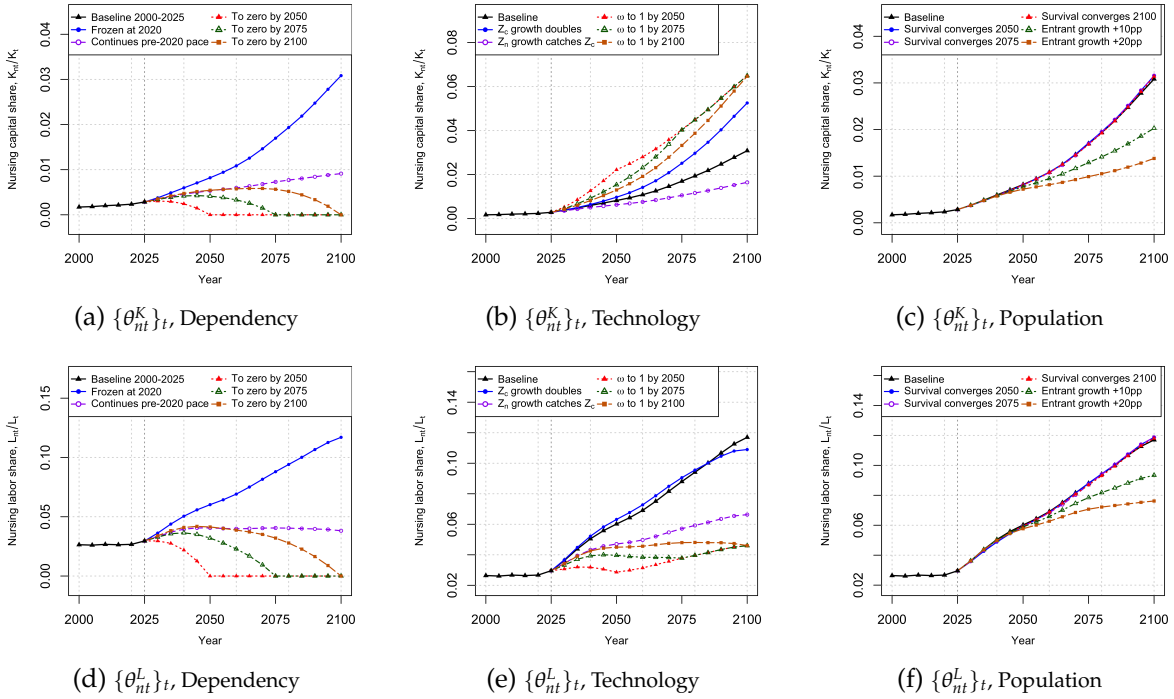
Figures D.2 and D.3 show how the two sets of drivers have shaped informal care over time. The rise in informal care relative to formal care at every recipient age requires both demographic and technological change. The demographic channel supplies more providers per dependent as the population ages, while cost disease raises the relative price of formal care, p_t. Cost disease works through the channel of Section 3.5: because $\rho_d < \rho_n$, a higher p_t cuts formal care and raises the value of informal care to helpers, who supply more of it. Note that either force alone delivers less informal care relative to formal care than the baseline (Figures D.3a and D.3d). The informal share of the value of elder care rises only when demography moves (see counterfactual (2) in Figure D.2). Cost disease, meanwhile, pulls labor into the formal sector, so the care-labor share, flat in the baseline, rises once technology and markups are frozen. The two forces pull in opposite directions on each provider with the pull amplified across the helper age profile (see Figure D.3f). With demography frozen there are fewer providers per dependent, so each provider supplies more care at nearly every age, but with technology and markups frozen formal care stays cheap, so each provides less. The baseline sits between the two, and the gap widens from 2005 to 2020 as both forces accumulate (compare Figures D.3c and D.3f).

D.3 Forward Projections

This appendix completes the projections of Section 5.3. Technical Appendix D.3.1 shows the aggregate series not shown there and the market-concentration scenario, and Technical Appendix D.3.2 shows informal care by age and the agent-level benefit–cost ratio.

D.3.1 Aggregates

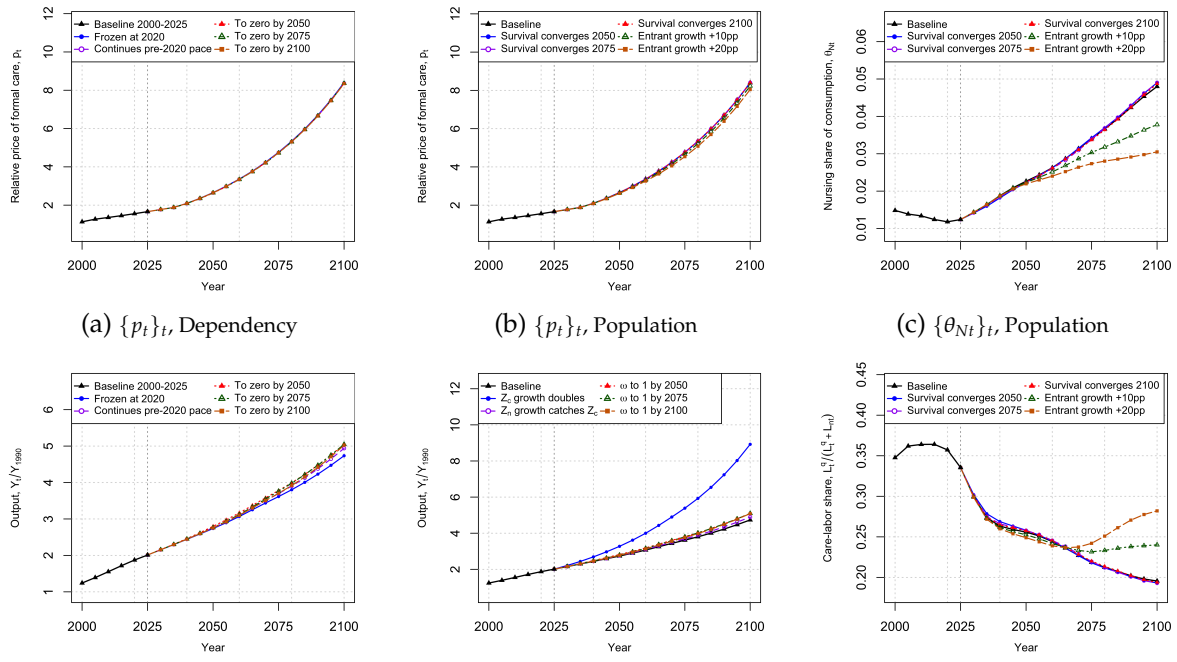
Figure D.4 shows the nursing sector's shares of capital and labor, and Figure D.5 the remaining series, under the scenarios of Figure 8. Figure D.6 shows every series under the market-concentration scenario.



(a) $\{\theta^K_{nt}\}_t$, Dependency (b) $\{\theta^K_{nt}\}_t$, Technology (c) $\{\theta^K_{nt}\}_t$, Population

(d) $\{\theta^L_{nt}\}_t$, Dependency (e) $\{\theta^L_{nt}\}_t$, Technology (f) $\{\theta^L_{nt}\}_t$, Population

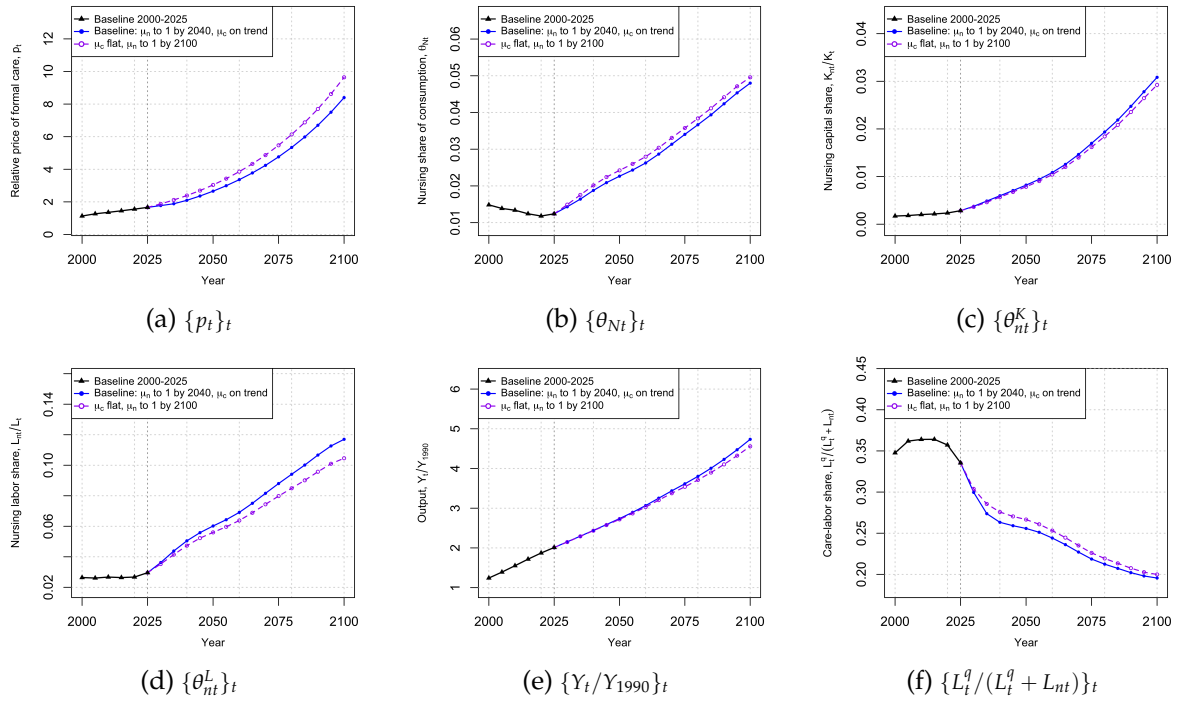
Figure D.4: Nursing-sector shares of capital and labor, 2000–2100, under the dependency-rate (Dependency), technology and sectoral wage catch-up (Technology), and survival and natality (Population) scenarios of Figure 8.



(a) $\{p_t\}_t$, Dependency (b) $\{p_t\}_t$, Population (c) $\{\theta_{Nt}\}_t$, Population

(d) $\{Y_t/Y_{1990}\}_t$, Dependency (e) $\{Y_t/Y_{1990}\}_t$, Technology (f) $\{L_t^q/(L_t^q + L_{nt})\}_t$, Population

Figure D.5: The remaining projected series, 2000–2100, under the dependency-rate (Dependency), technology and sectoral wage catch-up (Technology), and survival and natality (Population) scenarios of Figure 8.



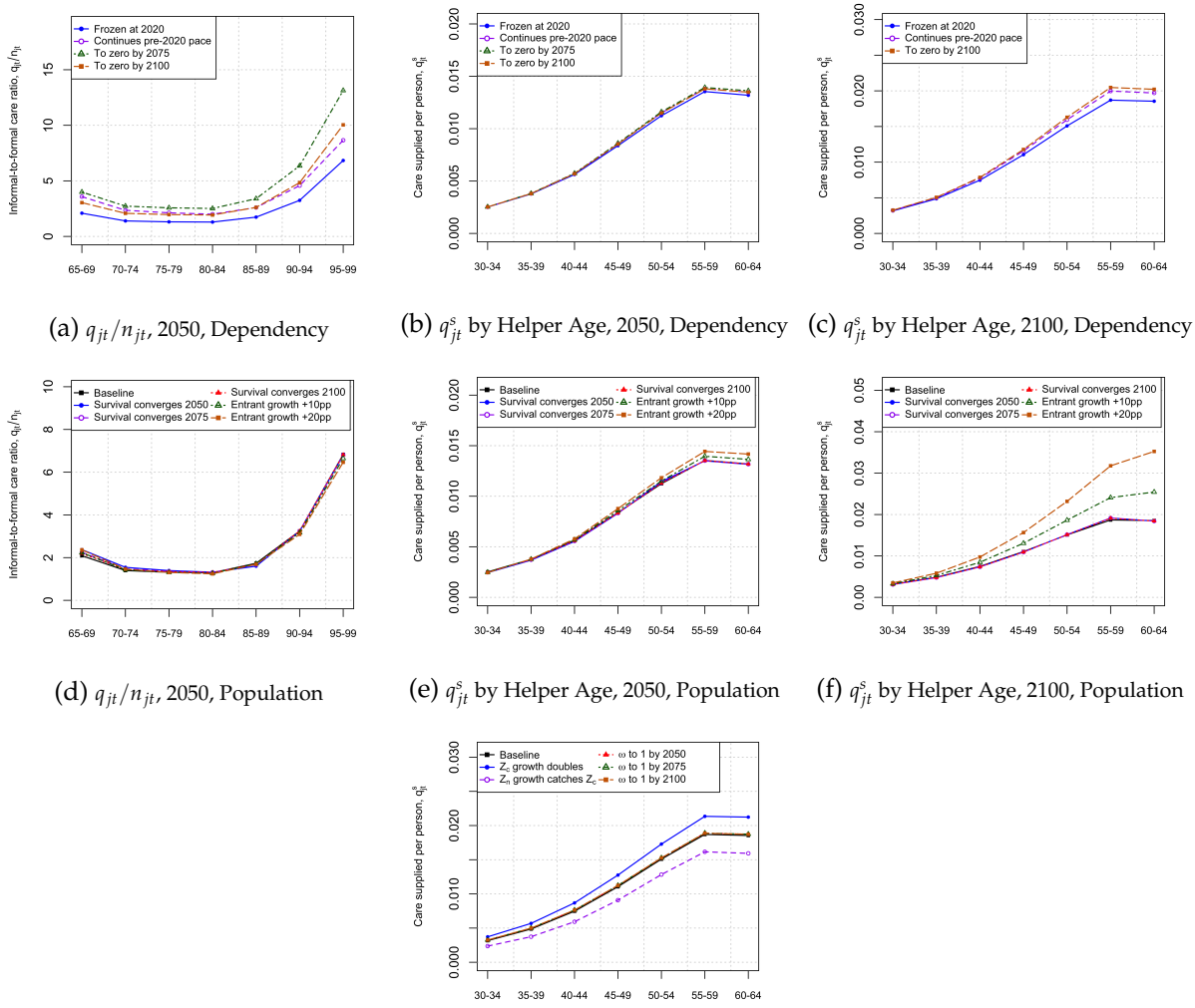
(a) $\{p_t\}_t$ (b) $\{\theta_{Nt}\}_t$ (c) $\{\theta_{nt}^K\}_t$

(d) $\{\theta_{nt}^L\}_t$ (e) $\{Y_t/Y_{1990}\}_t$ (f) $\{L_t^q/(L_t^q + L_{nt})\}_t$

Figure D.6: Projections under the market-concentration scenario, 2000–2100, against the baseline markup path. The black line is the baseline through the 2025 branch.

D.3.2 Informal Care

Figure D.7 shows informal care by age under the exact scenarios of Figure 9, and Figure D.8 under the market-concentration scenario. Projected forward, the agent-level benefit–cost ratio (D.2) remains below one at every recipient age, in every year through 2100 and in every scenario, though we do not show it here.



(a) q_{jt}/n_{jt}, 2050, Dependency

(b) q^s_{jt} by Helper Age, 2050, Dependency

(c) q^s_{jt} by Helper Age, 2100, Dependency

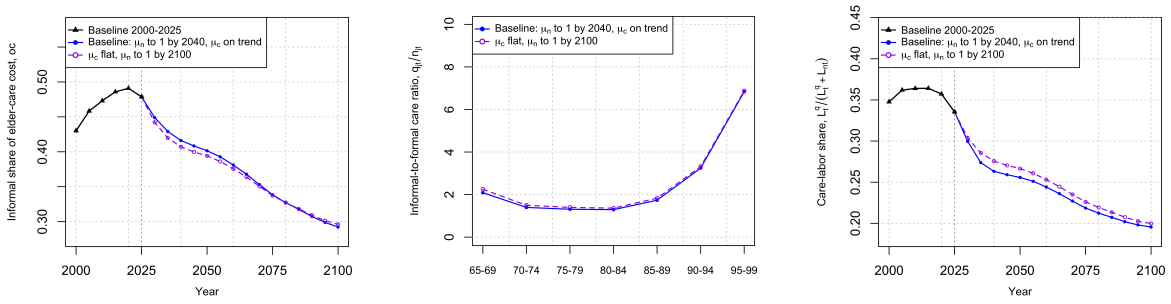
(d) q_{jt}/n_{jt}, 2050, Population

(e) q^s_{jt} by Helper Age, 2050, Population

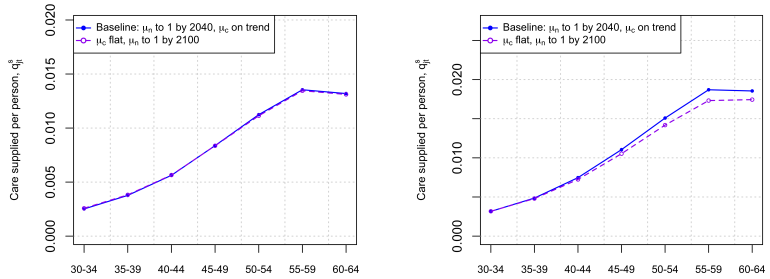
(f) q^s_{jt} by Helper Age, 2100, Population

(g) q^s_{jt} by Helper Age, 2100, Technology

Figure D.7: Informal care by age under the dependency-rate (Dependency, panels (a)–(c)) and survival and natality (Population, panels (d)–(f)) scenarios of Figure 9: the informal-to-formal care ratio by recipient age in 2050, and care supplied per person by helper age, q^s_{jt}, in 2050 and 2100. Panel (g) shows care supplied per person in 2100 under the technology scenarios, whose 2050 profiles are in Figure 9.



(a) Informal Share of Elder Care, oc (b) q_{jt}/n_{jt}, 2050 (c) $\{L_t^q/(L_t^q + L_{nt})\}_t$



(d) q_{jt}^s by Helper Age, 2050 (e) q_{jt}^s by Helper Age, 2100

Figure D.8: Informal care under the market-concentration scenarios, analogous to Figures D.2 and D.3. Panels (a) and (c) show, over 2000–2100, the informal share of the total cost of elder care under opportunity cost and the care-labor share; the black line is the baseline through the 2025 branch. Panel (b) shows the informal-to-formal care ratio by recipient age in 2050, and panels (d) and (e) care supplied per person by helper age, q_{jt}^s, in 2050 and 2100, each scenario against the baseline continuation. A scenario with no dependents in a given year is not drawn.

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